

Asymptotically Optimal Mixed-State Cloning

Jiani Fei* Jinzhao Wang*,[†]

Abstract

We determine asymptotic global fidelities of cloning generic mixed states with simple full-rank spectra. In fixed dimension d , let n inputs and m_n outputs satisfy $m_n/n \rightarrow \gamma > 1$. For a known simple full-rank spectrum $p_1 > \dots > p_d > 0$, we prove that the optimal root fidelity between the entire output and the product target state is

$$F_{\text{orb}}(\gamma, p) = \prod_{i < j} \frac{\sqrt{\gamma} + \sqrt{q_{ij}(\gamma - 1 + q_{ij})}}{\gamma + q_{ij}}, \quad q_{ij} = \frac{p_j}{p_i}.$$

When the spectrum is unknown, one spectrum-independent cloner attains

$$F_{\text{univ}}(\gamma, p) = \left(\frac{2\sqrt{\gamma}}{1 + \gamma} \right)^{(d-1)/2} F_{\text{orb}}(\gamma, p),$$

uniformly on compact subsets of the simple full-rank chamber and is minimax optimal. The two factors quantify spectral and eigenbasis uncertainty. Our channels combine classical processing of Schur–Weyl labels with a covariant Cartan-sector channel. Local asymptotic normality and a Gaussian fidelity bound against arbitrary channels yield the converses. For flat projector states $\rho = P/r$, with P a rank- r orthogonal projector, we separately prove the optimum $\gamma^{-r(d-r)/2}$. Furthermore, computing the exact asymptotic fidelity of purify–clone–trace (PCT) at simple full-rank spectra, and bounding it from above for projector states, we show that our cloners strictly outperform PCT at every fixed gain for simple full-rank spectra and rank- r projector states with $1 < r < d$.

Generative AI usage. The original idea for the cloner construction is due to the authors. We initially conjectured its asymptotic optimality but did not have a complete proof. We used OpenAI’s ChatGPT (Sol 5.6 and 6) and Anthropic’s Claude (Fable 5 and 5.1) to assist with the analysis: the evaluation of the cloning fidelity and the converse argument benefited from their suggestions. The same tools assisted with language editing, manuscript organization, and figure preparation. All AI-assisted material was reviewed and revised by the authors, who take full responsibility for the accuracy and content of the manuscript.

**Leinweber Institute for Theoretical Physics, Stanford University, Stanford, CA 94305, USA*

[†]*Zyphra, San Francisco, CA 94105, USA. Email: wang.jinzhao226@gmail.com*

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1 Introduction

The no-cloning theorem rules out perfect copying of an arbitrary unknown quantum state [WZ82, Die82]. Approximate cloning asks how well a channel can turn n identical inputs into m copies, at *gain* m/n . We compare the entire output with the product target $\rho^{\otimes m}$, rather than testing its marginals as in broadcasting [BCFJS96]. Optimal pure-state cloners were established in the 1990s [Wer98, KW99], but mixed-state cloning has remained open [SIGA05]: existing results treat restricted families, different loss criteria, or high-fidelity sample complexity, without determining the optimal global fidelity in our setting [BDMP05, BDMP06, DF07, FLS07, Kaz10, LTHC26, FGSS26, JSO26]. We resolve this fixed-gain global-fidelity problem in the generic mixed-state regime of simple full-rank spectra: the eigenvalues are strictly positive and pairwise distinct. Such states form an open dense subset of the state space.¹ We determine the known-spectrum orbit optimum and, when the spectrum is unknown, the compact-set minimax optimum for spectral sets with dense full-dimensional interior.

1.1 Results

Fix an integer $d \geq 2$ and positive integers m_n such that

$$\gamma_n := \frac{m_n}{n} \longrightarrow \gamma > 1. \quad (1)$$

In particular, $m_n > n$ for all sufficiently large n .² Let $\mathbf{A}_d := \{x \in \mathbb{R}^d : \sum_i x_i = 1\}$ and define

$$\Delta_d^\circ := \{p \in \mathbf{A}_d : p_1 > \cdots > p_d > 0\}. \quad (2)$$

For every ordered positive spectrum $p \in \mathbf{A}_d$ with $p_1 \geq \cdots \geq p_d > 0$, including repeated eigenvalues, set $\rho_p = \text{diag}(p_1, \dots, p_d)$ and, for $g \in \text{U}(d)$, $\rho_{p,g} = g\rho_p g^*$. For positive trace-class operators A, C , not necessarily normalized, we write

$$F(A, C) = \|\sqrt{A}\sqrt{C}\|_1. \quad (3)$$

For states, this is the root (unsquared) Uhlmann fidelity. For nonnegative functions or measures, the same notation denotes their Hellinger affinity. In particular, $F(P, Q) = \sum_x \sqrt{P(x)Q(x)}$ on a countable space. The target in every statement below is the entire product state $\rho_{p,g}^{\otimes m_n}$. For $p \in \Delta_d^\circ$, define

$$q_{ij} = \frac{p_j}{p_i}, \quad f_\gamma(q) = \frac{\sqrt{\gamma} + \sqrt{q(\gamma - 1 + q)}}{\gamma + q}, \quad (4)$$

and

$$F_{\text{orb}}(\gamma, p) = \prod_{i < j} f_\gamma(q_{ij}), \quad F_{\text{cl}}(\gamma, d) = \left(\frac{2\sqrt{\gamma}}{1 + \gamma} \right)^{(d-1)/2}, \quad F_{\text{univ}}(\gamma, p) = F_{\text{cl}}(\gamma, d) F_{\text{orb}}(\gamma, p). \quad (5)$$

The factor F_{orb} measures the cost of amplifying the unknown eigenbasis. The *classical fidelity factor* F_{cl} accounts for the additional cost of reproducing the unknown spectral fluctuations.

¹They also have full measure under Hilbert–Schmidt volume: the excluded states are zeros of the determinant or the characteristic-polynomial discriminant.

²At the finitely many exceptional indices with $m_n \leq n$, our cloners are defined as the exact partial trace channel (and the identity when $m_n = n$). This convention does not affect any limit, and all construction formulas below concern $m_n > n$.

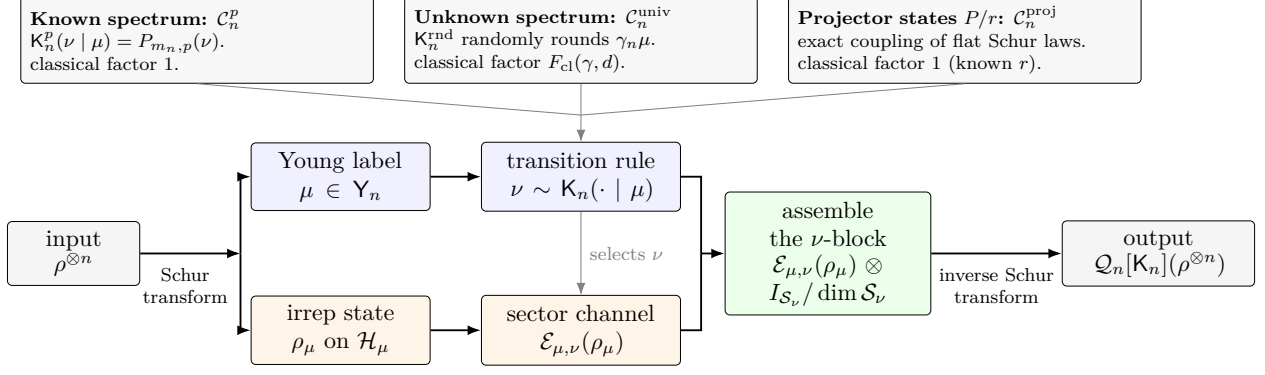


Figure 1: **Schur–Cartan cloner construction.** A Young-label transition rule selects ν , while the covariant channel $\mathcal{E}_{\mu,\nu}$ processes the conditional irrep state. Known-spectrum, unknown-spectrum, and projector cloners differ only in their label rules, shown above. Their asymptotic fidelities are given in [Theorems 1.1 to 1.3](#).

Known spectrum. For fixed p , the target Young-label distribution is known and can be sampled exactly at size m_n . Only the conditional irrep sectors encoding the eigenbasis need to be cloned.

Theorem 1.1 (Known-spectrum optimum). *For every $p \in \Delta_d^\circ$,*

$$\lim_{n \rightarrow \infty} \sup_{\mathcal{M}_n} \inf_{g \in \mathbf{U}(d)} F\left(\mathcal{M}_n(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}\right) = F_{\text{orb}}(\gamma, p), \quad (6)$$

where the supremum is over all quantum channels from n to m_n qudits. The explicit sequence of channels $\mathcal{C}_n^p$ defined by [equation \(25\)](#) asymptotically attains the displayed value. Its fidelity converges to $F_{\text{orb}}(\gamma, p)$ uniformly for p in every compact $K \Subset \Delta_d^\circ$ and $g \in \mathbf{U}(d)$.

Unknown spectrum. Without p , the target Young label law must instead be generated from the measured input Young label. Its $n^{-1/2}$ -scale spectral fluctuation must be dilated to the output scale, producing the factor $F_{\text{cl}}(\gamma, d)$.

Theorem 1.2 (Unknown-spectrum cloner fidelity and compact-set optimum). *There exists a sequence of spectrum-independent, $\mathbf{U}(d)$ -covariant channels $\mathcal{C}_n^{\text{univ}}$, defined by [equation \(28\)](#), with the following properties.*

(a) *For every compact $K \Subset \Delta_d^\circ$,*

$$\lim_{n \rightarrow \infty} \sup_{\substack{p \in K \\ g \in \mathbf{U}(d)}} \left| F\left(\mathcal{C}_n^{\text{univ}}(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}\right) - F_{\text{univ}}(\gamma, p) \right| = 0. \quad (7)$$

(b) *Let $K \Subset \Delta_d^\circ$ be nonempty and satisfy $K = \overline{\text{int}_{\mathbf{A}_d} K}^{\mathbf{A}_d}$. Then*

$$\lim_{n \rightarrow \infty} \sup_{\mathcal{M}_n} \inf_{\substack{p \in K \\ g \in \mathbf{U}(d)}} F\left(\mathcal{M}_n(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}\right) = \inf_{p \in K} F_{\text{univ}}(\gamma, p), \quad (8)$$

where the supremum is over all quantum channels from n to m_n qudits.

Part (a) evaluates one prescribed cloner, even on singleton or lower-dimensional sets. In part (b), the competing channel may depend on K , but not on the actual p or g . The interior is taken in the full $(d-1)$ -dimensional hyperplane $\mathbf{A}_d$: its spectral directions give the converse factor $F_{\text{cl}}(\gamma, d)$, and density extends the bound to the infimum over K . See [Section 3.3](#). The converse in part (b) is minimax: no competing channel sequence can asymptotically exceed $\inf_{p \in K} F_{\text{univ}}(\gamma, p)$ in worst-case fidelity over $K \times \text{U}(d)$, even though it may outperform $\mathcal{C}_n^{\text{univ}}$ at particular (p, g) . For the singleton $K = \{p\}$, which fails the closure condition in part (b), the known-spectrum channel $\mathcal{C}_n^p$ is admissible and attains $F_{\text{orb}}(\gamma, p) > F_{\text{univ}}(\gamma, p)$. Part (a) still evaluates the spectrum-independent cloner.

A natural example bounds the smallest eigenvalue and every adjacent gap away from zero. For $0 < \kappa < 2/[d(d+1)]$, take

$$\Delta_{d,\kappa} = \{p \in \mathbf{A}_d : p_d \geq \kappa, p_i - p_{i+1} \geq \kappa \ (i < d)\}.$$

This is a nonempty compact subset of Δ_d° and satisfies $\Delta_{d,\kappa} = \overline{\text{int}_{\mathbf{A}_d} \Delta_{d,\kappa}}^{\mathbf{A}_d}$, so (8) applies.³

[Figure 1](#) summarizes the common construction. Schur–Weyl duality separates a Young label, carrying spectral information, from a conditional irreducible state, carrying eigenbasis information [\[KW01\]](#). Each cloner processes the label classically and applies the same multiplicity-one Cartan-sector channel to the irreducible state. Only the label transition rule changes [\[PRV67, FH91\]](#). We develop the rules in [Section 2](#). For the converses, local asymptotic normality (LAN) gives a classical Gaussian shift and displaced thermal modes. A flat-prior Gaussian fidelity bound applies to arbitrary channels, while least-noise amplifiers attain the quantum factor. See [Section 3](#).

Although the preceding results do not cover rank-deficient or spectrally degenerate states, we also study flat projector states P/r , with P an unknown rank- r projector, and determine their optimal cloning fidelity.

Theorem 1.3 (Cloning projector states). *For fixed $1 \leq r < d$, let $\text{Gr}(r, d)$ denote the set of rank- r orthogonal projectors on $\mathbb{C}^d$. Then*

$$\lim_{n \rightarrow \infty} \sup_{\mathcal{M}_n} \inf_{P \in \text{Gr}(r, d)} F(\mathcal{M}_n((P/r)^{\otimes n}), (P/r)^{\otimes m_n}) = \gamma^{-r(d-r)/2}. \quad (9)$$

The limit is attained by our cloner $\mathcal{C}_n^{\text{proj}}$ constructed in [Section 4.2](#).

For $r = 1$, the value $\gamma^{-(d-1)/2}$ recovers the optimal asymptotic global fidelity for pure-state cloning [\[Wer98\]](#). For $r = d$, the family contains only I_d/d , and the value is one. The projector cloner uses the same Schur–Cartan architecture, with a different Young-label transition rule ([Fig. 1](#)). Its classical factor is 1 and its asymptotic sector factor is $\gamma^{-r(d-r)/2}$. Exact Schur-sector compression gives its fidelity, and a direct finite-dimensional converse in [Proposition 4.2](#) holds at every n . When $m_n = \gamma n + O(1)$, its upper bound approaches this limit at rate $O(n^{-1})$.

A natural idea for mixed-state cloning is to first purify the mixed states and then apply Werner’s optimal pure state cloner [\[Wer98\]](#). This approach was recently proposed and studied in [\[LTHC26, FGSS26, JSO26\]](#). Here we show that our optimal cloner has better performance.

For a rank bound $1 \leq r \leq d$, let $E = \mathbb{C}^r$, let $\mathcal{P}_n^{(r)}$ be the random-purification channel with purifying register E [\[TWZ25, GML25\]](#), and let $\mathcal{W}_{n,m}^{(dr)}$ be Werner’s pure-state n -to- m cloner in dimension dr . For $m \geq n$, the purify–clone–trace (PCT) channel is the following composition:

$$\mathcal{C}_{n,m}^{\text{PCT},r} := \text{Tr}_{E^{\otimes m}} \circ \mathcal{W}_{n,m}^{(dr)} \circ \mathcal{P}_n^{(r)}.$$

³Being convex with nonempty relative interior, $\Delta_{d,\kappa}$ is the closure of that interior. For a relative interior point, choose $\kappa' \in (\kappa, 2/[d(d+1)])$ and set $p_i = 1/d + ((d+1)/2 - i)\kappa'$. At the endpoint $\kappa = 2/[d(d+1)]$, the constraints force a singleton, so the compact-set theorem no longer applies.

For every state ρ of rank at most r , we compare $\mathcal{C}_{n,m}^{\text{PCT},r}(\rho^{\otimes n})$ with the entire product target $\rho^{\otimes m}$. The formulas for $\mathcal{P}_n^{(r)}$ and $\mathcal{W}_{n,m}^{(dr)}$ are (136) and (135) in Appendix D.1.

Theorem 1.4 (Comparison with purify-clone-trace). *Fix $\gamma = \lim_{n \rightarrow \infty} m_n/n > 1$, and let $\mathcal{C}_{n,m}^{\text{PCT},r}$ be the purify-clone-trace channel with rank bound r .*

(a) *For $p \in \Delta_d^\circ$, let $\rho_p = \text{diag}(p_1, \dots, p_d)$. Then PCT with rank bound d has*

$$\lim_{n \rightarrow \infty} F\left(\mathcal{C}_{n,m_n}^{\text{PCT},d}(\rho_p^{\otimes n}), \rho_p^{\otimes m_n}\right) = F_{\text{PCT}}(\gamma, p) < F_{\text{univ}}(\gamma, p), \quad (10)$$

where $q_{ij} = p_j/p_i$ and

$$F_{\text{PCT}}(\gamma, p) = \left(\frac{\sqrt{2\gamma-1}}{\gamma}\right)^{(d-1)/2} \prod_{i < j} \frac{\sqrt{\gamma + (\gamma-1)q_{ij}} + \sqrt{q_{ij}(\gamma-1 + \gamma q_{ij})}}{\gamma(1+q_{ij})}. \quad (11)$$

(b) *Let $1 < r < d$ and let P be a rank- r orthogonal projector on $\mathbb{C}^d$. Then PCT with rank bound r satisfies*

$$\limsup_{n \rightarrow \infty} F\left(\mathcal{C}_{n,m_n}^{\text{PCT},r}((P/r)^{\otimes n}), (P/r)^{\otimes m_n}\right) \leq \gamma^{-r(d-r)/2} \left(\frac{\sqrt{2\gamma-1}}{\gamma}\right)^{(r-1)/2} < \gamma^{-r(d-r)/2}. \quad (12)$$

For $r = 1$, PCT and our cloner both reduce to Werner's pure-state cloner.

See Section 5 for more details. The proof is provided in Appendix D.

Figure 2 illustrates the full-rank and projector-state comparisons using two examples.

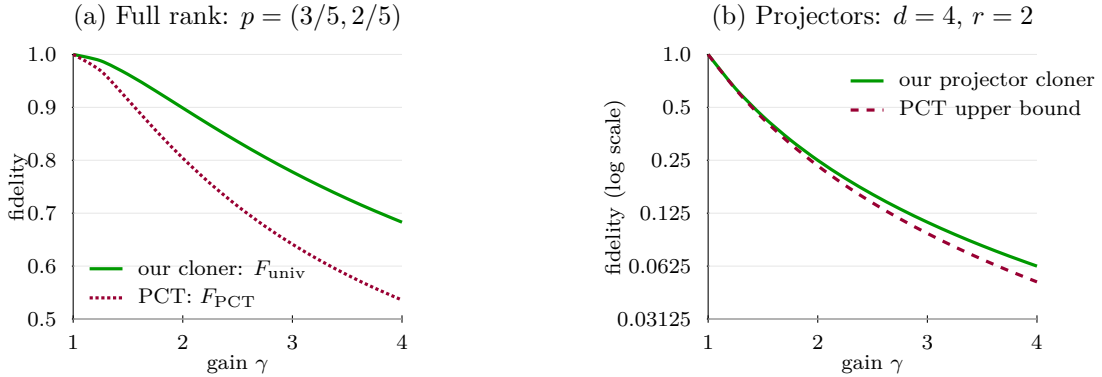


Figure 2: **Fixed-gain comparison with PCT.** (a) For $p = (3/5, 2/5)$, both curves are exact asymptotic fidelities for protocols without the spectrum (Theorems 1.2 and 1.4). (b) For rank-two projectors in $d = 4$, the solid curve is our cloner's exact fidelity γ^{-2} . The dashed curve is the PCT upper bound $\gamma^{-2}(\sqrt{2\gamma-1}/\gamma)^{1/2}$ (Theorems 1.3 and 1.4). The gap is strict for every simple full-rank spectrum and every $1 < r < d$.

1.2 Scope and limitations

Apart from flat projector states, our theorems require a simple full-rank spectrum. We do not treat nonflat rank-deficient spectra, spectral degeneracies, or neighborhoods in which the rank changes. The worst-case minimax value over all states and finite-copy optimality of our channels also remain open. Our asymptotic estimates hold at fixed gain $\gamma > 1$. Expected extensions to degenerate and rank-deficient spectra, and the noncommuting limits at degeneracies, are discussed in Section 6.4.

1.3 Earlier and related work

Pure-state cloning. Universal qubit cloners [BH96, GM97, BDE+98] were followed by Werner’s solution of global universal cloning in arbitrary dimension [Wer98] and Keyl–Werner’s optimality result for individual-clone tests [KW99]. State-dependent global cloning was developed in [BDE+98, CB99]. The many-output limit for individual clones is tied to estimation [MP95, BEM98, BA06, CD06]. For joint fidelity, optimal measure-and-prepare outputs need not be independent copies of an estimate [CY14]. This distinction also matters for coherent-state and phase-covariant cloning [CKNWW05, KC22]: Cerf et al. prove Gaussian optimality for joint fidelity, although non-Gaussian operations can improve single-clone fidelity. Reviews include [SIGA05, Chi11, FWJ+14].

Mixed-state cloning, broadcasting, and PCT. Exact broadcasting, and hence cloning, of a noncommuting family is impossible [BCFJS96, KH08]. Approximate results include global-fidelity, relative-error, and state-dependent bounds for finite ensembles [Ras03a, Ras03b, Ras03c], cloners for symmetric-subspace inputs and identical mixed qubits under one-site shrinking-factor criteria [Fan03, FLS07], a restricted mixed-qubit construction under averaged Hilbert–Schmidt loss [Kaz10], and entropic limitations on simultaneous approximate cloning and broadcasting [LW17]. Cirac–Ekert–Macchiavello solved mixed-qubit purification and a cloning/estimation problem under local fidelity [CEM99]. Keyl–Werner studied asymptotic purification rates under both one-site and joint-product tests [KW01p]. Multi-input broadcasting and superbroadcasting also optimize marginals and permit correlations [DMP05, BDMP05, BDMP06, DF07]. These criteria do not determine fidelity with the joint product target.

Random purification [TWZ25, GML25] also reduces mixed-state tomography to pure-state tomography [PSTW25]. Combined with pure-state cloning, it gives a global-fidelity guarantee for bounded-rank mixed states [LTHC26]. Concurrent work shows that PCT attains the optimal sample complexity up to constants for squared fidelity $1 - \epsilon$, uniformly over states of rank at most r [FGSS26, JSO26]. These worst-case high-fidelity results use a different order of limits from our exact fixed-gain values. See Section 6.3.

Gaussian cloning and LAN. Guță–Matsumoto solved cloning of displaced thermal oscillator states with known temperature under joint norm-distance loss, reducing it to amplification and proving Gaussian optimality by stochastic ordering [GM06]. Guță–Bowles–Adesso used LAN to derive trace-norm teleportation benchmarks for i.i.d. qubits and displaced thermal states [GBA10]. Bowles–Guță–Adesso used LAN for asymptotically optimal mixed-qubit purification and dilution under global trace-norm loss [BGA11]. Related methods solve qudit channel reversal [BGA12]. These norm-distance results do not determine our fidelity constants, separate spectral and orbital costs, or finite-dimensional cloners.

The relevant LAN framework was developed in [GJ07, GK06, KG09, YFG13]. Our fidelity amplification bound is attained by the least-noise amplifier [Cav82, GM06]. Low-rank LAN has a different mode structure [LN24]. We treat the boundary directly only for flat projector states.

1.4 Organization

Section 2 proves achievability for the Schur–Cartan cloners, using the sector and label estimates of Appendices A and B. Section 3 proves full-rank optimality via LAN and the Gaussian converse of Appendix C. Section 4 treats projector states independently. Section 5 compares our cloners with PCT. Its calculations are in Appendix D. Section 6 discusses spectral mixedness and sample complexity, then states conjectures and open problems.

Acknowledgments. We thank Patrick Hayden and Henry Purcell for early collaborations on mixed qubit cloning. JW thanks Daniel Bump and Cynthia Yan for discussions on covariant maps.

Glossary of notation

We use $U(d)$ for the unitary group and $\mathcal{T}_1(\mathcal{H})$ for trace-class operators. The dimension d is fixed, $\gamma_n = m_n/n \rightarrow \gamma > 1$ gives the gain and its limit, N is a generic sample size, and $r_d = \binom{d}{2}$. Fidelity is unsquared: $F(\rho, \sigma) = \|\sqrt{\rho}\sqrt{\sigma}\|_1$. Unless stated otherwise, limits are as $n \rightarrow \infty$ with d, γ fixed, and $o(1)$ is uniform only over the parameters specified in its statement. We usually say *Young label* for a partition $\lambda \vdash N$, identified with its Young diagram. *Partition* and *Young diagram* refer to the same label.

Symbols	Meaning
$A_d, \Delta_d^\circ; K \in \Delta_d^\circ$	Trace-one hyperplane, simple positive spectra, and compact spectral set. See (2) and Theorem 1.2.
$\rho_p, \rho_{p,g}; P \in \text{Gr}(r, d)$	Diagonal and rotated states, also for repeated positive eigenvalues. P/r is the rank- r projector state. See Section 1.1 and Theorem 1.3.
$q_{ij}, f_\gamma(q_{ij})$	Eigenvalue ratio and one-mode fidelity factor. See (4).
$F_{\text{orb}}, F_{\text{cl}}, F_{\text{univ}}$	Known-spectrum fidelity, extra factor for unknown spectra, and their product. See (5).
$Y_N; \mathcal{H}_\lambda, \mathcal{S}_\lambda$	Young labels and their representation spaces. See (13).
$P_{N,p}, \rho_{p,\lambda}; \mathsf{T}_N(p)$	Young-label law, conditional sector state, and typical-label set. See (14) and (15).
$K_n^p, K_n^{\text{rnd}}, K_n^{\text{proj}}; \mathcal{Q}_n[K_n]$	The three label rules and their Schur–Cartan lift. See Sections 2.1 and 4.2.
$\mathcal{C}_{\mu,\nu}, \mathcal{E}_{\mu,\nu}$	Sector channel and its extension to all label pairs. See (19) and (20).
$\mathcal{M}_n; \mathcal{C}_n^p, \mathcal{C}_n^{\text{univ}}, \mathcal{C}_n^{\text{proj}}$	Arbitrary cloner and our three constructions. See (25), (28), and (68).
$\mathsf{H}; h, z$	Eigenvalue-change space and local eigenvalue/eigenbasis coordinates. See Section 3.1.
$Y_p(z), g_{n,z}^{(p)}; \Omega_L$	Generator and unitary for a small eigenbasis rotation, bounded parameter box. See Section 3.1.
$\mathcal{F}, \mathcal{F}_{\text{orb}}; \hat{N}$	One- and r_d -oscillator spaces, and excitation-counting operator. See Section 3.1.
$\tau_p; \tau_p^{(\gamma)}$	Thermal state and its amplified version. See (40).
$\Phi_z^{(p)}, \Psi_z^{(p,\gamma)}$	Gaussian input and ideal target for eigenbasis changes. See (41).
$\Theta_{h,z}^{\text{in}}, \Theta_{h,z}^{\text{tar}}$	Gaussian input and ideal target including eigenvalue changes. See (44).

Symbols confined to one proof are defined there.

2 Cloner construction and achievability

The known- and unknown-spectrum protocols share the architecture in Fig. 1: a transition rule on Young labels lifts to a covariant quantum channel. We construct the two rules and combine their label estimates with a common sector analysis to evaluate the fidelities in Section 2.3.

Let $\mathbf{Y}_N = \{\lambda \vdash N : \ell(\lambda) \leq d\}$. Schur–Weyl duality gives

$$(\mathbb{C}^d)^{\otimes N} \cong \bigoplus_{\lambda \in \mathbf{Y}_N} \mathcal{H}_\lambda \otimes \mathcal{S}_\lambda, \quad (13)$$

where $\mathcal{H}_\lambda$ is the irreducible $U(d)$ -module of highest weight λ and $\mathcal{S}_\lambda$ is the corresponding Specht module. For $\rho_p = \text{diag}(p_1, \dots, p_d)$,

$$\rho_p^{\otimes N} = \bigoplus_{\lambda \in \mathbf{Y}_N} P_{N,p}(\lambda) \rho_{p,\lambda} \otimes \frac{I_{\mathcal{S}_\lambda}}{\dim \mathcal{S}_\lambda}, \quad P_{N,p}(\lambda) = \dim \mathcal{S}_\lambda s_\lambda(p), \quad \rho_{p,\lambda} = \frac{\pi_\lambda(\rho_p)}{s_\lambda(p)}. \quad (14)$$

Here π_λ is the irreducible polynomial representation of highest weight λ , acting on $\mathcal{H}_\lambda$, and $s_\lambda(p) = \text{Tr } \pi_\lambda(\rho_p)$ is the corresponding Schur polynomial. For $\rho_{p,g} = g\rho_p g^*$, replace $\rho_{p,\lambda}$ by $\rho_{p,g,\lambda} = U_\lambda(g)\rho_{p,\lambda}U_\lambda(g)^*$, where $U_\lambda(g) = \pi_\lambda(g)$. The Young law $P_{N,p}$ depends on p , but not on g .

Define the typical Young-label set

$$\mathbf{T}_N(p) = \left\{ \lambda \in \mathbf{Y}_N : \left\| \frac{\lambda}{N} - p \right\|_\infty \leq N^{-1/3} \right\}. \quad (15)$$

Lemma 2.1 (Uniform Young concentration). *For all sufficiently large N , depending only on d ,*

$$\sup_{\substack{p_1 \geq \dots \geq p_d > 0 \\ \sum_i p_i = 1}} P_{N,p}(\mathbf{T}_N(p)^c) \leq e^{-N^{1/3}/4}. \quad (16)$$

Equal positive eigenvalues are allowed.

Proof. The Weyl dimension formula gives $\dim \mathcal{H}_\lambda \leq (N+1)^{d(d-1)/2}$. Every weight ξ of $\mathcal{H}_\lambda$ has $\lambda - \xi$ equal to a nonnegative integral combination of positive roots. Since $p_j/p_i \leq 1$ for $i < j$, this gives $p^\xi \leq p^\lambda$. Hence $s_\lambda(p) \leq \dim \mathcal{H}_\lambda p^\lambda$. The standard bounds $\dim \mathcal{S}_\lambda \leq \binom{N}{\lambda_1, \dots, \lambda_d}$ and $\binom{N}{\lambda_1, \dots, \lambda_d} p^\lambda \leq \exp[-ND(\lambda/N \| p)]$ give

$$P_{N,p}(\lambda) \leq (N+1)^{d(d-1)/2} \exp[-ND(\lambda/N \| p)], \quad (17)$$

where D denotes classical relative entropy, using natural logarithms. This is the estimate underlying the Keyl–Werner spectrum large-deviation theorem [KW01].

There are at most $(N+1)^{d-1}$ partitions of N with at most d rows. If $\lambda \notin \mathbf{T}_N(p)$, then $\|\lambda/N - p\|_1 > N^{-1/3}$, and Pinsker’s inequality gives $D(\lambda/N \| p) \geq \frac{1}{2}N^{-2/3}$. Therefore

$$P_{N,p}(\mathbf{T}_N(p)^c) \leq (N+1)^{d(d-1)/2+d-1} e^{-N^{1/3}/2}.$$

For all sufficiently large N , the polynomial factor is at most $e^{N^{1/3}/4}$, which proves (16). □

2.1 Cartan-sector channel and label rules

Let λ, θ be dominant weights, with unit highest-weight vectors v_λ, v_θ . The Cartan component $\mathcal{H}_{\lambda+\theta}$ occurs with multiplicity one in $\mathcal{H}_\lambda \otimes \mathcal{H}_\theta$. Let

$$V_{\lambda,\theta} : \mathcal{H}_{\lambda+\theta} \longrightarrow \mathcal{H}_\lambda \otimes \mathcal{H}_\theta, \quad V_{\lambda,\theta} v_{\lambda+\theta} = v_\lambda \otimes v_\theta, \quad (18)$$

be the normalized intertwining isometry. For $\mu \in \mathbf{Y}_n$ and $\nu \in \mathbf{Y}_m$, write $\nu - \mu \vdash m - n$ when the coordinatewise difference $\nu - \mu$ is a partition of $m - n$, and call such a pair (μ, ν) *compatible*. If this holds, let $\omega = \nu - \mu$ and define

$$\mathcal{C}_{\mu,\nu}(X) = \frac{\dim \mathcal{H}_\mu}{\dim \mathcal{H}_\nu} V_{\mu,\omega}^* (X \otimes I_{\mathcal{H}_\omega}) V_{\mu,\omega}. \quad (19)$$

The partial trace $\text{Tr}_{\mathcal{H}_\omega}(V_{\mu,\omega} V_{\mu,\omega}^*)$ commutes with the irreducible $U(d)$ -action on $\mathcal{H}_\mu$. Hence Schur's lemma and a trace comparison give $\text{Tr}_{\mathcal{H}_\omega}(V_{\mu,\omega} V_{\mu,\omega}^*) = \frac{\dim \mathcal{H}_\nu}{\dim \mathcal{H}_\mu} I_{\mathcal{H}_\mu}$. The map in (19) is completely positive. The identity above makes it trace preserving, and equivariance of $V_{\mu,\omega}$ gives $U(d)$ -covariance.

When $\mu = (n, 0, \dots, 0)$ and $\nu = (m, 0, \dots, 0)$, the three irreps in (19) are symmetric, and the formula is Werner's universal pure-state n -to- m cloner [Wer98]. Thus the Cartan-sector channel extends Werner's construction from symmetric tensors to arbitrary compatible Schur sectors.

For every pair $(\mu, \nu) \in \mathbf{Y}_n \times \mathbf{Y}_m$, define a covariant channel $\mathcal{E}_{\mu,\nu} : \mathcal{T}_1(\mathcal{H}_\mu) \rightarrow \mathcal{T}_1(\mathcal{H}_\nu)$ by

$$\mathcal{E}_{\mu,\nu}(X) = \begin{cases} \mathcal{C}_{\mu,\nu}(X), & \nu - \mu \vdash m - n, \\ \text{Tr}(X) I_{\mathcal{H}_\nu} / \dim \mathcal{H}_\nu, & \text{otherwise.} \end{cases} \quad (20)$$

Let $K_n : \mathbf{Y}_n \rightsquigarrow \mathbf{Y}_{m_n}$ be any transition rule: for each input label μ , the output label ν is selected with probability $K_n(\nu \mid \mu) \geq 0$, where $\sum_\nu K_n(\nu \mid \mu) = 1$. Its Schur–Cartan lifting $\mathcal{Q}_n[K_n]$ acts as follows: apply the Schur transform, measure μ , discard the input Specht factor, sample ν from $K_n(\cdot \mid \mu)$, apply $\mathcal{E}_{\mu,\nu}$, adjoin $I_{\mathcal{S}_\nu} / \dim \mathcal{S}_\nu$, and apply the inverse Schur transform. Each step is completely positive and trace preserving, so their composition is a channel. Because the label measurement and sampling are invariant and every map $\mathcal{E}_{\mu,\nu}$ is covariant, $\mathcal{Q}_n[K_n]$ is $U(d)$ -covariant.

Proposition 2.2 (Fidelity of a lifted transition rule). *Fix $p \in \Delta_d^\circ$ and $g \in U(d)$. Let*

$$J_{n,p}(\mu, \nu) = P_{n,p}(\mu) K_n(\nu \mid \mu), \quad Q_{m_n,p}(\nu) = \sum_\mu J_{n,p}(\mu, \nu), \quad (21)$$

so $J_{n,p}$ is the joint input-output label law and $Q_{m_n,p}$ its output marginal. For $\nu \in \mathbf{Y}_{m_n}$, let

$$\Xi_{n,p,\nu} = \sum_{\mu \in \mathbf{Y}_n} J_{n,p}(\mu, \nu) \mathcal{E}_{\mu,\nu}(\rho_{p,\mu}) \quad (22)$$

be the unnormalized irreducible output block, so that $\text{Tr} \Xi_{n,p,\nu} = Q_{m_n,p}(\nu)$. The lifted channel has fidelity

$$F(\mathcal{Q}_n[K_n](\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}) = \sum_{\nu \in \mathbf{Y}_{m_n}} \sqrt{P_{m_n,p}(\nu)} F(\Xi_{n,p,\nu}, \rho_{p,\nu}). \quad (23)$$

Proof. By covariance of $\mathcal{Q}_n[K_n]$ and unitary invariance of fidelity, we may take $g = I$. Under the output Schur decomposition, the actual and target states are

$$\bigoplus_{\nu \in \mathbf{Y}_{m_n}} \Xi_{n,p,\nu} \otimes \frac{I_{\mathcal{S}_\nu}}{\dim \mathcal{S}_\nu}, \quad \bigoplus_{\nu \in \mathbf{Y}_{m_n}} P_{m_n,p}(\nu) \rho_{p,\nu} \otimes \frac{I_{\mathcal{S}_\nu}}{\dim \mathcal{S}_\nu}.$$

Fidelity is additive over orthogonal direct sums, multiplicative over tensor products, and satisfies $F(A, cB) = \sqrt{c} F(A, B)$ for $c \geq 0$. Since $F(I_{\mathcal{S}_\nu} / \dim \mathcal{S}_\nu, I_{\mathcal{S}_\nu} / \dim \mathcal{S}_\nu) = 1$, this proves (23). $\square$

Known spectrum: exact target-label sampling. Fix $p \in \Delta_d^\circ$. Since p is known, we can sample the target Young-label law $P_{m_n, p}$ directly, independently of the input label μ . Typical input and output labels are compatible, so the Cartan-sector channel applies. Define

$$\mathbf{K}_n^p(\nu \mid \mu) = P_{m_n, p}(\nu). \quad (24)$$

Set

$$\mathcal{C}_n^p := \mathcal{Q}_n[\mathbf{K}_n^p]. \quad (25)$$

Both cases of (20) output into the sampled ν -sector. Thus the output Young-label law on $\rho_{p, g}^{\otimes n}$ is

$$Q_{m_n, p}(\nu) = \sum_{\mu} P_{n, p}(\mu) \mathbf{K}_n^p(\nu \mid \mu) = P_{m_n, p}(\nu) \sum_{\mu} P_{n, p}(\mu) = P_{m_n, p}(\nu)$$

for every n and g .

Unknown spectrum: randomized rounding. For a typical input label $\mu \in \mathbf{Y}_n$, we want an output label near $\gamma_n \mu$, where $\gamma_n = m_n/n$: the dilation has total size m_n and preserves the empirical spectrum, since $\gamma_n \mu / m_n = \mu / n$. Its coordinates need not be integers, so we round it randomly.

For $N \in \mathbb{N}$, let

$$\mathbf{L}_N = \left\{ x \in \mathbb{Z}^d : \sum_i x_i = N \right\}.$$

Given $\mu \in \mathbf{Y}_n$, draw independent variables $y_1, \dots, y_{d-1}$, each uniform on $[-1/2, 1/2)$, and set

$$\tilde{\nu}_i = \left\lfloor \gamma_n(\mu_i + y_i) + \frac{1}{2} \right\rfloor \quad (1 \leq i < d), \quad \tilde{\nu}_d = m_n - \sum_{i < d} \tilde{\nu}_i. \quad (26)$$

Here $\lfloor x + 1/2 \rfloor$ rounds x to the nearest integer. Thus $\tilde{\nu} \in \mathbf{L}_{m_n}$. Let $R_n(\tilde{\nu} \mid \mu)$ denote its distribution. The shifts spread mass across the output lattice. To see why this matters, suppose $\gamma_n = \gamma$ is an integer and instead set $\nu = \gamma \mu$. This rule restricts the output to a sublattice of index γ^{d-1} in $\mathbf{L}_{m_n}$. The Young local limit gives label affinity $\gamma^{-(d-1)/2} F_{\text{cl}}(\gamma, d)$, rather than $F_{\text{cl}}(\gamma, d)$.⁴ Retain the rounded label when compatible, and otherwise use the one-row partition $(m_n, 0, \dots, 0)$ as the fallback:

$$\nu = \begin{cases} \tilde{\nu}, & \tilde{\nu} - \mu \vdash m_n - n, \\ (m_n, 0, \dots, 0), & \text{otherwise.} \end{cases} \quad (27)$$

The conditional law of ν defines a transition rule $\mathbf{K}_n^{\text{rnd}} : \mathbf{Y}_n \rightsquigarrow \mathbf{Y}_{m_n}$. Define

$$\mathcal{C}_n^{\text{univ}} := \mathcal{Q}_n[\mathbf{K}_n^{\text{rnd}}]. \quad (28)$$

Since the transition rule is independent of p , this is a spectrum-independent, $U(d)$ -covariant cloner.

2.2 Sector and label fidelity estimates

The block fidelity formula (23) leaves two tasks: evaluate the conditional state fidelity within typical output sectors, and compare the output Young-label law with the target law. The sector estimate must hold even when several input labels lead to the same output label. It applies to both cloners. Only the unknown-spectrum rule needs a separate label estimate.

⁴For $p = (3/5, 2/5)$ and $\gamma = 2$, the undithered and rounded label affinities are 0.6866 and 0.9710, respectively. After the common sector factor, the corresponding cloning fidelities are 0.6355 and 0.8988.

Proposition 2.3 (Uniform Cartan-sector fidelity). *Let $K \Subset \Delta_d^\circ$ and let $\delta_n \geq 0$ satisfy $\delta_n \rightarrow 0$. For all sufficiently large n , every $p \in K$, $\mu \in \mathbf{Y}_n$, and $\nu \in \mathbf{L}_{m_n}$ satisfying*

$$\left\| \frac{\mu}{n} - p \right\|_\infty \leq \delta_n, \quad \left\| \frac{\nu}{m_n} - p \right\|_\infty \leq \delta_n$$

also satisfies $\nu - \mu \vdash m_n - n$. In particular, $\nu \in \mathbf{Y}_{m_n}$. Moreover,

$$F(\mathcal{C}_{\mu,\nu}(\rho_{p,\mu}), \rho_{p,\nu}) = F_{\text{orb}}(\gamma, p) + o(1) \quad (29)$$

uniformly over all such p, μ, ν . More generally, for fixed p, ν as above and any probability distribution $(a_\mu)_\mu$ supported on input labels satisfying the same closeness condition,

$$F\left(\sum_\mu a_\mu \mathcal{C}_{\mu,\nu}(\rho_{p,\mu}), \rho_{p,\nu}\right) = F_{\text{orb}}(\gamma, p) + o(1), \quad (30)$$

uniformly over these distributions as well as p, ν .

The factor $F_{\text{orb}}(\gamma, p)$ naturally appears as the fidelity of the thermal states from [Section 3.1](#):

$$F_{\text{orb}}(\gamma, p) = F(\tau_p^{(\gamma)}, \tau_p).$$

To see this in the sector calculation, let $\alpha = e_i - e_j$ with $i < j$, $q_\alpha = p_j/p_i$, and $q_\alpha^{(\gamma)} = 1 - (1 - q_\alpha)/\gamma$. On weight spaces reached by a bounded number of lowerings from the highest weight ν , an orthonormalized Poincaré–Birkhoff–Witt (PBW) basis makes $\rho_{p,\nu}$ diagonal and $\mathcal{C}_{\mu,\nu}(\rho_{p,\mu})$ asymptotically diagonal. At a fixed basis index $\mathbf{r} = (r_\alpha)_\alpha$, their diagonal entries converge to the corresponding number-basis entries:

$$\begin{aligned} (\rho_{p,\nu})_{\mathbf{r},\mathbf{r}} &\longrightarrow \prod_\alpha (1 - q_\alpha) q_\alpha^{r_\alpha} = (\tau_p)_{\mathbf{r},\mathbf{r}}, \\ (\mathcal{C}_{\mu,\nu}(\rho_{p,\mu}))_{\mathbf{r},\mathbf{r}} &\longrightarrow \prod_\alpha (1 - q_\alpha^{(\gamma)}) (q_\alpha^{(\gamma)})^{r_\alpha} = (\tau_p^{(\gamma)})_{\mathbf{r},\mathbf{r}}. \end{aligned}$$

The powers of $q_\alpha^{(\gamma)}$ in the output entries come from the binomial expansion of the Cartan intertwiner: with $s_{\alpha,n} = (\mu_i - \mu_j)/(\nu_i - \nu_j) \rightarrow \gamma^{-1}$, its factor $(1 - s_{\alpha,n} + s_{\alpha,n} q_\alpha)^{r_\alpha}$ tends to $(q_\alpha^{(\gamma)})^{r_\alpha}$. A uniform tail bound extends this fixed-weight comparison to the full fidelity in (29). See [Appendix A](#).

The second estimate concerns only $\mathbf{K}_n^{\text{rnd}}$.

Proposition 2.4 (Randomized Young-label dilation). *There is a constant $c_{\text{rnd}} < \infty$, depending only on d and the fixed sequence (γ_n) , such that, for every compact $K \Subset \Delta_d^\circ$ and all sufficiently large n , every $p \in K$ and $\mu \in \mathbf{T}_n(p)$ satisfy*

$$\text{supp } \mathbf{K}_n^{\text{rnd}}(\cdot \mid \mu) \subseteq \{\nu \in \mathbf{Y}_{m_n} : \nu - \mu \vdash m_n - n, \quad \|\nu - \gamma_n \mu\|_\infty \leq c_{\text{rnd}}\}. \quad (31)$$

For every such K , the induced output-label law $Q_{m_n,p}$ from (21) with $\mathbf{K}_n = \mathbf{K}_n^{\text{rnd}}$,

$$F(Q_{m_n,p}, P_{m_n,p}) = F_{\text{cl}}(\gamma, d) + o(1) \quad \text{uniformly for } p \in K. \quad (32)$$

The randomized transition rule $\mathbf{K}_n^{\text{rnd}}$ is a discrete realization of the dilation $\mu \mapsto \gamma_n \mu$: the uniform shifts in (26) spread each input label over nearby output labels. Represent the input Young-label law $P_{n,p}$ by its piecewise-constant interpolation $\mathcal{I}_{n,p} P_{n,p}$, formed by centering at np , dividing by $\sqrt{n}$, and spreading each label mass over its lattice cell. If $\tilde{Q}_{m_n,p} := P_{n,p} \mathbf{R}_n$ is the law of the rounded

label $\tilde{\nu}$, the intersection-volume identity (109) identifies its embedded density $\mathcal{I}_{m_n,p}\tilde{Q}_{m_n,p}$ with the $\sqrt{\gamma_n}$ -dilation of $\mathcal{I}_{n,p}P_{n,p}$, averaged over each cell of the m_n -copy target-label lattice. Write

$$\Sigma_p := (\text{diag}(p) - pp^\top) \Big|_{\{x \in \mathbb{R}^d : \sum_i x_i = 0\}}. \quad (33)$$

The Young local limit says that $\mathcal{I}_{N,p}P_{N,p}$ converges to the density φ_{0,Σ_p} of $\mathbf{N}_{0,\Sigma_p}$, both for the input law at $N = n$ and the ideal output law at $N = m_n$. Thus the rounded output converges to $\mathbf{N}_{0,\gamma\Sigma_p}$, while the ideal output converges to $\mathbf{N}_{0,\Sigma_p}$. Their fidelity (Hellinger affinity) is

$$F(\mathbf{N}_{0,\gamma\Sigma_p}, \mathbf{N}_{0,\Sigma_p}) = \left(\frac{2\sqrt{\gamma}}{1+\gamma} \right)^{(d-1)/2} = F_{\text{cl}}(\gamma, d). \quad (34)$$

For typical input labels, every rounded output is compatible for all sufficiently large n . Fallback therefore changes only atypical inputs, whose total probability vanishes uniformly on compact spectral sets. The full proof is given in Appendix B.

2.3 Cloning fidelity

We use the following continuity estimate for positive trace-class operators A, C, A', C' with trace at most one:

$$|F(A, C) - F(A', C')| \leq \|A - A'\|_1^{1/2} + \|C - C'\|_1^{1/2}. \quad (35)$$

Indeed, the Powers–Størmer inequality bounds $\|\sqrt{A} - \sqrt{A'}\|_2$ by $\|A - A'\|_1^{1/2}$, and similarly for C, C' . Adding and subtracting $\sqrt{A'}\sqrt{C}$, the reverse triangle inequality and Schatten Hölder give (35), since $\|\sqrt{C}\|_2, \|\sqrt{A'}\|_2 \leq 1$.

The sector-mixture estimate gives a common factorization for both cloners.

Lemma 2.5 (Factorization of the lifted fidelity). *Fix $K \Subset \Delta_d^\circ$. For each $p \in K$, choose a transition rule $\mathbf{K}_n^{(p)} : \mathbf{Y}_n \rightsquigarrow \mathbf{Y}_{m_n}$. Use $\mathbf{J}_{n,p}, Q_{m_n,p}$ from (21) and $\Xi_{n,p,\nu}$ from (22) with $\mathbf{K}_n = \mathbf{K}_n^{(p)}$. $P_{m_n,p}$ is the target Young-label law from (14). Suppose that, for some $\delta_n \rightarrow 0$, the set of close label pairs*

$$G_{n,p} = \left\{ (\mu, \nu) \in \mathbf{Y}_n \times \mathbf{Y}_{m_n} : \left\| \frac{\mu}{n} - p \right\|_\infty \leq \delta_n, \left\| \frac{\nu}{m_n} - p \right\|_\infty \leq \delta_n \right\}$$

satisfies $\sup_{p \in K} \mathbf{J}_{n,p}(G_{n,p}^c) \rightarrow 0$. Then, uniformly for $p \in K$ and $g \in \mathbf{U}(d)$,

$$F\left(\mathcal{Q}_n[\mathbf{K}_n^{(p)}](\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}\right) = F_{\text{orb}}(\gamma, p)F(Q_{m_n,p}, P_{m_n,p}) + o(1). \quad (36)$$

Proof. By covariance, take $g = I$. Write $P = P_{m_n,p}$, $Q = Q_{m_n,p}$, and $G = G_{n,p}$. Omitting the maximally mixed Specht factors, the output and target are $O = \bigoplus_\nu \Xi_{n,p,\nu}$ and $T = \bigoplus_\nu P(\nu)\rho_{p,\nu}$. For large n , all pairs in G are compatible by Proposition 2.3. Retain only these pairs:

$$O_G = \bigoplus_\nu \Xi_\nu^G, \quad \Xi_\nu^G = \sum_{\mu: (\mu,\nu) \in G} \mathbf{J}_{n,p}(\mu, \nu) \mathcal{C}_{\mu,\nu}(\rho_{p,\mu}), \quad Q_G(\nu) = \text{Tr } \Xi_\nu^G.$$

Whenever $Q_G(\nu) > 0$, the mixture statement of Proposition 2.3 applies to $\Xi_\nu^G/Q_G(\nu)$. The direct-sum fidelity identity (23) and the fact that labels with $Q_G(\nu) = 0$ contribute zero to both fidelities, and $F(Q_G, P) \leq 1$ therefore give

$$|F(O_G, T) - F_{\text{orb}}(\gamma, p)F(Q_G, P)| \leq \eta_n, \quad \eta_n \longrightarrow 0,$$

uniformly on K . Positivity and trace preservation give $\|O - O_G\|_1 = \|Q - Q_G\|_{\ell^1} = J_{n,p}(G^c) =: \epsilon_{n,p}$. Applying (35) once to the states and once to the label laws yields

$$|F(O, T) - F_{\text{orb}}(\gamma, p)F(Q, P)| \leq \eta_n + 2\sqrt{\epsilon_{n,p}} = o(1),$$

which proves (36). $\square$

We now complete the achievability proofs of Theorems 1.1 and 1.2.

Known spectrum. Fix any compact $K \Subset \Delta_d^\circ$. For each $p \in K$, take the known-spectrum rule $\mathcal{K}_n^{(p)} = \mathcal{K}_n^p$ from (24). The joint law is $P_{n,p} \otimes P_{m_n,p}$ and $Q_{m_n,p} = P_{m_n,p}$. With $\delta_n = n^{-1/3} \geq m_n^{-1/3}$, $G_{n,p}$ contains $\mathbb{T}_n(p) \times \mathbb{T}_{m_n}(p)$, so Young concentration makes the discarded mass vanish uniformly on K , while the label fidelity is exactly one. Thus (36) gives

$$\sup_{\substack{p \in K \\ g \in \mathbb{U}(d)}} \left| F\left(\mathcal{C}_n^p(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}\right) - F_{\text{orb}}(\gamma, p) \right| \longrightarrow 0,$$

proving achievability in Theorem 1.1.

Unknown spectrum. For every p , take $\mathcal{K}_n^{(p)} = \mathcal{K}_n^{\text{rnd}}$ and $\delta_n = n^{-1/3} + c_{\text{rnd}}/m_n$. By (31), every pair in the support of $J_{n,p}$ with $\mu \in \mathbb{T}_n(p)$ lies in $G_{n,p}$, so the discarded mass is at most $P_{n,p}(\mathbb{T}_n(p)^c) = o(1)$ uniformly on K . Combining (36) and (32) yields

$$F\left(\mathcal{C}_n^{\text{univ}}(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}\right) = F_{\text{cl}}(\gamma, d)F_{\text{orb}}(\gamma, p) + o(1)$$

uniformly for $p \in K$ and $g \in \mathbb{U}(d)$, proving the achievability assertion of Theorem 1.2.

3 Optimality from LAN and Gaussian converses

We use two-way LAN to transport arbitrary cloners on local neighborhoods to channels between Gaussian models. After establishing the Gaussian bounds, we apply the transfer in Fig. 3 to complete the minimax proofs.

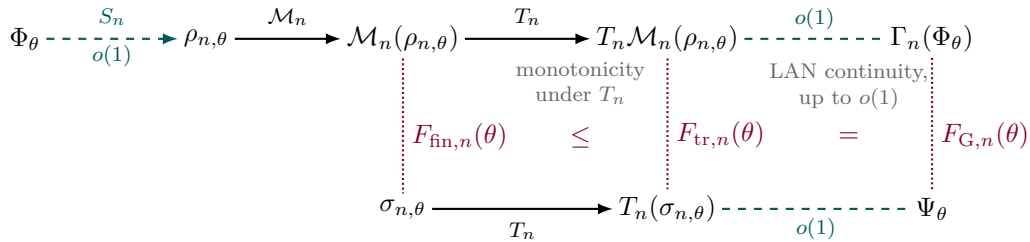


Figure 3: LAN transfer for $\Gamma_n = T_n \circ \mathcal{M}_n \circ S_n$ on a fixed compact local parameter set C . The top and bottom rows track output and target. Dotted segments indicate fidelity pairs, solid arrows exact channels, and dashed links uniform $o(1)$ trace-norm errors. Monotonicity and continuity give $F_{\text{fin},n}(\theta) \leq F_{\text{tr},n}(\theta) = F_{G,n}(\theta) + o(1)$ uniformly on C , as proved in Lemma 3.5. The LAN maps need not be inverses, and Γ_n need not be Gaussian or converge.

3.1 Local parametrization and Gaussian amplification

Fix a base spectrum $p \in \Delta_d^\circ$. Let

$$\mathbf{H} = \left\{ h \in \mathbb{R}^d : \sum_i h_i = 0 \right\}, \quad Y_p(z) = \sum_{j < k} \frac{z_{jk} E_{kj} - \bar{z}_{jk} E_{jk}}{\sqrt{p_j - p_k}}, \quad z \in \mathbb{C}^{r_d}, \quad r_d = \binom{d}{2},$$

where E_{jk} are the standard matrix units, and define

$$g_{n,z}^{(p)} = \exp\left(\frac{Y_p(z)}{\sqrt{n}}\right).$$

In $\rho_{p+h/\sqrt{n}, g_{n,z}^{(p)}}$, the coordinates h and z describe local changes of eigenvalues and eigenbasis, respectively. The denominators in Y_p use the fixed base spectrum p , even when the local spectrum is $p + h/\sqrt{n}$. For $L > 0$, set

$$\Omega_L = \{(h, z) \in \mathbf{H} \times \mathbb{C}^{r_d} : \|h\|_\infty \leq L, |\Re z_{ij}|, |\Im z_{ij}| \leq L \text{ for all } i < j\}.$$

For fixed p, L , $p + h/\sqrt{n} \in \Delta_d^\circ$ throughout Ω_L for large n . Under LAN, h becomes a classical Gaussian shift and z an oscillator displacement.

Let $\mathcal{F}$ be the one-mode Fock space with number basis $\{|k\rangle : k \geq 0\}$, creation and annihilation operators a^*, a , and number operator $\hat{N} = a^*a$. We use the Weyl convention

$$D(z) = \exp(za^* - \bar{z}a). \quad (37)$$

For $0 \leq q < 1$, define the number-diagonal thermal state

$$\tau_q = (1 - q) \sum_{k \geq 0} q^k |k\rangle\langle k|.$$

Since these states commute,

$$F(\tau_q, \tau_{q'}) = \frac{\sqrt{(1-q)(1-q')}}{1 - \sqrt{qq'}}. \quad (38)$$

For $G \geq 1$, let $q^{(G)} = 1 - (1 - q)/G$. The quantum-limited phase-insensitive amplifier $\mathcal{A}_G$ satisfies

$$\mathcal{A}_G(D(z)\tau_q D(z)^*) = D(\sqrt{G}z)\tau_{q^{(G)}} D(\sqrt{G}z)^*. \quad (39)$$

Equivalently, it is implemented by a two-mode squeezer with a vacuum idler.

On $\mathcal{F}_{\text{orb}} = \mathcal{F}^{\otimes r_d}$, let $q_{ij}^{(\gamma)} = 1 - (1 - q_{ij})/\gamma$ and define

$$\tau_p = \bigotimes_{i < j} \tau_{q_{ij}}, \quad \tau_p^{(\gamma)} = \bigotimes_{i < j} \tau_{q_{ij}^{(\gamma)}}. \quad (40)$$

Let $D(z) = \bigotimes_{i < j} D(z_{ij})$. The three relevant states are

$$\begin{aligned} \text{input:} \quad & \Phi_z^{(p)} = D(z)\tau_p D(z)^*, \\ \text{ideal gain-}\sqrt{\gamma} \text{ target:} \quad & \Psi_z^{(p, \gamma)} = D(\sqrt{\gamma}z)\tau_p D(\sqrt{\gamma}z)^*, \\ \text{amplifier output:} \quad & \mathcal{A}_\gamma^{\otimes r_d}(\Phi_z^{(p)}) = D(\sqrt{\gamma}z)\tau_p^{(\gamma)} D(\sqrt{\gamma}z)^*. \end{aligned} \quad (41)$$

By (38),

$$F(\tau_{q^{(\gamma)}}, \tau_q) = f_\gamma(q). \quad (42)$$

Multiplicativity then gives, for every $z \in \mathbb{C}^{r_d}$,

$$F\left(\mathcal{A}_\gamma^{\otimes r_d}(\Phi_z^{(p)}), \Psi_z^{(p,\gamma)}\right) = F_{\text{orb}}(\gamma, p). \quad (43)$$

Equip $\mathbf{H}$ with its induced Euclidean measure and recall Σ_p from (33). Write $\mathbf{N}_{h,\Sigma_p}$ for the Gaussian law on $\mathbf{H}$ with mean h and covariance Σ_p . A classical–quantum state is a measurable field $R(x)$ of positive trace-class operators on $\mathcal{F}_{\text{orb}}$, with $\int_{\mathbf{H}} \text{Tr } R(x) dx = 1$. A classical–quantum channel Λ is a completely positive map, in the sense specified in Appendix C, that preserves $\int \text{Tr } R(x) dx$. For two such states R, S ,

$$F(R, S) = \int_{\mathbf{H}} F(R(x), S(x)) dx.$$

This is the continuous direct-sum identity. With $\|R\|_1 := \int_{\mathbf{H}} \|R(x)\|_1 dx$, fidelity retains its monotonicity and joint-concavity properties.

Combining the classical spectral fluctuations with the quantum oscillator modes, define the input and ideal target states of the full Gaussian model by

$$\Theta_{h,z}^{\text{in}} = \mathbf{N}_{h,\Sigma_p} \otimes \Phi_z^{(p)}, \quad \Theta_{h,z}^{\text{tar}} = \mathbf{N}_{\sqrt{\gamma}h,\Sigma_p} \otimes \Psi_z^{(p,\gamma)}. \quad (44)$$

The target scales both local parameters (h, z) by $\sqrt{\gamma}$, while retaining the input covariance and thermal noise. For the orbital model, let Z_L be the box of oscillator displacements over which we average the fidelity, with Lebesgue measure on its real and imaginary coordinates:

$$Z_L = \{z \in \mathbb{C}^{r_d} : |\Re z_{ij}|, |\Im z_{ij}| \leq L \text{ for all } i < j\}.$$

Theorem 3.1 (Gaussian amplification optimum). *For every $p \in \Delta_d^\circ$,*

$$\lim_{L \rightarrow \infty} \sup_{\Lambda} \frac{1}{|Z_L|} \int_{Z_L} F\left(\Lambda(\Phi_z^{(p)}), \Psi_z^{(p,\gamma)}\right) dz = F_{\text{orb}}(\gamma, p), \quad (45)$$

and

$$\lim_{L \rightarrow \infty} \sup_{\Lambda} \frac{1}{|\Omega_L|} \int_{\Omega_L} F\left(\Lambda(\Theta_{h,z}^{\text{in}}), \Theta_{h,z}^{\text{tar}}\right) dh dz = F_{\text{univ}}(\gamma, p). \quad (46)$$

The first supremum is over all quantum channels on $\mathcal{F}_{\text{orb}}$. The second is over all classical–quantum channels. No Gaussianity, covariance, or product-structure assumption is imposed on the competing channels in either supremum.

The product amplifier $\mathcal{A}_\gamma^{\otimes r_d}$ attains $F_{\text{orb}}(\gamma, p)$ in the orbital model. Tensoring it with the classical pushforward $x \mapsto \sqrt{\gamma}x$ supplies the Gaussian affinity $F_{\text{cl}}(\gamma, d)$, attaining $F_{\text{univ}}(\gamma, p)$ in the full model.

For the converse, average a near-optimal channel over translates of the expanding flat-prior boxes and use a weighted-fidelity witness to reduce the problem to an oscillator moment. The weighted oscillator maps have a subsequential limit against compact observables that is completely positive and displacement covariant. Its output trace on normalized inputs is at most 1 in the orbital model and, by a Gaussian translation estimate, $F_{\text{cl}}(\gamma, d)$ in the full model. The least-noise inequality then gives the respective bounds $F_{\text{orb}}(\gamma, p)$ and $F_{\text{univ}}(\gamma, p)$. See Proposition C.5 in Appendix C.

3.2 Local asymptotic normality

Here LAN means that, uniformly for (h, z) in compact sets, the state families $\rho_{p+h/\sqrt{n}, g_{n,z}^{(p)}}^{\otimes n}$ and $\mathbf{N}_{h,\Sigma_p} \otimes \Phi_z^{(p)}$ are asymptotically interconvertible by channels with vanishing trace-norm error. The

Gaussian and oscillator factors represent spectral and eigenbasis fluctuations, respectively. Cloning scales (h, z) by $\sqrt{\gamma}$.

We fix the oscillator normalization directly from the local tangent and the collective fluctuation operators. Let $\Delta_{ij} = p_i - p_j$. The chart defined above satisfies

$$\frac{[Y_p(z), \rho_p]_{ji}}{\sqrt{\Delta_{ij}}} = z_{ij}, \quad i < j. \quad (47)$$

For $a_{ij,n} := (n\Delta_{ij})^{-1/2} \sum_{k=1}^n E_{ij}^{(k)}$, where $E_{ij}^{(k)}$ acts on the k -th tensor factor, the expectations read

$$\left\langle [a_{ij,n}, a_{ij,n}^*] \right\rangle_{\rho_p^{\otimes n}} = 1, \quad \left\langle a_{ij,n}^* a_{ij,n} \right\rangle_{\rho_p^{\otimes n}} = \frac{p_j}{\Delta_{ij}} = \frac{q_{ij}}{1 - q_{ij}}.$$

For the local product state $\rho_{p+h/\sqrt{n}, g_{n,z}^{(p)}}^{\otimes n}$, (47) gives $\langle a_{ij,n} \rangle = z_{ij} + o(1)$, uniformly on compact (h, z) -sets. The (i, j) oscillator factor of $\Phi_z^{(p)}$, defined in (41), is $D(z_{ij})\tau_{q_{ij}}D(z_{ij})^*$. With the convention (37), its mean is z_{ij} , and its occupation at zero displacement is $q_{ij}/(1 - q_{ij})$. These match the finite-dimensional moments above, fixing the displacement and thermal occupation parameters of the Gaussian model. The following two-way LAN statement follows from Kahn–Guță’s channel construction [KG09, Theorem 4.3, Lemma 6.4, and Section 7.3], in the oscillator coordinates fixed above and with the target-sample rescaling below.⁵

Theorem 3.2 (Finite-dimensional LAN and target-sample scaling). *Fix $p \in \Delta_d^\circ$. There are channels*

$$T_N^{\text{full}} : \mathcal{T}_1((\mathbb{C}^d)^{\otimes N}) \longrightarrow L^1(\mathbf{H}; \mathcal{T}_1(\mathcal{F}_{\text{orb}})), \quad S_N^{\text{full}} : L^1(\mathbf{H}; \mathcal{T}_1(\mathcal{F}_{\text{orb}})) \longrightarrow \mathcal{T}_1((\mathbb{C}^d)^{\otimes N})$$

such that, for every compact $C \subset \mathbf{H} \times \mathbb{C}^{rd}$,

$$\sup_{(h,z) \in C} \left\| T_n^{\text{full}} \left(\rho_{p+h/\sqrt{n}, g_{n,z}^{(p)}}^{\otimes n} \right) - \mathbf{N}_{h, \Sigma_p} \otimes \Phi_z^{(p)} \right\|_1 \longrightarrow 0, \quad (48)$$

$$\sup_{(h,z) \in C} \left\| S_n^{\text{full}} \left(\mathbf{N}_{h, \Sigma_p} \otimes \Phi_z^{(p)} \right) - \rho_{p+h/\sqrt{n}, g_{n,z}^{(p)}}^{\otimes n} \right\|_1 \longrightarrow 0, \quad (49)$$

and

$$\sup_{(h,z) \in C} \left\| T_{m_n}^{\text{full}} \left(\rho_{p+h/\sqrt{n}, g_{n,z}^{(p)}}^{\otimes m_n} \right) - \mathbf{N}_{\sqrt{\gamma}h, \Sigma_p} \otimes \Psi_z^{(p, \gamma)} \right\|_1 \longrightarrow 0. \quad (50)$$

Chart independence of the converses. The converses of Theorem 1.1 and Theorem 1.2(b) are chart-independent: an invertible real-linear reparametrization of (h, z) sends the local boxes to dilates of bounded polytopes, and the averaging in Proposition C.5 uses only $|(\Xi_L + \xi) \triangle \Xi_L|/|\Xi_L| \rightarrow 0$. The resulting Gaussian optimum is determined by Σ_p and the q_{ij} . The explicit displacement scale enters Theorem 1.4 through $\mathbb{E}|\beta_{ij}|^2$ in q_{ij}^{PCT} , and the fluctuation calculation above fixes it.

⁵We use the displacement normalization of the constructive Lemma 6.4 and Section 7.3 of the cited work. The displayed Eq. (4.19) in its arXiv version has a different factor and adjoint placement. The rotation chart, fluctuation calculation, and qubit check below fix our convention.

Coordinate identification and target scaling. For (48)–(49), identify the spectral coordinate $u \in \mathbb{R}^{d-1}$ of the cited theorem with $h = (u_1, \dots, u_{d-1}, -\sum_{i < d} u_i) \in \mathbf{H}$. Its covariance $V(p)_{ij} = \delta_{ij}p_i - p_i p_j$, $i, j < d$ [KG09, Eq. (3.3)], becomes $\Sigma_p = \text{diag}(p) - pp^\top$ under this embedding, giving $\mathbf{N}_{h, \Sigma_p}$. The normalization (47) identifies the oscillator factor with $\Phi_z^{(p)}$.

For (50), write $\gamma_n = m_n/n$ and use

$$\rho_{p+h/\sqrt{n}, g_{n,z}^{(p)}} = \rho_{p+\sqrt{\gamma_n}h/\sqrt{m_n}, g_{m_n, \sqrt{\gamma_n}z}^{(p)}}$$

with the same base p at both sample sizes. Apply (48) at sample size m_n on a common compact set containing $(\sqrt{\gamma_n}h, \sqrt{\gamma_n}z)$ for all sufficiently large n . Since $\gamma_n \rightarrow \gamma$, uniform trace-norm continuity of Gaussian translations and Weyl conjugation on that set gives the claimed target limit.

Remark 3.3 (Qubit normalization check). For $d = 2$, let $\Delta = p_1 - p_2 > 0$, $q = p_2/p_1$, and take $z = t \in \mathbb{R}$. The exact qubit fidelity identity gives

$$F\left(\rho_p^{\otimes n}, \rho_{p, g_{n,t}^{(p)}}^{\otimes n}\right) = \left(1 - \Delta^2 \sin^2 \frac{t}{\sqrt{n}\Delta}\right)^{n/2} \longrightarrow e^{-\Delta t^2/2}.$$

This equals $F(\tau_q, D(t)\tau_q D(t)^*) = \exp\left[-\frac{(1-q)t^2}{2(1+q)}\right]$, confirming the displacement normalization we use.

Corollary 3.4 (Fixed-spectrum LAN). *Fix $p \in \Delta_d^\circ$. There are channels*

$$T_N^{\text{orb}} : \mathcal{T}_1((\mathbb{C}^d)^{\otimes N}) \rightarrow \mathcal{T}_1(\mathcal{F}_{\text{orb}}), \quad S_N^{\text{orb}} : \mathcal{T}_1(\mathcal{F}_{\text{orb}}) \rightarrow \mathcal{T}_1((\mathbb{C}^d)^{\otimes N})$$

such that, uniformly for z in compact sets,

$$\|T_n^{\text{orb}}(\rho_{p, g_{n,z}^{(p)}}^{\otimes n}) - \Phi_z^{(p)}\|_1 \longrightarrow 0, \quad (51)$$

$$\|S_n^{\text{orb}}(\Phi_z^{(p)}) - \rho_{p, g_{n,z}^{(p)}}^{\otimes n}\|_1 \longrightarrow 0, \quad (52)$$

and

$$\|T_{m_n}^{\text{orb}}(\rho_{p, g_{n,z}^{(p)}}^{\otimes m_n}) - \Psi_z^{(p, \gamma)}\|_1 \longrightarrow 0. \quad (53)$$

Proof. Restrict Theorem 3.2 to $h = 0$ and set $T_N^{\text{orb}}(X) := \int_{\mathbf{H}} T_N^{\text{full}}(X)(x) dx$ and $S_N^{\text{orb}}(\sigma) := S_N^{\text{full}}(\mathbf{N}_{0, \Sigma_p} \otimes \sigma)$. Marginalization contracts trace norm, and tensoring with $\mathbf{N}_{0, \Sigma_p}$ preserves it, giving all three assertions. $\square$

Lemma 3.5 (LAN fidelity transfer). *Let $\{\rho_{n, \theta}\}_{\theta \in C}$ and $\{\sigma_{n, \theta}\}_{\theta \in C}$ be input and target models on a compact parameter set C . Let $\{\Phi_\theta\}$ and $\{\Psi_\theta\}$ be limiting models. Suppose channels S_n, T_n satisfy*

$$\delta_n := \sup_{\theta \in C} \|S_n(\Phi_\theta) - \rho_{n, \theta}\|_1 \rightarrow 0,$$

$$\eta_n := \sup_{\theta \in C} \|T_n(\sigma_{n, \theta}) - \Psi_\theta\|_1 \rightarrow 0.$$

For every channel $\mathcal{M}_n$, let $\Gamma_n = T_n \circ \mathcal{M}_n \circ S_n$. Then, for every $\theta \in C$,

$$F(\mathcal{M}_n(\rho_{n, \theta}), \sigma_{n, \theta}) \leq F(\Gamma_n(\Phi_\theta), \Psi_\theta) + \sqrt{\delta_n} + \sqrt{\eta_n}. \quad (54)$$

The error is uniform over both $\theta \in C$ and $\mathcal{M}_n$.

Proof. Trace-norm contractivity of $T_n \circ \mathcal{M}_n$ gives $\|\Gamma_n(\Phi_\theta) - T_n \mathcal{M}_n(\rho_{n, \theta})\|_1 \leq \delta_n$. Monotonicity under T_n , followed by (35) and the definition of η_n , therefore yields

$$F(\mathcal{M}_n(\rho_{n, \theta}), \sigma_{n, \theta}) \leq F(T_n \mathcal{M}_n(\rho_{n, \theta}), T_n(\sigma_{n, \theta})) \leq F(\Gamma_n(\Phi_\theta), \Psi_\theta) + \sqrt{\delta_n} + \sqrt{\eta_n}.$$

$\square$

3.3 Minimax converses

Known-spectrum converse and completion of Theorem 1.1. For the known-spectrum problem, set $u_n(p) := \sup_{\mathcal{M}_n} \inf_{g \in \mathcal{U}(d)} F(\mathcal{M}_n(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n})$. For every channel $\mathcal{M}_n$ and fixed L , set

$$\Gamma_n = T_{m_n}^{\text{orb}} \circ \mathcal{M}_n \circ S_n^{\text{orb}}.$$

The local orbit $\{g_{n,z}^{(p)} : z \in Z_L\}$ is contained in the full unitary orbit. Apply Lemma 3.5, average over Z_L , and take the supremum over $\mathcal{M}_n$. Since the transfer error is independent of $\mathcal{M}_n$,

$$u_n(p) \leq \sup_{\Lambda} \frac{1}{|Z_L|} \int_{Z_L} F(\Lambda(\Phi_z^{(p)}), \Psi_z^{(p,\gamma)}) dz + o(1),$$

where the supremum is over all channels on the Gaussian model and the error tends to zero for fixed L . Taking $\limsup_n$, then $L \rightarrow \infty$, (45) gives

$$\limsup_{n \rightarrow \infty} u_n(p) \leq F_{\text{orb}}(\gamma, p).$$

Together with $\liminf_n u_n(p) \geq F_{\text{orb}}(\gamma, p)$, proved in Section 2, this proves Theorem 1.1. $\square$

Fixing p would miss the cost of an unknown spectrum. Instead, we use $n^{-1/2}$ -neighborhoods of an interior spectrum and let the local window grow after the large- n limit.

Proof of part (b) of Theorem 1.2. Let $w_n(K)$ denote the quantity inside the limit on the left of (8). Part (a) gives $\liminf_n w_n(K) \geq \inf_{p \in K} F_{\text{univ}}(\gamma, p)$. For the reverse inequality, fix $p_0 \in \text{int}_{\mathbb{A}_d} K$ and L . For large n , all local spectra $p_0 + h/\sqrt{n}$, $(h, z) \in \Omega_L$, lie in K , so the worst-case fidelity of $\mathcal{M}_n$ on K is at most its average over this window. Repeating the known-spectrum argument with the full LAN maps at p_0 , $\Gamma_n = T_{m_n}^{\text{full}} \circ \mathcal{M}_n \circ S_n^{\text{full}}$, and Ω_L in place of Z_L gives

$$w_n(K) \leq \sup_{\Lambda} \frac{1}{|\Omega_L|} \int_{\Omega_L} F(\Lambda(\Theta_{h,z}^{\text{in}}, \Theta_{h,z}^{\text{tar}})) dh dz + o(1),$$

where both Gaussian models are based at p_0 and the supremum is over all classical-quantum channels. Taking $\limsup_n$ and then $L \rightarrow \infty$, (46) yields $\limsup_n w_n(K) \leq F_{\text{univ}}(\gamma, p_0)$. Since $\text{int}_{\mathbb{A}_d} K$ is dense in K and $F_{\text{univ}}(\gamma, \cdot)$ is continuous, taking the infimum over p_0 completes the proof. $\square$

4 Cloning projector states

A flat rank- r projector state has the form $\rho_P = P/r$, where P is a rank- r orthogonal projector. These projectors form the complex Grassmannian $\text{Gr}(r, d)$. Rank deficiency, and for $r > 1$ the degenerate positive eigenvalue, place these states outside the simple full-rank theorems. We construct $\mathcal{C}_n^{\text{proj}}$ and prove Theorem 1.3 directly, without a general rank-deficient LAN theorem. The compression lemma below holds for any positive spectrum on its support, but does not by itself establish optimal cloning beyond the flat case.

4.1 Exact rank compression

A rank- r input occupies only part of each Schur sector in ambient dimension d , while our Cartan channel acts on the full sector. To evaluate fidelity with a target on the same support, we compare this ambient channel with the Cartan channel in dimension r . The compression identity below

separates the mass retained on the target support from the fidelity within it. For flat projector states, the latter is one.

Superscripts (s) indicate ambient dimension. For $\ell(\lambda) \leq r$, write $D_s^\lambda = \dim \mathcal{H}_\lambda^{(s)}$ and define

$$R_\lambda := \frac{D_d^\lambda}{D_r^\lambda} = \prod_{\substack{1 \leq i \leq r \\ r < j \leq d}} \frac{\lambda_i + j - i}{j - i}. \quad (55)$$

Let $J_\lambda : \mathcal{H}_\lambda^{(r)} \rightarrow \mathcal{H}_\lambda^{(d)}$ be the isometric $U(r)$ -intertwiner mapping unit highest-weight vectors to each other, and let P_0 project onto the first r coordinates. Write $\Pi_{\lambda, P_0} = J_\lambda J_\lambda^*$ for the corresponding Schur-sector support projection.

For a compatible pair of r -row partitions $\mu \vdash n$ and $\nu \vdash m$, let $\omega = \nu - \mu$. Write $\mathcal{C}_{\mu, \nu}^{(s)}$ for the Cartan-sector channel (19) in ambient dimension $s \in \{r, d\}$. For a probability vector $x \in \mathbb{R}^s$, use the dimension- s polynomial representation $\pi_\lambda^{(s)}$ and Schur polynomial $s_\lambda^{(s)}$ to define

$$\rho_{x, \lambda}^{(s)} = \frac{\pi_\lambda^{(s)}(\text{diag}(x))}{s_\lambda^{(s)}(x)}$$

when $\ell(\lambda) \leq |\{i : x_i > 0\}|$, exactly the condition $s_\lambda^{(s)}(x) > 0$.

Lemma 4.1 (Exact rank-compression identity). *Let $p = (p_1, \dots, p_r)$ satisfy $p_i > 0$ and $\sum_i p_i = 1$, and let $\bar{p} = (p_1, \dots, p_r, 0, \dots, 0) \in \mathbb{R}^d$. For every r -row partition λ ,*

$$\rho_{\bar{p}, \lambda}^{(d)} = J_\lambda \rho_{p, \lambda}^{(r)} J_\lambda^*. \quad (56)$$

For every compatible r -row pair μ, ν and every trace-class operator X on $\mathcal{H}_\mu^{(r)}$,

$$\Pi_{\nu, P_0} \mathcal{C}_{\mu, \nu}^{(d)}(J_\mu X J_\mu^*) \Pi_{\nu, P_0} = \frac{R_\mu}{R_\nu} J_\nu \mathcal{C}_{\mu, \nu}^{(r)}(X) J_\nu^*. \quad (57)$$

Consequently,

$$F\left(\mathcal{C}_{\mu, \nu}^{(d)}(\rho_{\bar{p}, \mu}^{(d)}), \rho_{\bar{p}, \nu}^{(d)}\right) = \sqrt{\frac{R_\mu}{R_\nu}} F\left(\mathcal{C}_{\mu, \nu}^{(r)}(\rho_{p, \mu}^{(r)}), \rho_{p, \nu}^{(r)}\right). \quad (58)$$

Proof. Let $\iota : \mathbb{C}^r \hookrightarrow \mathbb{C}^d$ be the coordinate inclusion. Schur–Weyl naturality identifies $\iota^{\otimes N}$ with $\bigoplus_{\ell(\lambda) \leq r} (J_\lambda \otimes I_{S_\lambda})$. Comparing blocks in $(\text{diag } \bar{p})^{\otimes N} = \iota^{\otimes N} (\text{diag } p)^{\otimes N} (\iota^*)^{\otimes N}$ gives

$$\pi_\lambda^{(d)}(\text{diag } \bar{p}) = J_\lambda \pi_\lambda^{(r)}(\text{diag } p) J_\lambda^*.$$

Taking traces gives $s_\lambda^{(d)}(\bar{p}) = s_\lambda^{(r)}(p) > 0$. Division proves (56).

The maps $V_{\mu, \omega}^{(d)} J_\nu$ and $(J_\mu \otimes J_\omega) V_{\mu, \omega}^{(r)}$ are $U(r)$ -intertwiners that both send the unit highest-weight vector to $v_\mu \otimes v_\omega$. Hence they agree:

$$V_{\mu, \omega}^{(d)} J_\nu = (J_\mu \otimes J_\omega) V_{\mu, \omega}^{(r)}. \quad (59)$$

Inserting (59) into the two sector-channel formulas proves (57).

For $A, B \geq 0$ and an orthogonal projection Π with $\Pi B \Pi = B$, $F(A, B) = F(\Pi A \Pi, B)$. Apply this with $\Pi = \Pi_{\nu, P_0}$. (56), (57), isometric invariance, and $F(aA, B) = \sqrt{a} F(A, B)$ give (58). $\square$

For P_0 , the flat-spectrum conditional state in the λ -sector is

$$\rho_{P_0, \lambda} = J_\lambda \frac{I_{\mathcal{H}_\lambda^{(r)}}}{D_r^\lambda} J_\lambda^*. \quad (60)$$

By $U(r)$ -covariance and irreducibility, the r -dimensional Cartan channel maps the maximally mixed input to the maximally mixed output. Thus (58) gives, for every compatible pair,

$$F\left(\mathcal{C}_{\mu, \nu}^{(d)}(\rho_{P_0, \mu}), \rho_{P_0, \nu}\right) = \sqrt{\frac{R_\mu}{R_\nu}}. \quad (61)$$

For $\mu_i/n \rightarrow 1/r$ and $\nu_i/m_n \rightarrow 1/r$, this tends to the right-hand side of (9).

4.2 Achievability

We follow the Schur–Cartan lifting of Section 2.1: sample an output Young label and apply the Cartan-sector channel when the input and output labels are compatible. To choose the transition kernel, we couple the exact flat-spectrum Schur laws at sizes n and m_n so that their centered diagram shapes are close, then condition on the input label. This gives the target label law exactly and makes compatible pairs overwhelmingly likely.

Let

$$\mathbf{P}_N^{(r)}(\lambda) = \frac{\dim \mathcal{S}_\lambda D_r^\lambda}{r^N}, \quad \lambda \vdash N, \quad \ell(\lambda) \leq r, \quad (62)$$

be the Schur–Weyl law of the maximally mixed state on $\mathbb{C}^r$. The complete Schur decomposition is

$$(P/r)^{\otimes N} = \bigoplus_{\ell(\lambda) \leq r} \mathbf{P}_N^{(r)}(\lambda) \frac{\Pi_{\lambda, P}}{D_r^\lambda} \otimes \frac{I_{\mathcal{S}_\lambda}}{\dim \mathcal{S}_\lambda}. \quad (63)$$

Here $\Pi_{\lambda, P} = \pi_\lambda^{(d)}(P)$ is the rank- D_r^λ projection obtained by applying the polynomial representation to P . For $P = P_0$, it is the support projection defined above. Sample $\lambda \sim \mathbf{P}_N^{(r)}$ and pad it with zero rows to length r . The Schur–Weyl/random-word central limit theorem [Kup02, Joh01] gives

$$X_N := \frac{\lambda - (N/r)\mathbf{1}_r}{\sqrt{N}} \longrightarrow X \quad \text{in distribution,} \quad (64)$$

where X is a fixed scaling of the ordered eigenvalue vector of a traceless $r \times r$ GUE matrix. In particular, $\Pr(X_1 > \dots > X_r) = 1$.

Both X_n and X_{m_n} converge to the same law. By Strassen’s coupling theorem [Str65], there is a sufficiently slow sequence $\varepsilon_n \downarrow 0$ for which the minimum, over couplings of the exact marginals $\mathbf{P}_n^{(r)}$ and $\mathbf{P}_{m_n}^{(r)}$, of the probability that their centered shapes differ by more than ε_n tends to zero. Fix such a sequence and let $\mathbf{J}_n(\mu, \nu)$ be a minimizer of this finite transportation linear program. Under $\mathbf{J}_n$,

$$\left\| \frac{\mu - (n/r)\mathbf{1}_r}{\sqrt{n}} - \frac{\nu - (m_n/r)\mathbf{1}_r}{\sqrt{m_n}} \right\|_\infty \longrightarrow 0 \quad \text{in probability.} \quad (65)$$

This coupling is needed when $r > 1$: independent sampling of the two labels leaves their $O(\sqrt{n})$ row gaps unrelated. For $1 \leq i < r$, independent sampling would give

$$\frac{(\nu_i - \mu_i) - (\nu_{i+1} - \mu_{i+1})}{\sqrt{n}} \longrightarrow \sqrt{\gamma}(X'_i - X'_{i+1}) - (X_i - X_{i+1}) \quad \text{in distribution,}$$

where X' is an independent copy of X . This limit is negative with positive probability. The chosen minimizer need not have a closed-form rule. Every partition $\mu \vdash n$ with $\ell(\mu) \leq r$ has $\mathbf{P}_n^{(r)}(\mu) > 0$. We therefore condition the coupling on its input coordinate and define

$$\mathbf{K}_n^{\text{proj}}(\nu \mid \mu) := \frac{\mathbf{J}_n(\mu, \nu)}{\mathbf{P}_n^{(r)}(\mu)}, \quad \ell(\mu) \leq r. \quad (66)$$

Thus sampling $\mu \sim \mathbf{P}_n^{(r)}$ and then $\nu \sim \mathbf{K}_n^{\text{proj}}(\cdot \mid \mu)$ has joint law $\mathbf{J}_n$. In particular, the output label has exactly the target marginal:

$$\sum_{\ell(\mu) \leq r} \mathbf{P}_n^{(r)}(\mu) \mathbf{K}_n^{\text{proj}}(\nu \mid \mu) = \sum_{\ell(\mu) \leq r} \mathbf{J}_n(\mu, \nu) = \mathbf{P}_{m_n}^{(r)}(\nu). \quad (67)$$

For input diagrams with more than r rows, choose $\mathbf{K}_n^{\text{proj}}(\cdot \mid \mu)$ arbitrarily. Such diagrams have zero probability on rank- r projector inputs, but this extension makes the lifted map a channel on the entire input space. Set

$$\mathcal{C}_n^{\text{proj}} := \mathcal{Q}_n[\mathbf{K}_n^{\text{proj}}]. \quad (68)$$

Its dependence on the fixed rank r is suppressed in the notation. The channel depends on the chosen minimizer, but is independent of the unknown projector P .

Let $\omega = \nu - \mu$. Its leading term is $(m_n - n)/r$ in every coordinate, while its fluctuations are of order $\sqrt{n}$. Hence $\omega_i > 0$ for every i with probability $1 - o(1)$. For adjacent rows,

$$\frac{\omega_i - \omega_{i+1}}{\sqrt{n}} = \sqrt{\gamma_n} \left[\frac{\nu_i - \nu_{i+1}}{\sqrt{m_n}} \right] - \left[\frac{\mu_i - \mu_{i+1}}{\sqrt{n}} \right] \longrightarrow (\sqrt{\gamma} - 1)(X_i - X_{i+1}) \quad \text{in distribution.}$$

The limit is strictly positive almost surely. Thus the compatibility event

$$C_n = \{(\mu, \nu) : \nu - \mu \vdash m_n - n\}$$

has probability $1 - o(1)$. The cloner $\mathcal{C}_n^{\text{proj}}$ uses the ambient Cartan-sector channel on C_n and, otherwise, the covariant replacement channel into the same sampled ν -block. Write $\mathbf{1}_{C_n}$ for the indicator of C_n : it equals one on compatible pairs and zero on fallback pairs. Under the coupling $(\mu, \nu) \sim \mathbf{J}_n$, the central limit theorem also gives $\mu_i/n, \nu_i/m_n \rightarrow 1/r$ in probability. By (55), the random variable $\mathbf{1}_{C_n} \sqrt{R_\mu/R_\nu}$ therefore satisfies

$$\mathbf{1}_{C_n} \sqrt{\frac{R_\mu}{R_\nu}} \longrightarrow \gamma^{-r(d-r)/2} \quad \text{in probability.} \quad (69)$$

On C_n , compatibility gives $\nu_i \geq \mu_i$, hence $0 \leq \mathbf{1}_{C_n} \sqrt{R_\mu/R_\nu} \leq 1$. This uniform bound upgrades the convergence in probability in (69) to convergence of expectations. Since the output-label marginal is exact, the block identity (23) (valid also for r -row label laws), monotonicity and concavity of F in its first argument, and (61) give

$$F\left(\mathcal{C}_n^{\text{proj}}((P/r)^{\otimes n}), (P/r)^{\otimes m_n}\right) \geq \mathbb{E}_{\mathbf{J}_n} \left[\mathbf{1}_{C_n} \sqrt{\frac{R_\mu}{R_\nu}} \right] \longrightarrow \gamma^{-r(d-r)/2}. \quad (70)$$

This proves achievability in Theorem 1.3.

4.3 Converse

The unknown projector P selects an r -dimensional support inside $\mathbb{C}^d$. Averaging a cloner over unitaries and output permutations cannot worsen its worst-case fidelity, so we may work with a symmetric channel. Replacing each input support projection by the identity gives one P -independent operator B_n that dominates every projector input. In the μ -block, this multiplies its trace by R_μ . The symmetric output on B_n is maximally mixed within each Schur block, while the target support occupies only a fraction $1/R_\nu$ of the ν -block. Cauchy–Schwarz combines these factors into the bound below.

Proposition 4.2 (Projector converse). *For $1 \leq r < d$ and every n with $m_n \geq n$,*

$$\sup_{\mathcal{M}_n} \inf_{P \in \text{Gr}(r,d)} F(\mathcal{M}_n((P/r)^{\otimes n}), (P/r)^{\otimes m_n}) \leq \left[\left(\sum_{\substack{\mu \vdash n \\ \ell(\mu) \leq r}} \mathbf{P}_n^{(r)}(\mu) R_\mu \right) \left(\sum_{\substack{\nu \vdash m_n \\ \ell(\nu) \leq r}} \frac{\mathbf{P}_{m_n}^{(r)}(\nu)}{R_\nu} \right) \right]^{1/2}. \quad (71)$$

Here $\mathbf{P}_N^{(r)}$ is the flat Schur law from (62), and R_λ is the dimension ratio from (55). The supremum is over channels from n to m_n qudits.

Proof. Covariantize an arbitrary cloner over $U(d)$ and twirl its output over permutations. Neither decreases its worst-case fidelity: the first follows from joint concavity and unitary invariance, the second from monotonicity under the output twirl, which fixes every target. Write $\mathcal{M}_n$ for the resulting channel.

Replace each support projection in (63) by the identity to obtain the invariant positive operator

$$B_n = \bigoplus_{\substack{\mu \vdash n \\ \ell(\mu) \leq r}} \mathbf{P}_n^{(r)}(\mu) \frac{I_{\mathcal{H}_\mu^{(d)}}}{D_r^\mu} \otimes \frac{I_{\mathcal{S}_\mu}}{\dim \mathcal{S}_\mu}.$$

For every P ,

$$(P/r)^{\otimes n} \leq B_n, \quad \text{Tr } B_n = \sum_{\mu} \mathbf{P}_n^{(r)}(\mu) R_\mu.$$

Covariance and the output twirl imply that

$$\mathcal{M}_n(B_n) = \bigoplus_{\substack{\nu \vdash m_n \\ \ell(\nu) \leq d}} z_\nu \frac{I_{\mathcal{H}_\nu^{(d)}}}{D_d^\nu} \otimes \frac{I_{\mathcal{S}_\nu}}{\dim \mathcal{S}_\nu}, \quad z_\nu \geq 0, \quad \sum_{\nu} z_\nu = \text{Tr } B_n.$$

Fidelity increases in either positive argument: $A \leq B$ implies $\text{Tr} \sqrt{\sqrt{T} A \sqrt{T}} \leq \text{Tr} \sqrt{\sqrt{T} B \sqrt{T}}$. Positivity of $\mathcal{M}_n$ and the direct-sum formula therefore give

$$F(\mathcal{M}_n((P/r)^{\otimes n}), (P/r)^{\otimes m_n}) \leq F(\mathcal{M}_n(B_n), (P/r)^{\otimes m_n}) = \sum_{\ell(\nu) \leq r} \sqrt{\frac{z_\nu \mathbf{P}_{m_n}^{(r)}(\nu)}{R_\nu}}.$$

By Cauchy–Schwarz and $\sum_{\ell(\nu) \leq r} z_\nu \leq \text{Tr } B_n$,

$$\left(\sum_{\ell(\nu) \leq r} \sqrt{\frac{z_\nu \mathbf{P}_{m_n}^{(r)}(\nu)}{R_\nu}} \right)^2 \leq (\text{Tr } B_n) \sum_{\ell(\nu) \leq r} \frac{\mathbf{P}_{m_n}^{(r)}(\nu)}{R_\nu} = \left(\sum_{\ell(\mu) \leq r} \mathbf{P}_n^{(r)}(\mu) R_\mu \right) \left(\sum_{\ell(\nu) \leq r} \frac{\mathbf{P}_{m_n}^{(r)}(\nu)}{R_\nu} \right).$$

This gives (71) for the covariantized channel, hence the minimax bound for the original one. $\square$

For the asymptotics, let $a = r(d-r)$ and $c_{r,d} = \prod_{i \leq r < j \leq d} [r(j-i)]^{-1}$. Since $\mathbf{P}_N^{(r)}$ is the Young law of the flat spectrum $(1/r, \dots, 1/r)$, [Lemma 2.1](#) in dimension r shows that $\|\lambda/N - \mathbf{1}_r/r\|_\infty \leq N^{-1/3}$ outside an event of probability at most $e^{-N^{1/3/4}}$. Write $u_i = \lambda_i - N/r$, so $\sum_i u_i = 0$. On this typical set, the leading linear terms in u_i cancel in the product (55) because $\sum_i u_i = 0$, and Taylor expansion gives

$$R_\lambda = c_{r,d} N^a \left[1 + O_{r,d} \left(N^{-1} + \frac{\|u\|_2^2}{N^2} \right) \right], \quad \frac{1}{R_\lambda} = c_{r,d}^{-1} N^{-a} \left[1 + O_{r,d} \left(N^{-1} + \frac{\|u\|_2^2}{N^2} \right) \right].$$

The flat Young law has $\mathbb{E}\|u\|_2^2 = O_r(N)$. Indeed, the quadratic $U(r)$ Casimir has eigenvalue $\sum_i \lambda_i(\lambda_i + r + 1 - 2i)$. On $(\mathbb{C}^r)^{\otimes N}$, the Casimir operator equals $NrI + 2 \sum_{1 \leq j < k \leq N} \text{Swap}_{jk}$, where Swap_{jk} exchanges tensor factors j, k . Since $\text{Tr}[\text{Swap}_{12}(I_r/r)^{\otimes 2}] = 1/r$, its expectation under $\mathbf{P}_N^{(r)}$ is $Nr + N(N-1)/r$. Moreover,

$$\sum_i \lambda_i(\lambda_i + r + 1 - 2i) = \frac{N^2}{r} + \|u\|_2^2 + \sum_{i < j} (\lambda_i - \lambda_j),$$

so $\mathbb{E}\|u\|_2^2 \leq N(r - 1/r)$. Globally $1 \leq R_\lambda \leq (N+1)^a$, so the atypical event contributes less than any inverse power of N . Hence

$$\mathbb{E}_{\mathbf{P}_N^{(r)}} R_\lambda = c_{r,d} N^a (1 + O_{r,d}(N^{-1})), \quad \mathbb{E}_{\mathbf{P}_N^{(r)}} \frac{1}{R_\lambda} = c_{r,d}^{-1} N^{-a} (1 + O_{r,d}(N^{-1})). \quad (72)$$

Applying (72) at $N = n$ and $N = m_n$, the product under the square root in (71) is

$$\mathbb{E}_{\mathbf{P}_n^{(r)}} R_\mu \mathbb{E}_{\mathbf{P}_{m_n}^{(r)}} R_\nu^{-1} = n^a m_n^{-a} (1 + O_{r,d}(n^{-1}))(1 + O_{r,d}(m_n^{-1})) = \gamma_n^{-a} (1 + O_{r,d}(n^{-1})).$$

Taking its square root, the right-hand side of (71) is $\gamma_n^{-a/2} (1 + O_{r,d}(n^{-1}))$. If $m_n = \gamma n + O(1)$, this equals $\gamma^{-a/2} + O_{r,d,\gamma}(n^{-1})$. The assumption $\gamma_n \rightarrow \gamma$ alone gives no rate relative to that limit. Substitution yields

$$\limsup_{n \rightarrow \infty} \sup_{\mathcal{M}_n} \inf_{P \in \text{Gr}(r,d)} F(\mathcal{M}_n((P/r)^{\otimes n}), (P/r)^{\otimes m_n}) \leq \gamma^{-r(d-r)/2}, \quad (73)$$

which completes the converse and the proof of [Theorem 1.3](#).

5 Fixed-gain comparison with PCT

For the PCT channel defined before [Theorem 1.4](#), the published data-processing guarantee [[LTHC26](#), [FGSS26](#), [JSO26](#)] is, for every ρ of rank at most r ,

$$F\left(C_{n,m}^{\text{PCT},r}(\rho^{\otimes n}), \rho^{\otimes m}\right) \geq \left(\frac{\binom{n+dr-1}{dr-1}}{\binom{m+dr-1}{dr-1}}\right)^{1/2} \longrightarrow \gamma^{-(dr-1)/2} \quad (m = m_n). \quad (74)$$

The bound is exact for each purified component before averaging and discarding the purifying registers, but is only a lower bound for the final output.

Here and in [Appendix D](#), we analyze the PCT cloning fidelity directly. [Figure 2](#) illustrates the comparisons. At every fixed gain $\gamma > 1$, our cloners asymptotically outperform PCT on simple full-rank states and non-pure flat projector states, while both agree on pure states.

Simple full-rank spectra. Fix $p_1 > \dots > p_d > 0$. PCT uses rank bound $r = d$ and purification dimension d^2 . Neither it nor $\mathcal{C}_n^{\text{univ}}$ is supplied p . [Theorem 1.4\(a\)](#) gives PCT's exact limiting fidelity and its strict gap from $F_{\text{univ}}(\gamma, p)$. The loss for PCT comes from larger spectral fluctuations and less accurate reproduction of the eigenbasis. [Appendix D.2](#) gives its exact fidelity and proves the strict comparison.

Projector states. For $\rho = P/r$ with $P \in \text{Gr}(r, d)$ and $1 < r < d$, apply PCT with rank bound r . [Theorem 1.4\(b\)](#) bounds its limiting fidelity strictly below the projector optimum $\gamma^{-r(d-r)/2}$. The extra factor is visible even after conditioning on every output lying in $\text{ran } P$: counts of outcomes in a basis of that space fluctuate more under PCT than under the target $(P/r)^{\otimes m_n}$. [Appendix D.3](#) proves the bound.

6 Discussion

6.1 Dependence on spectral mixedness

For fixed $\gamma > 1$, the product formula shows how fidelity changes with spectral mixedness:

$$f'_\gamma(q) = \frac{(\gamma - 1)^2}{2\sqrt{q(\gamma - 1 + q)}(\sqrt{\gamma(\gamma - 1 + q)} + \sqrt{q})^2} > 0, \quad 0 < q < 1.$$

Consequently, if $p, p' \in \Delta_d^\circ$ satisfy

$$\frac{p'_j}{p'_i} \geq \frac{p_j}{p_i} \quad \text{for every } i < j, \quad (75)$$

that is, if p' is more mixed than p in that every logarithmic spectral gap contracts in passing from p to p' , then

$$F_{\text{orb}}(\gamma, p') \geq F_{\text{orb}}(\gamma, p), \quad F_{\text{univ}}(\gamma, p') \geq F_{\text{univ}}(\gamma, p), \quad (76)$$

with strict inequality if any ratio is strictly larger.⁶

6.2 Worst-case cloning over all states

The optimal worst-case fidelity over all d -dimensional states remains unknown. At fixed gain $\gamma > 1$, a single channel must work without spectral or rank information. Write its optimal guaranteed fidelity as

$$v_n := \sup_{\mathcal{M}_n} \inf_{\substack{\rho \geq 0 \\ \text{Tr } \rho = 1}} F(\mathcal{M}_n(\rho^{\otimes n}), \rho^{\otimes m_n}). \quad (77)$$

The PCT guarantee gives $\liminf_n v_n \geq \gamma^{-(d^2-1)/2}$. Our compact-spectrum optimum [\(8\)](#) gives

$$\limsup_n v_n \leq \inf_{p \in \Delta_d^\circ} F_{\text{univ}}(\gamma, p) = F_{\text{cl}}(\gamma, d) \gamma^{-d(d-1)/4}. \quad (78)$$

The equality follows by taking simple spectra with all ratios p_j/p_i , $i < j$, tending to zero. At large gain, these lower and upper bounds scale respectively as $\gamma^{-(d^2-1)/2}$ and $\gamma^{-d(d-1)/4}$, up to constants, leaving a factor of two between the polynomial exponents. Whether PCT attains the asymptotic all-state minimax value also remains open.

⁶Condition [\(75\)](#) implies that p' is majorized by p , but majorization alone does not ensure larger fidelity when $d \geq 3$. For example, $p' = (0.8, 0.16, 0.04)$ is majorized by $p = (0.9, 0.06, 0.04)$, yet $F_{\text{orb}}(10, p') \approx 0.0772 < 0.0791 \approx F_{\text{orb}}(10, p)$: the decrease in the $(2, 3)$ factor outweighs the increases in the other two.

6.3 High-fidelity sample complexity

In this subsection we follow the sample-complexity literature, using squared fidelity, N inputs, and M *additional* outputs, so that

$$m = N + M, \quad \delta = \frac{M}{N}, \quad \gamma = 1 + \delta. \quad (79)$$

The asymptotic family-specific thresholds below use the iterated limit: first $N, M \rightarrow \infty$ at fixed $\delta > 0$, then $\delta \downarrow 0$. We have not controlled approximation errors uniformly as $\gamma_n \downarrow 1$, so these thresholds are not finite-sample bounds for a joint high-fidelity regime and do not improve the rank-bounded worst-case $1/\epsilon$ scaling. The PCT sufficient condition (84) is the finite-sample statement here. The limiting fidelities give two high-fidelity behaviors. At a fixed simple full-rank spectrum, the ratio N/M needed for squared infidelity ϵ scales as $\epsilon^{-1/2}$, with a smaller leading constant for our cloner than for PCT. On non-pure projector states, both have the leading ratio $r(d-r)/\epsilon$. Our strict fidelity advantage first appears at order δ^2 . The published PCT guarantee below gives only a looser sufficient ratio.

For $1 \leq r \leq d$ and sufficiently small $\epsilon > 0$, the concurrent papers [FGSS26, JSO26] prove that cloning with squared fidelity at least $1 - \epsilon$ for every state of rank at most r has optimal worst-case sample complexity

$$N = \Theta\left(\frac{Mrd}{\epsilon}\right). \quad (80)$$

Their upper bounds use PCT, and their lower bounds use projectors of rank $\ell = \Theta(r)$, possibly $\ell < r$.

Simple full-rank spectra. For fixed p and $\bullet \in \{\text{univ}, \text{PCT}\}$, expand the squared limiting fidelity at $\gamma = 1 + \delta$ and invert it at $1 - \epsilon$. The resulting asymptotic threshold satisfies

$$F_{\bullet}(1 + \delta, p)^2 = 1 - A_{\bullet}(p)\delta^2 + O(\delta^3), \quad \frac{N}{M} \sim \sqrt{\frac{A_{\bullet}(p)}{\epsilon}}, \quad (81)$$

where

$$A_{\text{univ}}(p) = \frac{d-1}{8} + \frac{1}{4} \sum_{i < j} \frac{p_i}{p_j}, \quad A_{\text{PCT}}(p) = \frac{d-1}{2} + \frac{1}{4} \sum_{i < j} \frac{(p_i + p_j)^2}{p_i p_j}. \quad (82)$$

For each pair, $(p_i + p_j)^2/(p_i p_j) = p_i/p_j + 2 + p_j/p_i > p_i/p_j$. The constant term in A_{PCT} is larger as well. Thus $A_{\text{PCT}}(p) > A_{\text{univ}}(p)$. Since $\delta = M/N$, the squared infidelity is asymptotic to $A_{\bullet}(p)(M/N)^2$. To reach squared infidelity ϵ , PCT therefore needs a larger leading ratio N/M than our cloner.⁷

Projector states. For $1 < r < d$, inverting our limiting squared fidelity gives

$$\frac{N}{M} = \frac{1}{(1 - \epsilon)^{-1/[r(d-r)]} - 1} \sim \frac{r(d-r)}{\epsilon}. \quad (83)$$

The published guarantee (74) certifies PCT squared fidelity at least $1 - \epsilon$ whenever

$$\frac{N}{M} \geq \frac{1}{(1 - \epsilon)^{-1/(rd-1)} - 1} \sim \frac{rd-1}{\epsilon}, \quad (84)$$

⁷The constant hidden in $O(\delta^3)$ depends on p , and expanding $f_{1+\delta}(q_{ij})$ requires $\delta \ll q_{ij}$, hence $\delta \ll p_d/p_1$ for all pairs. As the smallest eigenvalue approaches zero, this window shrinks and both $A_{\text{univ}}(p)$ and $A_{\text{PCT}}(p)$ diverge. When $q_{ij} \ll \delta$, instead $f_{1+\delta}(q_{ij})^2 = (1 + \delta)^{-1} + o(\delta)$, so that pair contributes infidelity of order δ . This helps explain why the fixed-spectrum $1/\sqrt{\epsilon}$ scaling does not hold uniformly over the rank-bounded family, whose worst-case scaling is $1/\epsilon$.

since, with $n = N$ and $m = N + M$, its binomial ratio is $\prod_{j=1}^{rd-1} (N+j)/(N+M+j) \geq [N/(N+M)]^{rd-1}$. The coefficient $rd-1$ describes only this sufficient condition. The two thresholds have the subleading expansion

$$\frac{1}{(1-\epsilon)^{-1/a}-1} = \frac{a}{\epsilon} - \frac{a+1}{2} + O(\epsilon), \quad (85)$$

with $a = r(d-r)$ in (83) and $a = rd-1$ in (84). The latter expansion does not determine PCT's actual subleading threshold. Both the lim inf and the lim sup of PCT's actual squared infidelity equal $r(d-r)\delta + O_{r,d}(\delta^2)$ (Proposition D.3), so PCT shares the leading sample ratio $N/M \sim r(d-r)/\epsilon$. With $\gamma = 1 + \delta$, squaring (12) bounds the lim sup of PCT's squared fidelity by our value $(1 + \delta)^{-r(d-r)}$ times

$$\left(\frac{\sqrt{1+2\delta}}{1+\delta} \right)^{r-1} = 1 - \frac{r-1}{2}\delta^2 + O_r(\delta^3).$$

Thus the asymptotic fidelity gap is at least $(r-1)\delta^2/2 + O_{r,d}(\delta^3)$, while Proposition D.3 bounds it above by $O_{r,d}(\delta^2)$. Our advantage appears at second order, with its exact coefficient undetermined.

6.4 Conjectures for degenerate and rank-deficient spectra

Beyond the flat projector family, optimal cloning remains open for nonflat rank-deficient states and spectra with degeneracies. We conjecture separate optimal cloning fidelities for two families: on a known-spectrum orbit only the eigenbasis varies, while in a local fixed-rank family the positive eigenvalues may vary as well. Suppose a rank- r state ρ has distinct positive eigenvalues $u_1 > \dots > u_s > 0$, with multiplicities $r_1, \dots, r_s$, where $\sum_a r_a = r$ and $\sum_a r_a u_a = 1$. We conjecture that the known-spectrum optimum on its unitary orbit, defined as in (6), is

$$F_{\text{orb}}(\gamma, \rho) = \gamma^{-r(d-r)/2} \prod_{a < b} f_\gamma \left(\frac{u_b}{u_a} \right)^{r_a r_b}. \quad (86)$$

Rotations within degenerate eigenspaces are stabilizers and do not contribute. Since $f_\gamma(0) = \gamma^{-1/2}$, the first factor accounts for the $r(d-r)$ support-kernel directions. The simple full-rank case $r = d$, $s = d$, $r_a = 1$ recovers (5). The flat projector case $s = 1$ recovers Theorem 1.3, including Werner's pure-state value at $r = 1$. The exact compression identity in Lemma 4.1 establishes this support-kernel factorization for our sector channel, but does not by itself prove the conjecture for the remaining spectra.

The known-spectrum optimum need not be continuous at a degeneracy, because the orbit loses dimensions. For qubits, let $q = p_2/p_1$, so $F_{\text{orb}}(\gamma, p) = f_\gamma(q)$. As $p \rightarrow (1/2, 1/2)$ through simple spectra, $q \uparrow 1$, and

$$\lim_{\substack{p \rightarrow (1/2, 1/2) \\ p_1 > p_2}} F_{\text{orb}}(\gamma, p) = \lim_{q \uparrow 1} \frac{\sqrt{\gamma} + \sqrt{q(\gamma-1+q)}}{\gamma+q} = \frac{2\sqrt{\gamma}}{1+\gamma} < 1,$$

whereas the known-spectrum family at $p = (1/2, 1/2)$ contains only $I_2/2$ and can be cloned perfectly. For each fixed n , the constant channel preparing $(I_2/2)^{\otimes m_n}$ has fidelity tending to one as $p \rightarrow (1/2, 1/2)$, uniformly over the orbit. Thus the degeneracy and large- n limits do not commute.

For the local unknown-state problem, consider the smooth manifold of rank- r density matrices, denoted as $\mathcal{D}_r^\circ$. If P is the support projection of ρ , its tangent space is

$$T_\rho \mathcal{D}_r^\circ = \{X = X^* : \text{Tr } X = 0, (I - P)X(I - P) = 0\}. \quad (87)$$

Let Exp_ρ be the exponential map for the induced Hilbert–Schmidt metric. For fixed L and sufficiently large n , set $\rho_{n,X} = \text{Exp}_\rho(X/\sqrt{n})$ and define

$$v_{n,L}^{(r)}(\rho) := \sup_{\mathcal{M}_n} \inf_{\substack{X \in T_\rho \mathcal{D}_r^\circ \\ \|X\|_2 \leq L}} F\left(\mathcal{M}_n(\rho_{n,X}^{\otimes n}), \rho_{n,X}^{\otimes m_n}\right). \quad (88)$$

The channel may depend on the fixed center ρ , but not on the unknown displacement X . We conjecture that the limsup and liminf of $v_{n,L}^{(r)}(\rho)$, first as $n \rightarrow \infty$ and then as $L \rightarrow \infty$, agree. Writing their common value as $F_{\text{univ}}^{(r)}(\gamma, \rho)$, the conjectured formula is

$$F_{\text{univ}}^{(r)}(\gamma, \rho) = \left(\frac{2\sqrt{\gamma}}{1+\gamma}\right)^{(\sum_{a=1}^s r_a^2 - 1)/2} F_{\text{orb}}(\gamma, \rho). \quad (89)$$

Unlike v_n in (77), this problem tests only $n^{-1/2}$ -scale perturbations of a fixed rank- r center ρ . Thus (89) is a conjectured local value, whereas (78) is a proved upper bound for the all-state minimax value. The classical exponent in (89) counts perturbations within the positive-eigenvalue blocks: an $r_a \times r_a$ Hermitian block has r_a^2 real parameters, and the trace constraint removes one overall. A classical Gaussian shift with k real parameters contributes the fidelity factor $(2\sqrt{\gamma}/(1+\gamma))^{k/2}$. At fixed rank, the zero block has no independent parameters. Support–kernel rotations are already counted in $F_{\text{orb}}(\gamma, \rho)$. The GUE limit for flat-spectrum Young-label fluctuations [Kup02, Joh01] motivates treating the within-block directions as classical Gaussian variables, but does not establish the conjecture.

When the r positive eigenvalues are distinct (every $r_a = 1$), the fixed-rank LAN theorem of [LN24, Theorem 3.3] identifies $r - 1$ classical Gaussian coordinates, $\binom{r}{2}$ shifted thermal modes (one per pair of positive eigenvectors), and $r(d - r)$ shifted pure modes (one per support–kernel pair). The classical coordinates give the exponent $(r - 1)/2$ in (89). The thermal and pure modes correspond to the $f_\gamma(u_b/u_a)$ factors and $\gamma^{-r(d-r)/2}$, respectively, in (86). These mode counts support the conjecture but do not prove the optimal cloning fidelities.

The formula is consistent under positive-eigenvalue collisions at fixed rank: each colliding pair contributes $f_\gamma(1) = 2\sqrt{\gamma}/(1+\gamma)$, and $(r - 1)/2 + \sum_a \binom{r_a}{2} = (\sum_a r_a^2 - 1)/2$. When also $r = d$ and the spectrum is simple, the formula reduces to the full-rank value $F_{\text{univ}}(\gamma, p)$ in (5).

At a flat projector state, $s = 1$, the local model still allows the positive eigenvalues to vary, unlike the projector family in Theorem 1.3, where they are fixed at $1/r$. This uncertainty gives the conjectured classical factor for $r > 1$. The two values coincide for $r = 1$. Neither model allows rank changes, which remains an open question.

We further conjecture that, at every fixed $\gamma > 1$ and with the same prior spectral or rank information available to both protocols, Schur–Cartan cloners with transition rules adapted to degenerate or rank-deficient spectra outperform PCT on the corresponding families, and that a single spectrum-independent extension strictly beats PCT with rank bound d in the all-state problem (77). The latter would require control near the boundary of the spectral simplex, beyond the pointwise comparisons proved here.

A Exact-weight PBW calculation of the sector fidelity

We prove Proposition 2.3 in three steps. On fixed-height weight spaces, the Gram matrices of normalized PBW vectors approach the identity. The Cartan inclusion then splits lowering operators with asymptotically binomial weights. Uniform character normalization and a bound on the target mass above the cutoff yield the thermal fidelity $F_{\text{orb}}(\gamma, p)$.

Throughout, $O_L(\cdot)$ may depend on L, d, c, C in $cN \leq \Delta_\alpha^\lambda \leq CN$ for $\alpha \in \mathbf{R}_+$, but is uniform in N and all dominant λ satisfying these inequalities.

A.1 PBW vectors on complete weight spaces

Identify the positive roots of $\mathbf{U}(d)$ with

$$\mathbf{R}_+ = \{e_i - e_j : 1 \leq i < j \leq d\}.$$

For $\alpha = e_i - e_j$, set

$$F_\alpha = E_{ji}, \quad E_\alpha = E_{ij}, \quad H_\alpha = [E_\alpha, F_\alpha] = E_{ii} - E_{jj}, \quad \Delta_\alpha^\lambda = \lambda_i - \lambda_j.$$

Fix a total order on $\mathbf{R}_+$. For $\mathbf{r} \in \mathbb{Z}_{\geq 0}^{\mathbf{R}_+}$, write

$$F^\mathbf{r} = \prod_{\alpha \in \mathbf{R}_+}^< F_\alpha^{r_\alpha}, \quad |\mathbf{r}| = \sum_{\alpha} r_\alpha.$$

For a nonnegative simple-root combination η , let $\text{ht}(\eta)$ be the sum of its coefficients in the simple roots $e_i - e_{i+1}$, $i < d$, and let

$$\mathbf{P}(\eta) = \left\{ \mathbf{r} : \sum_{\alpha \in \mathbf{R}_+} r_\alpha \alpha = \eta \right\}.$$

The set $\mathbf{P}(\eta)$ is finite, and the PBW theorem implies that $\{F^\mathbf{r} v_\lambda : \mathbf{r} \in \mathbf{P}(\eta)\}$ spans $\mathcal{H}_\lambda[\lambda - \eta]$ [FH91]. Here $\mathcal{H}_\lambda[\lambda - \eta]$ denotes the weight space of weight $\lambda - \eta$. Define

$$\mathcal{K}_{\lambda, L} = \bigoplus_{\text{ht}(\eta) \leq L} \mathcal{H}_\lambda[\lambda - \eta]. \quad (90)$$

Let $P_{\lambda, L}$ denote the orthogonal projection onto $\mathcal{K}_{\lambda, L}$.

Lemma A.1 (PBW Gram matrices at fixed height). *Let $\lambda = \lambda^{(N)}$ range over dominant weights satisfying*

$$cN \leq \Delta_\alpha^\lambda \leq CN \quad (\alpha \in \mathbf{R}_+)$$

for fixed $0 < c < C < \infty$. Define

$$\tilde{e}_\mathbf{r}^\lambda = \prod_{\alpha \in \mathbf{R}_+}^< \frac{F_\alpha^{r_\alpha}}{\sqrt{r_\alpha!} (\Delta_\alpha^\lambda)^{r_\alpha/2}} v_\lambda. \quad (91)$$

For every fixed L , uniformly over $\text{ht}(\eta) \leq L$ and $\mathbf{r}, \mathbf{s} \in \mathbf{P}(\eta)$,

$$\langle \tilde{e}_\mathbf{r}^\lambda, \tilde{e}_\mathbf{s}^\lambda \rangle = \delta_{\mathbf{r}, \mathbf{s}} + O_L(N^{-1/2}). \quad (92)$$

Consequently, for all sufficiently large N , the full Gram matrix $G_{\lambda, \eta}$ on every such exact-weight block is invertible. If

$$(e_\mathbf{r}^\lambda)_{\mathbf{r} \in \mathbf{P}(\eta)} = (\tilde{e}_\mathbf{r}^\lambda)_{\mathbf{r} \in \mathbf{P}(\eta)} G_{\lambda, \eta}^{-1/2}, \quad (93)$$

then these vectors form an orthonormal basis of $\mathcal{H}_\lambda[\lambda - \eta]$, and

$$e_\mathbf{r}^\lambda = \tilde{e}_\mathbf{r}^\lambda + O_L(N^{-1/2}) \quad (94)$$

uniformly for $\text{ht}(\eta) \leq L$ and $\mathbf{r} \in \mathbf{P}(\eta)$.

Proof. For $\mathbf{r} \in \mathbf{P}(\eta)$, every positive root has positive simple-root height, so $|\mathbf{r}| \leq \text{ht}(\eta)$. Let

$$\widehat{F}_\alpha = \frac{F_\alpha}{\sqrt{\Delta_\alpha^\lambda}}, \quad \widehat{E}_\alpha = \frac{E_\alpha}{\sqrt{\Delta_\alpha^\lambda}}.$$

The word inside each unnormalized Gram entry $\langle v_\lambda, (\widehat{F}^{\mathbf{r}})^* \widehat{F}^{\mathbf{s}} v_\lambda \rangle$ contains at most $2L$ normalized root operators. Reordering them does not increase the number of factors. Since each root has height at most $d-1$, every intermediate vector lies in

$$\mathcal{K}_{\lambda, R_L}, \quad R_L := 2L(d-1).$$

The normalized root operators are uniformly bounded on each fixed cutoff. Indeed, let $h = \text{ht}(\alpha)$, set $\mathcal{K}_{\lambda, R} = \{0\}$ for $R < 0$, and write

$$M_R = \|\widehat{F}_\alpha|_{\mathcal{K}_{\lambda, R}}\|^2.$$

Because each cutoff $\mathcal{K}_{\lambda, R}$ contains complete weight spaces, $\widehat{E}_\alpha$ maps $\mathcal{K}_{\lambda, R}$ into $\mathcal{K}_{\lambda, R-h}$. It is the adjoint of $\widehat{F}_\alpha : \mathcal{K}_{\lambda, R-h} \rightarrow \mathcal{K}_{\lambda, R}$, so its squared operator norm is M_{R-h} . On $\mathcal{K}_{\lambda, R}$, $[\widehat{E}_\alpha, \widehat{F}_\alpha] = H_\alpha/\Delta_\alpha^\lambda = I + O_R(N^{-1})$. Applying this identity to unit vectors gives

$$M_R \leq M_{R-h} + 1 + O_R(N^{-1}) \leq \lfloor R/h \rfloor + 1 + O_R(N^{-1}) = O_R(1),$$

where the second inequality follows by induction from $M_R = 0$ for $R < 0$. This also bounds $\widehat{E}_\alpha$. All operator norms below are taken after restriction to $\mathcal{K}_{\lambda, R_L}$, with the codomain norm inherited from $\mathcal{H}_\lambda$. For $0 \leq R \leq R_L = 2L(d-1)$, the O_R bounds above can be written O_L .

The matrix-unit commutator formula now gives

$$[\widehat{E}_\alpha, \widehat{F}_\beta] = \delta_{\alpha\beta} I + O_L(N^{-1/2}). \quad (95)$$

For $\alpha = \beta$, the left side is $H_\alpha/\Delta_\alpha^\lambda = I + O_L(N^{-1})$ on the cutoff. For $\alpha \neq \beta$, the unnormalized commutator is zero or, up to sign, a root operator E_γ or F_γ . In the nonzero case, the preceding bound gives $\|[\widehat{E}_\alpha, \widehat{F}_\beta]\| = O_L(\sqrt{\Delta_\gamma^\lambda}/\sqrt{\Delta_\alpha^\lambda \Delta_\beta^\lambda}) = O_L(\sqrt{N}/N) = O_L(N^{-1/2})$.

Commute every $\widehat{E}_\alpha$ to the right in $\langle v_\lambda, (\widehat{F}^{\mathbf{r}})^* \widehat{F}^{\mathbf{s}} v_\lambda \rangle$. The scalar commutators in (95) contribute only when $\mathbf{r} = \mathbf{s}$, in which case they give $\prod_\alpha r_\alpha!$. Every other term vanishes or contains an $O_L(N^{-1/2})$ remainder. Only $O_L(1)$ terms occur, and all remaining operator factors have norm $O_L(1)$. Consequently,

$$\langle v_\lambda, (\widehat{F}^{\mathbf{r}})^* \widehat{F}^{\mathbf{s}} v_\lambda \rangle = \delta_{\mathbf{r}, \mathbf{s}} \prod_\alpha r_\alpha! + O_L(N^{-1/2}).$$

Dividing by the factorials in (91) proves (92).

The number and sizes of the exact-weight blocks are bounded in terms of L , so $G_{\lambda, \eta} = I + O_L(N^{-1/2})$ in operator norm. It is therefore invertible for large N , and the PBW spanning vectors form a basis. Functional calculus gives $G_{\lambda, \eta}^{-1/2} = I + O_L(N^{-1/2})$, proving (93) and (94). $\square$

A.2 Character normalization and fixed-height matrices

For $a = (a_1, \dots, a_d)$, write $p^a = \prod_i p_i^{a_i}$.

Lemma A.2 (Uniform character normalization). *Let $K \Subset \Delta_d^\circ$, let $\delta_N \rightarrow 0$, and suppose $p \in K$, $\lambda \in \mathbf{Y}_N$, and $\|\lambda/N - p\|_\infty \leq \delta_N$. Then, uniformly over this family,*

$$\frac{p^\lambda}{s_\lambda(p)} = \prod_{i < j} \left(1 - \frac{p_j}{p_i}\right) + o(1). \quad (96)$$

Proof. Write the numerator in the Weyl character formula as

$$\sum_{\sigma \in \mathfrak{S}_d} \text{sgn}(\sigma) A_\sigma, \quad A_\sigma = \prod_i p_i^{\lambda_{\sigma(i)} + d - \sigma(i)}.$$

For each nonidentity permutation, strict rearrangement makes $\sum_i (p_i - p_{\sigma(i)}) \log p_i$ positive. Compactness of K and finiteness of $\mathfrak{S}_d$ make this bound uniform for all sufficiently large N :

$$\frac{1}{N} \log \frac{A_{\text{id}}}{A_\sigma} = \sum_i (p_i - p_{\sigma(i)}) \log p_i + O_K(\delta_N + N^{-1}) \geq c_K > 0.$$

Thus the numerator is $A_{\text{id}}(1 + O_K(e^{-c_K N}))$. Cancelling $\prod_i p_i^{d-i}$ against the Weyl denominator gives

$$s_\lambda(p) = p^\lambda \prod_{i < j} \left(1 - \frac{p_j}{p_i}\right)^{-1} (1 + O_K(e^{-c_K N})).$$

This proves (96). $\square$

Index the number basis of $\mathcal{F}_{\text{orb}}$ by $\mathbf{r} \in \mathbb{Z}_{\geq 0}^{R_+}$, writing $\{|\mathbf{r}\rangle\}$, and define the finite-rank projection

$$P_L^{\mathcal{F}} = \sum_{\text{ht}(\sum_\alpha r_\alpha \alpha) \leq L} |\mathbf{r}\rangle \langle \mathbf{r}|. \quad (97)$$

For every fixed L and all sufficiently large N , each highest weight λ satisfying the gap bounds in Lemma A.1 has the orthonormal basis $e_{\mathbf{r}}^\lambda$ of $\mathcal{K}_{\lambda,L}$ constructed there. Define

$$J_{\lambda,L} : \mathcal{K}_{\lambda,L} \longrightarrow P_L^{\mathcal{F}} \mathcal{F}_{\text{orb}}, \quad J_{\lambda,L} e_{\mathbf{r}}^\lambda = |\mathbf{r}\rangle. \quad (98)$$

The common index set makes $J_{\lambda,L}$ unitary on its stated domain.

Proof of Proposition 2.3. Fix K and δ_n as in the proposition, write $m = m_n$, and let $\omega = \nu - \mu$. Since $m/n = \gamma_n \rightarrow \gamma > 1$, we have $m/n - 1 \geq (\gamma - 1)/2$ for all sufficiently large n . Uniformly over the allowed parameters,

$$\left\| \frac{\omega}{m - n} - p \right\|_\infty \leq \frac{m/n + 1}{m/n - 1} \delta_n = o(1).$$

Every coordinate and every adjacent gap is uniformly positive on K . Hence $\omega \vdash m - n$ for all sufficiently large n , and $\nu = \mu + \omega \in \mathbf{Y}_m$. For $\alpha = e_i - e_j$, set

$$s_{\alpha,n} := \frac{\Delta_\alpha^\mu}{\Delta_\alpha^\nu} = \frac{n}{m} \frac{\mu_i/n - \mu_j/n}{\nu_i/m - \nu_j/m} = \frac{1}{\gamma} + o(1), \quad 1 - s_{\alpha,n} = \frac{\Delta_\alpha^\omega}{\Delta_\alpha^\nu}. \quad (99)$$

Consequently, there are constants $0 < c_K < C_K < \infty$, depending only on K and γ , such that, for all large n ,

$$c_K n \leq \Delta_\alpha^\mu, \Delta_\alpha^\nu, \Delta_\alpha^\omega \leq C_K n \quad (\alpha \in \mathbf{R}_+). \quad (100)$$

Let $V = V_{\mu,\omega} : \mathcal{H}_\nu \rightarrow \mathcal{H}_\mu \otimes \mathcal{H}_\omega$, and write $F_\alpha^{(\lambda)} = d\pi_\lambda(F_\alpha)$ for the action of the root operator on $\mathcal{H}_\lambda$. Intertwining and the highest-weight normalization give

$$VF_\alpha^{(\nu)} = (F_\alpha^{(\mu)} \otimes I + I \otimes F_\alpha^{(\omega)})V.$$

Expanding each $(F_\alpha^{(\mu)} \otimes I + I \otimes F_\alpha^{(\omega)})^{r_\alpha}$ binomially gives the exact identity

$$V\tilde{e}_\mathbf{r}^\nu = \sum_{\mathbf{a} \leq \mathbf{r}} \beta_{\mathbf{r},\mathbf{a}}^{(n)} \tilde{e}_\mathbf{a}^\mu \otimes \tilde{e}_{\mathbf{r}-\mathbf{a}}^\omega, \quad (101)$$

where

$$\beta_{\mathbf{r},\mathbf{a}}^{(n)} = \prod_{\alpha \in \mathbf{R}_+} \sqrt{\binom{r_\alpha}{a_\alpha} s_{\alpha,n}^{a_\alpha} (1 - s_{\alpha,n})^{r_\alpha - a_\alpha}}. \quad (102)$$

If $\text{ht}(\sum_\alpha r_\alpha \alpha) \leq L$, then (101) has $O_L(1)$ terms, all of height at most L , with $0 \leq \beta_{\mathbf{r},\mathbf{a}}^{(n)} \leq 1$. Since $n, m, m-n$ are uniformly comparable, the gap bounds (100) and Lemma A.1 give $e_\mathbf{b}^\lambda = \tilde{e}_\mathbf{b}^\lambda + O_L(n^{-1/2})$ for $\lambda \in \{\mu, \nu, \omega\}$ and all relevant $\mathbf{b}$. Substituting these relations into (101) and using $\|Vx\| = \|x\|$ gives

$$V e_\mathbf{r}^\nu = \sum_{\mathbf{a} \leq \mathbf{r}} \beta_{\mathbf{r},\mathbf{a}}^{(n)} e_\mathbf{a}^\mu \otimes e_{\mathbf{r}-\mathbf{a}}^\omega + O_L(n^{-1/2}) \quad (103)$$

uniformly in norm for $\text{ht}(\sum_\alpha r_\alpha \alpha) \leq L$.

Let

$$\sigma_{\mu,\nu} = \mathcal{C}_{\mu,\nu}(\rho_{p,\mu}), \quad c_\lambda(p) = \frac{p^\lambda}{s_\lambda(p)}, \quad q_\alpha = \frac{p_j}{p_i} \quad (\alpha = e_i - e_j).$$

For $\lambda \in \{\mu, \nu\}$, the weight of $e_\mathbf{b}^\lambda$ is $\lambda - \sum_\alpha b_\alpha \alpha$, and hence

$$\rho_{p,\lambda} e_\mathbf{b}^\lambda = c_\lambda(p) \prod_\alpha q_\alpha^{b_\alpha} e_\mathbf{b}^\lambda.$$

The sector channel gives the local formula

$$\sigma_{\mu,\nu} = \frac{\dim \mathcal{H}_\mu}{\dim \mathcal{H}_\nu} V^*(\rho_{p,\mu} \otimes I_{\mathcal{H}_\omega}) V.$$

All estimates below are uniform over the allowed family. By Lemma A.2,

$$c_\lambda(p) = \prod_\alpha (1 - q_\alpha) + o(1) \quad (\lambda = \mu, \nu),$$

while the Weyl dimension formula, (100), and (99) give

$$\frac{\dim \mathcal{H}_\mu}{\dim \mathcal{H}_\nu} = \prod_{\alpha=e_i-e_j} \frac{\Delta_\alpha^\mu + j - i}{\Delta_\alpha^\nu + j - i} = \prod_{\alpha \in \mathbf{R}_+} (s_{\alpha,n} + O(n^{-1})) = \gamma^{-r_d} + o(1).$$

Substitute (103) into the sector formula. The binomial theorem and $\|\rho_{p,\mu}\| \leq 1$ therefore give, for fixed L ,

$$\begin{aligned} \langle e_\mathbf{r}^\nu, \sigma_{\mu,\nu} e_\mathbf{s}^\nu \rangle &= \frac{\dim \mathcal{H}_\mu}{\dim \mathcal{H}_\nu} \sum_{\substack{\mathbf{a} \leq \mathbf{r} \\ \mathbf{b} \leq \mathbf{s}}} \beta_{\mathbf{r},\mathbf{a}}^{(n)} \beta_{\mathbf{s},\mathbf{b}}^{(n)} \langle e_\mathbf{a}^\mu, \rho_{p,\mu} e_\mathbf{b}^\mu \rangle \langle e_{\mathbf{r}-\mathbf{a}}^\omega, e_{\mathbf{s}-\mathbf{b}}^\omega \rangle + O_L(n^{-1/2}) \\ &= \delta_{\mathbf{r},\mathbf{s}} \frac{\dim \mathcal{H}_\mu}{\dim \mathcal{H}_\nu} c_\mu(p) \prod_{\alpha \in \mathbf{R}_+} (1 - s_{\alpha,n} + s_{\alpha,n} q_\alpha)^{r_\alpha} + O_L(n^{-1/2}), \end{aligned} \quad (104)$$

$$\langle e_\mathbf{r}^\nu, \rho_{p,\nu} e_\mathbf{s}^\nu \rangle = \delta_{\mathbf{r},\mathbf{s}} c_\nu(p) \prod_{\alpha \in \mathbf{R}_+} q_\alpha^{r_\alpha}. \quad (105)$$

With $s_{\alpha,n} = \gamma^{-1} + o(1)$ and $q_{\alpha}^{(\gamma)} = 1 - (1 - q_{\alpha})/\gamma$, the output entries in (104) equal $\delta_{\mathbf{r},\mathbf{s}} \prod_{\alpha} (1 - q_{\alpha}^{(\gamma)})(q_{\alpha}^{(\gamma)})^{r_{\alpha}} + o(1)$ uniformly for fixed L . The target entries in (105) have the same form with q_{α} in place of $q_{\alpha}^{(\gamma)}$. These are the number-basis entries of $\tau_p^{(\gamma)}$ and τ_p , respectively. The cutoff space has fixed finite dimension, so this uniform entrywise convergence gives

$$\left\| J_{\nu,L} P_{\nu,L} \sigma_{\mu,\nu} P_{\nu,L} J_{\nu,L}^* - P_L^{\mathcal{F}} \tau_p^{(\gamma)} P_L^{\mathcal{F}} \right\|_1 \rightarrow 0. \quad (106)$$

$$\left\| J_{\nu,L} P_{\nu,L} \rho_{p,\nu} P_{\nu,L} J_{\nu,L}^* - P_L^{\mathcal{F}} \tau_p P_L^{\mathcal{F}} \right\|_1 \rightarrow 0. \quad (107)$$

To obtain the fidelity assertions, let $(a_{\mu})_{\mu}$ be a probability distribution supported on labels $\mu \in Y_n$ with $\|\mu/n - p\|_{\infty} \leq \delta_n$. All such pairs (μ, ν) are compatible for large n , by the first part of the proof. Let

$$\bar{\sigma}_{p,\nu} = \sum_{\mu} a_{\mu} \sigma_{\mu,\nu}, \quad \sigma_{\mu,\nu} = \mathcal{C}_{\mu,\nu}(\rho_{p,\mu}).$$

For fixed L , let $\varepsilon_{n,L} \rightarrow 0$ bound both trace-norm errors in (106) and (107), uniformly over the allowed parameters. Since the output frame $J_{\nu,L} P_{\nu,L}$ is independent of μ , convexity gives the same bound for $\bar{\sigma}_{p,\nu}$, uniformly over all mixture weights.

Both $\bar{\sigma}_{p,\nu}$ and $\rho_{p,\nu}$ commute with the diagonal torus. The second assertion is immediate. The first follows from torus invariance of each $\rho_{p,\mu}$ and covariance of $\mathcal{C}_{\mu,\nu}$, and persists under the mixture. Consequently, $P_{\nu,L}$ reduces both states, and the direct-sum identity gives

$$F(\bar{\sigma}_{p,\nu}, \rho_{p,\nu}) = F(P_{\nu,L} \bar{\sigma}_{p,\nu} P_{\nu,L}, P_{\nu,L} \rho_{p,\nu} P_{\nu,L}) + F((I - P_{\nu,L}) \bar{\sigma}_{p,\nu} (I - P_{\nu,L}), (I - P_{\nu,L}) \rho_{p,\nu} (I - P_{\nu,L})).$$

The trace bound $F(A, B) \leq \sqrt{\text{Tr } A \text{ Tr } B}$ shows that the second term is at most $\sqrt{\text{Tr}[(I - P_{\nu,L}) \rho_{p,\nu}]}$, because $\text{Tr}[(I - P_{\nu,L}) \bar{\sigma}_{p,\nu}] \leq 1$.

Compactness of $K \subseteq \Delta_d^{\circ}$ gives $b_K := \sup_{p \in K, \alpha \in \mathbb{R}_+} q_{\alpha} < 1$. Let $k_L := \lfloor L/((d-1)r_d) \rfloor + 1$. Since there are r_d roots, each of height at most $d-1$, a number vector outside $P_L^{\mathcal{F}}$ has $r_{\alpha} \geq k_L$ for some α . Under τ_p , each r_{α} has the geometric law $\Pr_{\tau_p}(r_{\alpha} = j) = (1 - q_{\alpha}) q_{\alpha}^j$, so $\Pr_{\tau_p}(r_{\alpha} \geq k_L) = q_{\alpha}^{k_L} \leq b_K^{k_L}$. A union bound gives

$$t_L := \sup_{p \in K} \text{Tr}[(I - P_L^{\mathcal{F}}) \tau_p] \leq r_d b_K^{k_L} \rightarrow 0 \quad (L \rightarrow \infty). \quad (108)$$

Taking traces in (107) now gives $\text{Tr}[(I - P_{\nu,L}) \rho_{p,\nu}] \leq t_L + \varepsilon_{n,L}$ uniformly for fixed L . The states $\tau_p^{(\gamma)}$ and τ_p are number diagonal, so $P_L^{\mathcal{F}}$ reduces them. The same trace bound makes their complementary fidelity at most $\sqrt{t_L}$. Consequently,

$$\begin{aligned} \left| F(\bar{\sigma}_{p,\nu}, \rho_{p,\nu}) - F(\tau_p^{(\gamma)}, \tau_p) \right| &\leq \left| F(P_{\nu,L} \bar{\sigma}_{p,\nu} P_{\nu,L}, P_{\nu,L} \rho_{p,\nu} P_{\nu,L}) - F(P_L^{\mathcal{F}} \tau_p^{(\gamma)} P_L^{\mathcal{F}}, P_L^{\mathcal{F}} \tau_p P_L^{\mathcal{F}}) \right| \\ &\quad + \sqrt{\text{Tr}[(I - P_{\nu,L}) \rho_{p,\nu}]} + \sqrt{\text{Tr}[(I - P_L^{\mathcal{F}}) \tau_p]}. \end{aligned}$$

By (35) and (106)–(107), together with the mixture bound above, for every fixed L the cutoff-fidelity difference on the right is at most $2\sqrt{\varepsilon_{n,L}}$ uniformly. The preceding tail estimates therefore give

$$\left| F(\bar{\sigma}_{p,\nu}, \rho_{p,\nu}) - F(\tau_p^{(\gamma)}, \tau_p) \right| \leq 2\sqrt{\varepsilon_{n,L}} + \sqrt{t_L + \varepsilon_{n,L}} + \sqrt{t_L}.$$

First let $n \rightarrow \infty$ at fixed L , and then let $L \rightarrow \infty$, using $\varepsilon_{n,L} \rightarrow 0$ uniformly and (108). This proves uniform convergence. Finally,

$$F(\tau_p^{(\gamma)}, \tau_p) = \prod_{i < j} F(\tau_{q_{ij}^{(\gamma)}}, \tau_{q_{ij}}) = F_{\text{orb}}(\gamma, p)$$

by multiplicativity of the trace norm under tensor products, (38), and (42). This proves the mixture assertion of Proposition 2.3. Taking a point mass gives the individual-sector assertion. $\square$

B Young local limit and randomized rounding

To prove [Proposition 2.4](#), we represent each Young-label law as a density constant on lattice cells. In this representation, randomized rounding is a dilation followed by cell averaging. A uniform Gaussian local limit then yields $F_{\text{cl}}(\gamma, d)$.

Define the half-open fundamental cell

$$\mathbf{C} = \left\{ \sum_{i=1}^{d-1} t_i(e_i - e_d) : -\frac{1}{2} \leq t_i < \frac{1}{2} \right\} \subset \mathbf{H}.$$

Volumes are taken with the induced $(d-1)$ -dimensional Lebesgue measure. Except on cell boundaries, which have probability zero, the rounding rule [\(26\)](#) selects the unique $\tilde{\nu} \in \mathbf{L}_{m_n}$ for which

$$\gamma_n(\mu + y) \in \tilde{\nu} + \mathbf{C}, \quad y = \sum_{i=1}^{d-1} y_i(e_i - e_d).$$

The first $d-1$ coordinates follow the nearest-integer rule. The final coordinate is determined by the required sum. Since y is uniform on $\mathbf{C}$,

$$\mathbf{R}_n(\tilde{\nu} \mid \mu) = \frac{|\gamma_n(\mu + \mathbf{C}) \cap (\tilde{\nu} + \mathbf{C})|}{\gamma_n^{d-1} |\mathbf{C}|}. \quad (109)$$

Extend every label law on $\mathbf{Y}_N$ by zero to $\mathbf{L}_N$.

For $\lambda \in \mathbf{L}_N$, define the recentered and rescaled lattice cell

$$\mathbf{C}_{N,p}(\lambda) = \frac{\lambda - Np + \mathbf{C}}{\sqrt{N}}.$$

For any $R : \mathbf{L}_N \rightarrow \mathbb{R}$ satisfying $\sum_{\lambda \in \mathbf{L}_N} |R(\lambda)| < \infty$, define its piecewise-constant interpolation by

$$(\mathcal{I}_{N,p}R)(x) = \frac{N^{(d-1)/2}}{|\mathbf{C}|} R(\lambda) \quad (x \in \mathbf{C}_{N,p}(\lambda)).$$

These cells tile $\mathbf{H}$ up to boundaries of zero Lebesgue measure. Each has volume $|\mathbf{C}|N^{-(d-1)/2}$, which cancels the density prefactor $N^{(d-1)/2}/|\mathbf{C}|$ in $\mathcal{I}_{N,p}$. Hence $\mathcal{I}_{N,p}$ preserves ℓ^1 -distance, $\|\mathcal{I}_{N,p}R - \mathcal{I}_{N,p}S\|_1 = \|R - S\|_{\ell^1}$, and, for nonnegative R, S , Hellinger affinity, $\int_{\mathbf{H}} \sqrt{(\mathcal{I}_{N,p}R)(\mathcal{I}_{N,p}S)} \, dx = \sum_{\lambda \in \mathbf{L}_N} \sqrt{R(\lambda)S(\lambda)}$.

Lemma B.1 (Uniform Young local limit). *For every compact $K \Subset \Delta_d^\circ$,*

$$\sup_{p \in K} \|\mathcal{I}_{N,p}P_{N,p} - \varphi_{0,\Sigma_p}\|_1 \longrightarrow 0. \quad (110)$$

Proof. Write $f_{N,p} = \mathcal{I}_{N,p}P_{N,p}$, and fix a ball $B_R = \{x \in \mathbf{H} : \|x\| \leq R\}$. For every cell $\mathbf{C}_{N,p}(\lambda)$ meeting B_R , its center $x_\lambda = (\lambda - Np)/\sqrt{N}$ lies in a bounded set independent of N, p . Compactness of $K \Subset \Delta_d^\circ$ gives positive lower bounds on p_d and all adjacent gaps, so these λ belong to $\mathbf{Y}_N$ for all sufficiently large N . The Frobenius dimension formula and [Lemma A.2](#) give, uniformly on these cells,

$$P_{N,p}(\lambda) = \frac{N! \prod_i p_i^{\lambda_i}}{\prod_i \lambda_i!} \prod_{i < j} \frac{\lambda_i - \lambda_j + j - i}{N(p_i - p_j)} \prod_i \prod_{a=1}^{d-i} \frac{Np_i}{\lambda_i + a} (1 + o(1)).$$

Each factor in the two last products is $1 + O_{K,R}(N^{-1/2})$. Stirling's formula and a Taylor expansion of the first factor therefore yield

$$P_{N,p}(\lambda) = \frac{\exp\left[-\frac{1}{2} \sum_i x_{\lambda,i}^2 / p_i\right]}{(2\pi N)^{(d-1)/2} \sqrt{\prod_i p_i}} (1 + o(1)).$$

The normalization agrees with the Gaussian density on $\mathbf{H}$:

$$|\mathbf{C}| = \sqrt{d}, \quad \det_{\mathbf{H}} \Sigma_p = d \prod_i p_i, \quad \langle x, \Sigma_p^{-1} x \rangle = \sum_i \frac{x_i^2}{p_i} \quad (x \in \mathbf{H}).$$

Indeed, the Gram matrix of $(e_i - e_d)_{i < d}$ has determinant d . The ambient covariance $\text{diag}(p) - pp^\top$ has kernel $\text{span}\{(1, \dots, 1)\}$. Deleting row and column i leaves a principal minor of determinant $(\prod_{j \neq i} p_j)(1 - \sum_{j \neq i} p_j) = \prod_j p_j$. The product of the nonzero eigenvalues is the sum of these d principal minors, giving $\det_{\mathbf{H}} \Sigma_p = d \prod_i p_i$. Its inverse on $\mathbf{H}$ sends x to $\text{diag}(p)^{-1}x$ up to a multiple of $(1, \dots, 1)$, giving the quadratic identity. For cells meeting B_R , the Gaussian log-density has gradient norm $\|\nabla \log \varphi_{0,\Sigma_p}(z)\| = \|\Sigma_p^{-1}z\| = O_{K,R}(1)$, while each cell has diameter $O(N^{-1/2})$. Thus its density changes by a factor $1 + O_{K,R}(N^{-1/2})$ within a cell. Together with the cell-center estimate for $P_{N,p}(\lambda)$, this gives

$$\sup_{p \in K} \sup_{x \in B_R} |f_{N,p}(x) - \varphi_{0,\Sigma_p}(x)| \longrightarrow 0.$$

Both densities have integral one. Consequently,

$$\|f_{N,p} - \varphi_{0,\Sigma_p}\|_1 \leq 2 \int_{B_R} |f_{N,p} - \varphi_{0,\Sigma_p}| dx + 2 \int_{B_R^c} \varphi_{0,\Sigma_p}(x) dx.$$

Let $N \rightarrow \infty$, then $R \rightarrow \infty$. The Gaussian tails vanish uniformly on compact K , proving (110). $\square$

Lemma B.2 (Continuous limit of randomized dilation). *Let $\tilde{Q}_{m_n,p} = P_{n,p} \mathbf{R}_n$ be the law obtained by applying $\mathbf{R}_n$ to an input drawn from $P_{n,p}$, before the fallback replacement in (27). For every compact $K \subseteq \Delta_d^\circ$,*

$$\sup_{p \in K} \|\mathcal{I}_{m_n,p} \tilde{Q}_{m_n,p} - \varphi_{0,\gamma \Sigma_p}\|_1 \longrightarrow 0. \quad (111)$$

$$\sup_{p \in K} |F(\tilde{Q}_{m_n,p}, P_{m_n,p}) - F_{\text{cl}}(\gamma, d)| \longrightarrow 0. \quad (112)$$

Proof. For $a > 0$ and $f \in L^1(\mathbf{H})$, set

$$(\text{dil}_a f)(z) = a^{-(d-1)/2} f(z/\sqrt{a}),$$

and define the target-cell averaging operator by

$$(\text{Av}_{m_n,p} f)(x) := \frac{m_n^{(d-1)/2}}{|\mathbf{C}|} \int_{\mathbf{C}_{m_n,p}(\nu)} f(z) dz, \quad x \in \mathbf{C}_{m_n,p}(\nu).$$

If $X = (\mu - np + y)/\sqrt{n}$, where $\mu \sim P_{n,p}$ and y is independent of μ and uniform on $\mathbf{C}$, then X has density $\mathcal{I}_{n,p} P_{n,p}$. Because $m_n = \gamma_n n$, $\sqrt{\gamma_n} X = [\gamma_n(\mu + y) - m_n p]/\sqrt{m_n}$. Thus the condition $\sqrt{\gamma_n} X \in \mathbf{C}_{m_n,p}(\nu)$ is equivalent to the rounding event $\gamma_n(\mu + y) \in \nu + \mathbf{C}$. Hence the dilated density assigns mass $\tilde{Q}_{m_n,p}(\nu)$ to that target cell. Averaging over each target cell gives

$$\text{Av}_{m_n,p} \text{dil}_{\gamma_n}(\mathcal{I}_{n,p} P_{n,p}) = \mathcal{I}_{m_n,p} \tilde{Q}_{m_n,p}. \quad (113)$$

Cell averaging is an L^1 -contraction and dilation an L^1 -isometry. Since $\text{dil}_{\gamma_n} \varphi_{0,\Sigma_p} = \varphi_{0,\gamma_n \Sigma_p}$, (113) yields

$$\|\mathcal{I}_{m_n,p} \tilde{Q}_{m_n,p} - \varphi_{0,\gamma \Sigma_p}\|_1 \leq \|\mathcal{I}_{n,p} P_{n,p} - \varphi_{0,\Sigma_p}\|_1 + \|\text{Av}_{m_n,p} \varphi_{0,\gamma_n \Sigma_p} - \varphi_{0,\gamma \Sigma_p}\|_1 + \|\varphi_{0,\gamma_n \Sigma_p} - \varphi_{0,\gamma \Sigma_p}\|_1.$$

The first term tends uniformly to zero by Lemma B.1, and the last does so because $\gamma_n \rightarrow \gamma$. For the middle term, the cellwise L^1 Poincaré inequality gives, for smooth f with integrable gradient,

$$\|\text{Av}_{m_n,p} f - f\|_1 \leq C_d m_n^{-1/2} \|\nabla f\|_1.$$

Gaussian gradients have uniformly bounded L^1 -norm on the compact covariance family $\gamma_n \Sigma_p$, so this term is $O_K(m_n^{-1/2})$. This proves (111).

Applying Lemma B.1 at sample size m_n gives

$$\mathcal{I}_{m_n,p} P_{m_n,p} \longrightarrow \varphi_{0,\Sigma_p}$$

uniformly in L^1 . The embedding preserves Hellinger affinity, and (35) also holds for nonnegative L^1 densities. Therefore

$$F(\tilde{Q}_{m_n,p}, P_{m_n,p}) \longrightarrow F(\mathbf{N}_{0,\gamma \Sigma_p}, \mathbf{N}_{0,\Sigma_p}) = F_{\text{cl}}(\gamma, d)$$

uniformly on K . $\square$

Proof of Proposition 2.4. Write $e = \tilde{\nu} - \gamma_n \mu$. The nearest-integer rule gives $|e_i| \leq (\gamma_n + 1)/2$ for $i < d$, and $e_d = -\sum_{i < d} e_i$. Thus $\|e\|_\infty \leq c_{\text{rnd}}$, where

$$c_{\text{rnd}} = \frac{d-1}{2} \left(1 + \sup_n \gamma_n \right) < \infty,$$

where finiteness follows from $\gamma_n \rightarrow \gamma$.

For $p \in K$ and a typical input label $\mu \in \mathbf{T}_n(p)$, the identity $\tilde{\nu}/m_n = \mu/n + e/m_n$ gives

$$\left\| \frac{\tilde{\nu}}{m_n} - p \right\|_\infty \leq n^{-1/3} + \frac{c_{\text{rnd}}}{m_n}.$$

Apply the compatibility assertion of Proposition 2.3 with $\delta_n = n^{-1/3} + c_{\text{rnd}}/m_n$. For all sufficiently large n , it yields $\tilde{\nu} - \mu \vdash m_n - n$, so the fallback rule leaves every typical rounded label unchanged. This proves (31).

Coupling the raw and final labels by the fallback rule therefore gives

$$\|Q_{m_n,p} - \tilde{Q}_{m_n,p}\|_{\ell^1} \leq 2P_{n,p}(\mathbf{T}_n(p)^c) = o(1)$$

uniformly on K , by Lemma 2.1. Combining (112) with (35) proves (32). $\square$

C Unified Gaussian flat-prior converse

We prove the two converses in Theorem 3.1 through the common flat-prior bound in Proposition C.5. Its proof combines weighted fidelity, limits tested on compact observables, a least-noise oscillator moment, and Gaussian translation.

The classical parameter space has dimension $k = 0$ in the orbital model and $k = d - 1$ in the full model. All trace-class-valued integrals below are Bochner integrals in trace norm. For classical-quantum maps, complete positivity means positivity of $\text{id}_{M_a} \otimes \Lambda$ for every $a \geq 1$, with the matrix factor acting on the quantum trace-class fibers. The classical algebra is abelian.

C.1 Weighted fidelity and compact limits

Lemma C.1 (Weighted fidelity bound). *Let R, T be positive trace-class operators, and let W be bounded, positive, and injective. Define*

$$\mathrm{Tr}(TW^{-1}) := \lim_{\varepsilon \downarrow 0} \mathrm{Tr}(T(W + \varepsilon I)^{-1}) \in [0, \infty].$$

If this quantity is finite, then

$$F(R, T)^2 \leq \mathrm{Tr}(RW) \mathrm{Tr}(TW^{-1}). \quad (114)$$

Let $\mathbf{E}$ be a finite-dimensional Euclidean space, let $R \in L^1(\mathbf{E}; \mathcal{T}_1(\mathcal{F}_{\mathrm{orb}}))$ be positive, let g be a probability density on $\mathbf{E}$, and let $\chi > 0$ be measurable. For positive R, S in this space, write $F(R, S) = \int_{\mathbf{E}} F(R(y), S(y)) \, dy$, in agreement with the hybrid-state convention in [Section 3](#). The notation gT means the density $y \mapsto g(y)T$. Whenever the right-hand side below is finite,

$$F(R, gT)^2 \leq \left(\int_{\mathbf{E}} \chi(y) \mathrm{Tr}[R(y)W] \, dy \right) \left(\int_{\mathbf{E}} \frac{g(y)}{\chi(y)} \, dy \right) \mathrm{Tr}(TW^{-1}). \quad (115)$$

Proof. For $W_\varepsilon = W + \varepsilon I$, the Schatten Hölder inequality gives

$$F(R, T) = \|(R^{1/2}W_\varepsilon^{1/2})(W_\varepsilon^{-1/2}T^{1/2})\|_1 \leq \sqrt{\mathrm{Tr}(RW_\varepsilon)} \sqrt{\mathrm{Tr}(TW_\varepsilon^{-1})}.$$

Squaring and letting $\varepsilon \downarrow 0$, using $\mathrm{Tr}(RW_\varepsilon) = \mathrm{Tr}(RW) + \varepsilon \mathrm{Tr} R$ and monotone convergence for the second trace, proves (114). For the hybrid state,

$$F(R, gT) = \int_{\mathbf{E}} \sqrt{g(y)} F(R(y), T) \, dy.$$

Apply the first inequality pointwise and then Cauchy–Schwarz to the factors $\sqrt{\chi(y) \mathrm{Tr}[R(y)W]}$ and $\sqrt{g(y)/\chi(y)}$. This gives (115). $\square$

Lemma C.2 (Compact-observable limits of completely positive maps). *Let $\mathcal{H}_0, \mathcal{H}_1$ be separable Hilbert spaces and let $\Gamma_j : \mathcal{T}_1(\mathcal{H}_0) \rightarrow \mathcal{T}_1(\mathcal{H}_1)$ be completely positive maps with $\sup_j \|\Gamma_j\|_{1 \rightarrow 1} < \infty$. There are a subsequence and a completely positive map $\Gamma : \mathcal{T}_1(\mathcal{H}_0) \rightarrow \mathcal{T}_1(\mathcal{H}_1)$ such that*

$$\mathrm{Tr}[\Gamma_j(X)O] \longrightarrow \mathrm{Tr}[\Gamma(X)O] \quad (116)$$

for every $X \in \mathcal{T}_1(\mathcal{H}_0)$ and every compact operator O on $\mathcal{H}_1$. If $\limsup_j \mathrm{Tr} \Gamma_j(X) \leq c \mathrm{Tr} X$ for every $X \geq 0$, then $\mathrm{Tr} \Gamma(X) \leq c \mathrm{Tr} X$.

Suppose in addition that $U_g \in \mathrm{U}(\mathcal{H}_0)$ and $V_g \in \mathrm{U}(\mathcal{H}_1)$ are families of unitaries indexed by a set G , and that, for every fixed $g \in G$ and $X \in \mathcal{T}_1(\mathcal{H}_0)$,

$$\|\Gamma_j(U_g X U_g^*) - V_g \Gamma_j(X) V_g^*\|_1 \longrightarrow 0.$$

Then

$$\Gamma(U_g X U_g^*) = V_g \Gamma(X) V_g^* \quad (117)$$

for every $g \in G$ and $X \in \mathcal{T}_1(\mathcal{H}_0)$.

Proof. Let $M = \sup_j \|\Gamma_j\|_{1 \rightarrow 1}$, and let $\mathcal{D}$ be a countable norm-dense set of compact operators on $\mathcal{H}_1$. Since $\mathcal{T}_1(\mathcal{H}_0)$ is separable, a diagonal weak-* compactness argument yields a subsequence along which $\Gamma_j^*(O)$ converges weak-* in $\mathcal{B}(\mathcal{H}_0)$ for every $O \in \mathcal{D}$. The bound $\|\Gamma_j^*(O)\| \leq M\|O\|$ extends both the limit and the convergence to every compact O , producing a bounded complex-linear map Γ^* from compact operators on $\mathcal{H}_1$ to $\mathcal{B}(\mathcal{H}_0)$.

For every positive block matrix $[O_{ab}]$ of compact operators, $[\Gamma_j^*(O_{ab})] \geq 0$, so its entrywise weak-* limit $[\Gamma^*(O_{ab})]$ is positive. Thus Γ^* is completely positive.

Since the dual of the compact operators on $\mathcal{H}_1$ is $\mathcal{T}_1(\mathcal{H}_1)$, define $\Gamma(X)$ by

$$\mathrm{Tr}[\Gamma(X)O] = \mathrm{Tr}[X\Gamma^*(O)] \quad (O \text{ compact}).$$

Then Γ is completely positive and (116) holds. If $P_N \uparrow I$ are finite-rank projections and $X \geq 0$, then

$$\mathrm{Tr} \Gamma(X) = \sup_N \lim_j \mathrm{Tr}[\Gamma_j(X)P_N] \leq \lim_j \sup \mathrm{Tr} \Gamma_j(X) \leq c \mathrm{Tr} X.$$

Finally, for fixed g, X and compact O , the operator $V_g^* O V_g$ is compact. Passing to the limit gives

$$\mathrm{Tr}[\Gamma(U_g X U_g^*)O] = \mathrm{Tr}[V_g \Gamma(X) V_g^* O].$$

Compact operators separate trace-class operators, so the limiting map is covariant. $\square$

C.2 The least-noise oscillator moment

Fix $\gamma > 1$. For $s \geq 1$ and $q_1, \dots, q_s \in (0, 1)$, let

$$q_\ell^{(\gamma)} = 1 - \frac{1 - q_\ell}{\gamma}, \quad \tau = \bigotimes_{\ell=1}^s \tau_{q_\ell}, \quad \tau^{(\gamma)} = \bigotimes_{\ell=1}^s \tau_{q_\ell^{(\gamma)}}.$$

Lemma C.3 (Covariant-amplifier representation). *Let $G_1, \dots, G_s > 1$, and write*

$$\sqrt{\mathbf{G}}z = (\sqrt{G_\ell} z_\ell)_{\ell=1}^s.$$

Every s -mode channel Λ satisfying

$$\Lambda(D(z)X D(z)^*) = D(\sqrt{\mathbf{G}}z) \Lambda(X) D(\sqrt{\mathbf{G}}z)^*$$

admits an idler realization

$$a_{\ell, \text{out}} = \sqrt{G_\ell} a_{\ell, \text{in}} + \sqrt{G_\ell - 1} b_\ell^*, \quad \ell = 1, \dots, s. \quad (118)$$

Here $a_{\ell, \text{in}}$ and b_ℓ annihilate the input and idler modes, respectively, and b_ℓ^ creates an idler excitation. For some joint idler state σ , a product U of two-mode squeezers implements $\Lambda(X) = \mathrm{Tr}_{\text{idler}}[U(X \otimes \sigma)U^*]$, with $a_{\ell, \text{out}} = U^*(a_{\ell, \text{in}} \otimes I)U$ in the Heisenberg picture. The channel uniquely determines the joint idler state, which may be non-Gaussian, nondiagonal in the number basis, and correlated across modes.*

Proof. Covariance and Weyl irreducibility give $\Lambda^*(D(w)) = f(w)D(\sqrt{\mathbf{G}}w)$. Vacuum matrix elements show that f is continuous at zero, so Λ is a linear bosonic channel. See [Hol13, Ch. 12] for the bosonic-channel formalism. Its defect form is $\bigoplus_{\ell=1}^s (1 - G_\ell) \Omega_2$, where $\Omega_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. This form is invertible, so [LSW18, Lemma 7] gives an exact dilation with s idler modes. Choose the noise matrix $\bigoplus_{\ell=1}^s \sqrt{G_\ell - 1} \mathrm{diag}(1, -1)$ and complete these blocks with a product of two-mode squeezers, yielding (118). The noise matrix is invertible, so f determines the idler characteristic function and hence its state. $\square$

Proposition C.4 (Least-noise moment). *Let Γ be a completely positive trace-nonincreasing map on the s -mode trace class satisfying*

$$\Gamma(D(z)XD(z)^*) = D(\sqrt{\gamma}z)\Gamma(X)D(\sqrt{\gamma}z)^* \quad (z \in \mathbb{C}^s).$$

Let $W(\hat{\mathbf{N}}) := \prod_{\ell=1}^s w_\ell(\hat{N}_\ell)$, where every w_ℓ is bounded, nonnegative, and nonincreasing on $\mathbb{Z}_{\geq 0}$. Then

$$\text{Tr}[\Gamma(\tau)W(\hat{\mathbf{N}})] \leq \text{Tr}[\Gamma(\tau)] \text{Tr}[\tau^{(\gamma)}W(\hat{\mathbf{N}})]. \quad (119)$$

Proof. Displacement covariance makes $\Gamma^*(I)$ commute with every Weyl displacement. Irreducibility therefore gives $\Gamma^*(I) = cI$, with $c = \text{Tr} \Gamma(\tau) \in [0, 1]$. The case $c = 0$ is immediate. Otherwise $\Lambda = c^{-1}\Gamma$ is a channel with the same covariance.

The quantum-limited amplifier $\mathcal{A}_G$ sends vacuum to $\tau_{1-1/G}$. Thus

$$\tau = \left(\bigotimes_{\ell=1}^s \mathcal{A}_{1/(1-q_\ell)} \right) (|\mathbf{0}\rangle\langle\mathbf{0}|). \quad (120)$$

The composite $\mathcal{B} = \Lambda \circ \bigotimes_{\ell} \mathcal{A}_{1/(1-q_\ell)}$ therefore sends joint vacuum to $\Lambda(\tau)$, with modewise gains

$$G_\ell = \frac{\gamma}{1 - q_\ell}, \quad \lambda_\ell = 1 - G_\ell^{-1} = q_\ell^{(\gamma)}.$$

Apply [Lemma C.3](#) to $\mathcal{B}$, and let σ be its joint idler state.

For a single two-mode squeezer, normal-ordered disentangling [\[BR97\]](#) gives, with $\lambda = 1 - G^{-1}$,

$$U_G|0, l\rangle = (1 - \lambda)^{(l+1)/2} \sum_{k \geq 0} \sqrt{\binom{k+l}{l}} \lambda^{k/2} |k, k+l\rangle. \quad (121)$$

For fixed $\mathbf{k}$, tracing out the idler requires $\mathbf{k} + \mathbf{l} = \mathbf{k} + \mathbf{l}'$, hence $\mathbf{l} = \mathbf{l}'$. Thus only $a_{\mathbf{l}} = \langle \mathbf{l} | \sigma | \mathbf{l} \rangle$ enters the output number probabilities:

$$\langle \mathbf{k} | \Lambda(\tau) | \mathbf{k} \rangle = \sum_{\mathbf{l} \in \mathbb{Z}_{\geq 0}^s} a_{\mathbf{l}} \prod_{\ell=1}^s \binom{k_\ell + l_\ell}{l_\ell} (1 - \lambda_\ell)^{l_\ell+1} \lambda_\ell^{k_\ell}. \quad (122)$$

Conditional on $\mathbf{l}$, the coordinates are independent $\text{NB}(l_\ell + 1, \lambda_\ell)$ variables. The mixing law $a_{\mathbf{l}}$ may be correlated. No product or phase-invariance assumption on the idler is needed.

Let $\mathbf{L} = (L_1, \dots, L_s)$ satisfy $\Pr(\mathbf{L} = \mathbf{l}) = a_{\mathbf{l}}$. Independently, draw geometric variables $X_{\ell,j}$, mutually independent over ℓ and j , with $\Pr(X_{\ell,j} = k) = (1 - \lambda_\ell)\lambda_\ell^k$ for $j, k \geq 0$. The coupling

$$N_\ell = \sum_{j=0}^{L_\ell} X_{\ell,j} \geq X_{\ell,0} \quad (\ell = 1, \dots, s)$$

has the output law [\(122\)](#), while $(X_{\ell,0})_\ell$ has the number law of $\tau^{(\gamma)}$. Since each w_ℓ is nonnegative and nonincreasing,

$$\text{Tr}[\Lambda(\tau)W] = \mathbb{E} \prod_{\ell} w_\ell(N_\ell) \leq \mathbb{E} \prod_{\ell} w_\ell(X_{\ell,0}) = \text{Tr}[\tau^{(\gamma)}W].$$

Multiplying by $c = \text{Tr} \Gamma(\tau)$ proves [\(119\)](#). □

Specialize to $s = r_d$, $q_\ell = q_{ij}$, $\tau = \tau_p$, and $\tau^{(\gamma)} = \tau_p^{(\gamma)}$. For $\mathbf{k} = (k_{ij})_{i < j} \in \mathbb{Z}_{\geq 0}^{r_d}$, let $t_{\mathbf{k}}$ and $t_{\mathbf{k}}^{(\gamma)}$ be the corresponding number-basis eigenvalues, and define

$$W_\gamma = \sum_{\mathbf{k}} \sqrt{\frac{t_{\mathbf{k}}}{t_{\mathbf{k}}^{(\gamma)}}} |\mathbf{k}\rangle \langle \mathbf{k}|. \quad (123)$$

Equivalently,

$$W_\gamma = \prod_{i < j} w_{ij}(\hat{N}_{ij}), \quad w_{ij}(n) = \sqrt{\frac{1 - q_{ij}}{1 - q_{ij}^{(\gamma)}}} \left(\frac{q_{ij}}{q_{ij}^{(\gamma)}} \right)^{n/2}.$$

Here $\hat{N}_{ij}$ is the number operator of the mode indexed by $i < j$, and $|\mathbf{k}| := \sum_{i < j} k_{ij}$ is the total occupation number. Since $q_{ij} < q_{ij}^{(\gamma)}$, each $w_{ij}(n)$ is positive, bounded, and strictly decreasing, and the eigenvalues of W_γ tend to zero as $|\mathbf{k}| \rightarrow \infty$. Thus W_γ is bounded, injective, and compact, with

$$\text{Tr}(\tau_p^{(\gamma)} W_\gamma) = \text{Tr}(\tau_p W_\gamma^{-1}) = \sum_{\mathbf{k}} \sqrt{t_{\mathbf{k}} t_{\mathbf{k}}^{(\gamma)}} = F_{\text{orb}}(\gamma, p), \quad (124)$$

where the inverse is understood as in [Lemma C.1](#).

C.3 One flat-prior converse for both Gaussian models

Proposition C.5 (Flat-prior converse inequality). *Let $k \in \{0, d-1\}$. For $k = 0$, let $\mathbf{E} = \{0\}$, omit the classical component, and interpret h -integration as evaluation at 0. For $k = d-1$, let $\mathbf{E} = \mathbf{H}$ with its induced Euclidean measure. Let*

$$\mathbf{E}_L = \{h \in \mathbf{E} : \|h\|_\infty \leq L\}, \quad \Xi_L = \mathbf{E}_L \times \mathbf{Z}_L,$$

with $|\mathbf{E}_L| = 1$ when $k = 0$. Define

$$R_{0,z}^{(0)} = \Phi_z^{(p)}, \quad S_{0,z}^{(0)} = \Psi_z^{(p,\gamma)}, \quad R_{h,z}^{(d-1)} = \Theta_{h,z}^{\text{in}}, \quad S_{h,z}^{(d-1)} = \Theta_{h,z}^{\text{tar}}, \quad c_k = \left(\frac{2\sqrt{\gamma}}{1+\gamma} \right)^{k/2}.$$

For $k = 0$, the supremum below is over quantum channels on $\mathcal{F}_{\text{orb}}$. For $k = d-1$, it is over classical-quantum channels. Then

$$\limsup_{L \rightarrow \infty} \sup_{\Lambda} \frac{1}{|\Xi_L|} \int_{\Xi_L} F(\Lambda(R_{h,z}^{(k)}), S_{h,z}^{(k)}) \, dh \, dz \leq c_k F_{\text{orb}}(\gamma, p). \quad (125)$$

Proof. The classical weight below captures the spectral affinity, while the compact witness W_γ controls the oscillator output. For $k = 0$, set $g_1 = g_\gamma = \chi = 1$ on the one-point space. For $k = d-1$, let g_1 and g_γ be the centered Gaussian densities with covariances Σ_p and $\gamma \Sigma_p$, and set

$$\chi(y) = \sqrt{\frac{g_1(y)}{g_\gamma(y)}} = \gamma^{k/4} \exp \left[-\frac{\gamma-1}{4\gamma} \langle y, \Sigma_p^{-1} y \rangle \right].$$

This function is bounded and strictly positive. In both cases,

$$\int_{\mathbf{E}} \frac{g_1(y)}{\chi(y)} \, dy = \int_{\mathbf{E}} \sqrt{g_1(y) g_\gamma(y)} \, dy = c_k. \quad (126)$$

For $k = d - 1$, completing the square also gives

$$\int_{\mathbb{E}} g_1(h) \chi(v + \sqrt{\gamma}h) dh = c_k \exp \left[-\frac{\gamma - 1}{2\gamma(\gamma + 1)} \langle v, \Sigma_p^{-1}v \rangle \right] \leq c_k. \quad (127)$$

For $\xi = (h, z)$, define

$$(\mathcal{U}_\xi^{\text{in}} R)(x) = D(z)R(x - h)D(z)^*, \quad (\mathcal{U}_\xi^{\text{out}} R)(y) = D(\sqrt{\gamma}z)R(y - \sqrt{\gamma}h)D(\sqrt{\gamma}z)^*.$$

Then $R_\xi^{(k)} = \mathcal{U}_\xi^{\text{in}}(g_1\tau_p)$ and $S_\xi^{(k)} = \mathcal{U}_\xi^{\text{out}}(g_1\tau_p)$. For a channel Λ , define its average pointwise by

$$\bar{\Lambda}_L(R) = \frac{1}{|\Xi_L|} \int_{\Xi_L} \mathcal{U}_{-\eta}^{\text{out}}(\Lambda(\mathcal{U}_\eta^{\text{in}} R)) d\eta.$$

For each fixed input R , the integrand is trace-norm continuous and bounded in norm by $\|R\|_1$, so the Bochner integral is well-defined. Because F is invariant under the output isometries $\mathcal{U}_\xi^{\text{out}}$ and jointly concave, the finite-average inequality follows. Fidelity is concave and trace-norm continuous in its first argument, so Riemann sums pass this inequality to the Bochner average:

$$F_L(\Lambda) = \frac{1}{|\Xi_L|} \int_{\Xi_L} F(\Lambda(R_\xi^{(k)}), S_\xi^{(k)}) d\xi \leq F(\bar{\Lambda}_L(g_1\tau_p), g_1\tau_p). \quad (128)$$

Let $F_* = \limsup_{L \rightarrow \infty} \sup_{\Lambda} F_L(\Lambda)$. Choose $L_j \rightarrow \infty$ with $\sup_{\Lambda} F_{L_j}(\Lambda) \rightarrow F_*$, and channels $\tilde{\Lambda}_j$ satisfying $F_{L_j}(\tilde{\Lambda}_j) \geq \sup_{\Lambda} F_{L_j}(\Lambda) - j^{-1}$. Set

$$\Lambda_j = \overline{\tilde{\Lambda}_{L_j}}, \quad R_j = \Lambda_j(g_1\tau_p).$$

Then

$$F_{L_j}(\tilde{\Lambda}_j) \rightarrow F_*, \quad F_{L_j}(\tilde{\Lambda}_j) \leq F(R_j, g_1\tau_p). \quad (129)$$

Write

$$A_{j,u} = \mathcal{U}_{-u}^{\text{out}} \circ \tilde{\Lambda}_j \circ \mathcal{U}_u^{\text{in}}.$$

The group law and the change of variables $u = \eta + \xi$ give

$$\begin{aligned} \Lambda_j(\mathcal{U}_\xi^{\text{in}} R) &= \mathcal{U}_\xi^{\text{out}} \frac{1}{|\Xi_{L_j}|} \int_{\Xi_{L_j} + \xi} A_{j,u}(R) du, \\ \mathcal{U}_\xi^{\text{out}} \Lambda_j(R) &= \mathcal{U}_\xi^{\text{out}} \frac{1}{|\Xi_{L_j}|} \int_{\Xi_{L_j}} A_{j,u}(R) du. \end{aligned}$$

The integrals over the intersection of the two averaging sets cancel. On their symmetric difference, each integrand has trace norm at most $\|R\|_1$: the output translations are isometries, and each $A_{j,u}$ is a channel and hence a trace-norm contraction.

$$\|\Lambda_j(\mathcal{U}_\xi^{\text{in}} R) - \mathcal{U}_\xi^{\text{out}} \Lambda_j(R)\|_1 \leq b_j(\xi) \|R\|_1, \quad b_j(\xi) := \frac{|(\Xi_{L_j} + \xi) \triangle \Xi_{L_j}|}{|\Xi_{L_j}|}. \quad (130)$$

Here $0 \leq b_j \leq 2$, and $b_j(\xi) = O_\xi(L_j^{-1}) \rightarrow 0$ for fixed ξ , since Ξ_1 is a bounded polytope and $\Xi_L = L\Xi_1$.

Weighting and integrating the classical output now produces a quantum map. The Gaussian translation bound controls its trace. Define

$$\Gamma_j^\chi(X) = \int_{\mathbb{E}} \chi(y) [\Lambda_j(g_1X)](y) dy.$$

These maps are completely positive and satisfy $\|\Gamma_j^\chi\|_{1 \rightarrow 1} \leq \|\chi\|_\infty$. We first show the sharper trace bound

$$\mathrm{Tr} \Gamma_j^\chi(X) \leq (c_k + e_j) \mathrm{Tr} X \quad (X \geq 0), \quad e_j := \|\chi\|_\infty \int_{\mathbb{E}} g_1(h) b_j(h, 0) dh \longrightarrow 0. \quad (131)$$

For $k = 0$, this is trace preservation, with $c_0 = 1$ and $e_j = 0$. For $k = d - 1$, take a nonnegative probability approximate identity u_ε on $\mathbb{E}$ and let $B_{j,\varepsilon} = \Lambda_j(u_\varepsilon X)$, whose total trace is $\mathrm{Tr} X$. Expressing $g_1 * u_\varepsilon$ as an average of translates of u_ε , (130) gives

$$\left\| \Lambda_j((g_1 * u_\varepsilon)X) - \int_{\mathbb{E}} g_1(h) \mathcal{U}_{(h,0)}^{\mathrm{out}} B_{j,\varepsilon} dh \right\|_1 \leq \mathrm{Tr} X \int_{\mathbb{E}} g_1(h) b_j(h, 0) dh.$$

Pairing with χ , changing variables, and using (127) therefore yields

$$\int_{\mathbb{E}} \chi(y) \mathrm{Tr}[\Lambda_j((g_1 * u_\varepsilon)X)(y)] dy \leq c_k \int_{\mathbb{E}} \mathrm{Tr} B_{j,\varepsilon}(v) dv + e_j \mathrm{Tr} X = (c_k + e_j) \mathrm{Tr} X.$$

Letting $\varepsilon \downarrow 0$ proves (131), since $g_1 * u_\varepsilon \rightarrow g_1$ in L^1 . Dominated convergence gives $e_j \rightarrow 0$.

Taking $\xi = (0, z)$ in (130) and integrating against χ gives

$$\|\Gamma_j^\chi(D(z)X D(z)^*) - D(\sqrt{\gamma}z) \Gamma_j^\chi(X) D(\sqrt{\gamma}z)^*\|_1 \leq \|\chi\|_\infty b_j(0, z) \|X\|_1 \longrightarrow 0.$$

Hence, after passing to a subsequence, Lemma C.2 and (131) give a completely positive, displacement-covariant limit Γ^χ with $\mathrm{Tr} \Gamma^\chi(X) \leq c_k \mathrm{Tr} X$ for $X \geq 0$, such that

$$\mathrm{Tr}[\Gamma_j^\chi(X)K] \longrightarrow \mathrm{Tr}[\Gamma^\chi(X)K] \quad (132)$$

for trace-class X and compact K . Since $c_k \leq 1$, the limit is trace nonincreasing. Define M_j by

$$M_j = \int_{\mathbb{E}} \chi(y) \mathrm{Tr}[R_j(y)W_\gamma] dy = \mathrm{Tr}[\Gamma_j^\chi(\tau_p)W_\gamma].$$

Because W_γ is compact, (132) passes this moment to the limit:

$$M_j \longrightarrow \mathrm{Tr}[\Gamma^\chi(\tau_p)W_\gamma]. \quad (133)$$

Moreover, Lemma C.1, (126), and (124) give

$$F(R_j, g_1 \tau_p)^2 \leq M_j c_k F_{\mathrm{orb}}(\gamma, p). \quad (134)$$

By Proposition C.4,

$$\lim_j M_j = \mathrm{Tr}[\Gamma^\chi(\tau_p)W_\gamma] \leq \mathrm{Tr} \Gamma^\chi(\tau_p) F_{\mathrm{orb}}(\gamma, p) \leq c_k F_{\mathrm{orb}}(\gamma, p).$$

Combining (129) with (134) yields $F_* \leq c_k F_{\mathrm{orb}}(\gamma, p)$, proving both cases. $\square$

D Comparison with purify–clone–trace

We compute the exact asymptotic PCT fidelity at simple full-rank spectra and derive a strict upper bound for flat projector states with $1 < r < d$, proving the comparisons stated in Section 5.

D.1 Common preliminaries

For $D \geq 1$, write $d_D(N) = \dim \text{Sym}^N(\mathbb{C}^D) = \binom{N+D-1}{D-1}$, and let $\Pi_{\text{sym},N}^{(D)}$ project onto this symmetric subspace. On inputs supported on $\text{Sym}^n(\mathbb{C}^D)$, Werner's pure-state cloner [Wer98] is

$$\mathcal{W}_{n,m}^{(D)}(X) = \frac{d_D(n)}{d_D(m)} \Pi_{\text{sym},m}^{(D)}(X \otimes I_D^{\otimes(m-n)}) \Pi_{\text{sym},m}^{(D)}, \quad m \geq n. \quad (135)$$

If ρ has rank at most r , choose a purification $\psi \in \mathbb{C}^d \otimes E$, $E = \mathbb{C}^r$. The random-purification channel acts on product inputs as [TWZ25] (see also [GML25, Eq. (50) and Theorem 22])

$$\mathcal{P}_n^{(r)}(\rho^{\otimes n}) = \int_{U(r)} ((I_d \otimes U)|\psi\rangle\langle\psi|(I_d \otimes U)^*)^{\otimes n} dU. \quad (136)$$

Here dU is normalized Haar measure. All purifications in this space differ by a unitary on E , so the average is independent of the chosen purification. Covariance of Werner's cloner and invariance of partial trace remove this average:

$$\mathcal{C}_{n,m}^{\text{PCT},r}(\rho^{\otimes n}) = \text{Tr}_{E^m} \mathcal{W}_{n,m}^{(dr)}(|\psi\rangle\langle\psi|^{\otimes n}). \quad (137)$$

Occupation and coherent-state limits. Fix $D \geq 2$, a unit vector $\psi \in \mathbb{C}^D$, and an orthonormal basis $e_1, \dots, e_{D-1}$ of $\psi^\perp$. For $\mathbf{k} = (k_1, \dots, k_{D-1})$, let $|\mathbf{k}\rangle_m$ be the normalized symmetric vector with k_j particles in e_j and the remaining $m - |\mathbf{k}|$ in ψ .

Lemma D.1 (Werner occupation law). *The output $\mathcal{W}_{n,m}^{(D)}(|\psi\rangle\langle\psi|^{\otimes n})$ is diagonal in this occupation basis, with eigenvalue*

$$\pi_{n,m}^{(D)}(\mathbf{k}) = \frac{\binom{m-|\mathbf{k}|}{n}}{\binom{m+D-1}{n+D-1}}, \quad |\mathbf{k}| \leq m - n. \quad (138)$$

Recall that $\tau_q = (1-q) \sum_{j \geq 0} q^j |j\rangle\langle j|$ is the one-mode thermal state. If $m_n/n \rightarrow \gamma > 1$, the output's pullback ω_{n,m_n} to the symmetric Fock space $\Gamma_s(\psi^\perp)$, with occupation basis $|\mathbf{k}\rangle$, satisfies

$$\left\| \omega_{n,m_n} - \tau_{(\gamma-1)/\gamma}^{\otimes(D-1)} \right\|_1 \rightarrow 0. \quad (139)$$

Proof. On an occupation vector with $t = |\mathbf{k}|$, the expectation of $\Pi_{\text{sym},m}^{(D)}(|\psi\rangle\langle\psi|^{\otimes n} \otimes I) \Pi_{\text{sym},m}^{(D)}$ is $\binom{m-t}{n} / \binom{m}{n}$. Multiplying by $d_D(n)/d_D(m)$ gives (138). The rotation $\psi \mapsto e^{i\theta}\psi$ fixes the input state and acts on the t -excitation subspace by $e^{i(m-t)\theta}$, so the output has no cross- t blocks. The stabilizer $U(D-1)$ acts irreducibly on each $\text{Sym}^t(\psi^\perp)$, making each block scalar. For each fixed $\mathbf{k}$,

$$\pi_{n,m_n}^{(D)}(\mathbf{k}) \rightarrow \gamma^{-(D-1)} \left(\frac{\gamma-1}{\gamma} \right)^{|\mathbf{k}|}.$$

The limit is a normalized product of geometric laws. The discrete Scheffé lemma therefore gives ℓ^1 convergence, which is (139). $\square$

For the rest of this appendix, L denotes a total product length. The M of Section 6.3 counts additional outputs. Let $\Pi_{\leq L}$ project onto Fock occupations of total number at most L , and let $J_L|\mathbf{k}\rangle = |\mathbf{k}\rangle_L$ on this subspace. The trace-preserving occupation embedding is

$$\mathcal{J}_L(X) = J_L \Pi_{\leq L} X \Pi_{\leq L} J_L^* + \text{Tr}[(I - \Pi_{\leq L})X] |\psi\rangle\langle\psi|^{\otimes L}. \quad (140)$$

For $Z \in \psi^\perp$, distinguish its coherent Fock vector $|\text{coh}(Z)\rangle = e^{-\|Z\|^2/2} \bigoplus_{t \geq 0} Z^{\otimes t} / \sqrt{t!}$ from the one-particle vector $|Z\rangle$, and let

$$|\psi_{Z,L}\rangle = \frac{|\psi\rangle + L^{-1/2}|Z\rangle}{\sqrt{1 + \|Z\|^2/L}}.$$

For fixed Z ,

$$\left\| \mathcal{J}_L(|\text{coh}(Z)\rangle\langle\text{coh}(Z)|) - |\psi_{Z,L}\rangle\langle\psi_{Z,L}|^{\otimes L} \right\|_1 \longrightarrow 0. \quad (141)$$

For $Z \neq 0$, both vectors have excitations only in the direction $Z/\|Z\|$ (the case $Z = 0$ is immediate). With $t = \|Z\|^2$, their number amplitudes after pulling back the product vector by J_L^* are the square roots of

$$B_L(j) = \binom{L}{j} \frac{(t/L)^j}{(1 + t/L)^L}, \quad \text{Poist}(j) = e^{-t} \frac{t^j}{j!}.$$

Pointwise convergence and the discrete Scheffé lemma give

$$\sum_{j \geq 0} (\sqrt{B_L(j)} - \sqrt{\text{Poist}(j)})^2 \leq \sum_{j \geq 0} |B_L(j) - \text{Poist}(j)| \longrightarrow 0.$$

Thus the pure states converge in trace norm. Contractivity of $\mathcal{J}_L$ proves (141).

For $s > 0$, let μ_s be the circular complex Gaussian measure on $\psi^\perp$ with covariance sI , so each orthonormal coordinate has $\mathbb{E}|Z_j|^2 = s$. Gaussian integration in the number basis gives the positive coherent-state representation

$$\tau_{s/(1+s)}^{\otimes(D-1)} = \int_{\psi^\perp} |\text{coh}(Z)\rangle\langle\text{coh}(Z)| d\mu_s(Z), \quad d\mu_s(Z) = \frac{e^{-\|Z\|^2/s}}{(\pi s)^{D-1}} dZ. \quad (142)$$

Specifically, the integral is diagonal with eigenvalue $s^{|\mathbf{k}|}/(1+s)^{|\mathbf{k}|+D-1}$. Combining (137), (139), and (141) therefore gives, for any fixed purification $\psi \in \mathbb{C}^d \otimes E$,

$$\left\| \mathcal{C}_{n,m_n}^{\text{PCT},r}(\rho^{\otimes n}) - \int_{\psi^\perp} \rho_{Z,m_n}^{\otimes m_n} d\mu_{\gamma^{-1}}(Z) \right\|_1 \longrightarrow 0, \quad \rho_{Z,L} = \text{Tr}_E |\psi_{Z,L}\rangle\langle\psi_{Z,L}|. \quad (143)$$

Here the finite Werner output equals $\mathcal{J}_{m_n}(\omega_{n,m_n})$. Contractivity controls its replacement by the thermal state, and dominated convergence controls the coherent-state integral, whose trace-distance integrand is bounded by 2. No uniform estimate over unbounded Z is needed.

D.2 Simple full-rank spectra

Fix $p \in \Delta_d^\circ$, so that $\rho_p = \text{diag}(p_1, \dots, p_d)$ with $p_1 > \dots > p_d > 0$, and recall $q_{ij} = p_j/p_i$. Define

$$q_{ij}^{\text{PCT}} = \frac{\gamma - 1 + \gamma q_{ij}}{\gamma + (\gamma - 1)q_{ij}}. \quad (144)$$

The asymptotic PCT fidelity is

$$F_{\text{PCT}}(\gamma, p) = \left(\frac{\sqrt{2\gamma - 1}}{\gamma} \right)^{(d-1)/2} \prod_{i < j} F(\tau_{q_{ij}}, \tau_{q_{ij}^{\text{PCT}}}). \quad (145)$$

The equivalent closed form is displayed in (11). PCT here uses rank bound d , hence purification dimension d^2 .

Proof of Theorem 1.4(a). The value F_{univ} is attained by Theorem 1.2(a). For the PCT limit, apply interior LAN to the mixture (143) and compare the resulting factors.

Step 1: The Gaussian output. Choose

$$|\psi_p\rangle = \sum_i \sqrt{p_i} |ii\rangle.$$

Identify the tangent vector $|Z\rangle = \sum_{i,j} Z_{ij} |ij\rangle$ with its coefficient matrix. Partial trace has differential

$$D_p(Z) = Z\sqrt{\rho_p} + \sqrt{\rho_p}Z^*, \quad \rho_{Z,L} = \rho_p + L^{-1/2}D_p(Z) + O_Z(L^{-1}). \quad (146)$$

Its diagonal and orbital LAN coordinates are

$$h_i(Z) = 2\sqrt{p_i} \Re Z_{ii}, \quad \sum_i h_i(Z) = 0, \quad (147)$$

$$\beta_{ij}(Z) = \frac{D_p(Z)_{ji}}{\sqrt{p_i - p_j}} = \frac{\sqrt{p_i}Z_{ji} + \sqrt{p_j}\overline{Z_{ij}}}{\sqrt{p_i - p_j}}, \quad i < j. \quad (148)$$

The derivative of the local chart from Section 3.1 is $(u, w) \mapsto \text{diag}(u) + [Y_p(w), \rho_p]$, a real linear isomorphism onto the traceless Hermitian matrices. The inverse function theorem and (146) give exact LAN coordinates:

$$\rho_{Z,L} = g_{L,z_L}^{(p)} \text{diag}(p + L^{-1/2}h_L)(g_{L,z_L}^{(p)})^*, \quad (h_L, z_L) = (h(Z), \beta(Z)) + O_Z(L^{-1/2}).$$

Let T_L^p, S_L^p be the forward and reverse interior LAN channels centered at ρ_p , in the normalization of (47), and set

$$\Theta(Z) = \mathbf{N}_{h(Z), \Sigma_p} \otimes \bigotimes_{i < j} D(\beta_{ij}(Z)) \tau_{q_{ij}} D(\beta_{ij}(Z))^*. \quad (149)$$

The bounded-parameter LAN estimates (48)–(49) and trace-norm continuity of Gaussian shifts imply, for each fixed Z ,

$$\|T_L^p(\rho_{Z,L}^{\otimes L}) - \Theta(Z)\|_1 \rightarrow 0, \quad \|\rho_{Z,L}^{\otimes L} - S_L^p(\Theta(Z))\|_1 \rightarrow 0.$$

To average $\Theta(Z)$ under $Z \sim \mu_{\gamma-1}$, use the orthogonal decomposition:

$$\psi_p^\perp = \left\{ \sum_i z_i |ii\rangle : \sum_i \sqrt{p_i} z_i = 0 \right\} \oplus \bigoplus_{i < j} \text{span}\{|ij\rangle, |ji\rangle\}$$

The diagonal and pair variables are independent, and (147)–(148) give

$$\text{Cov}(h) = 2(\gamma - 1)\Sigma_p, \quad \mathbb{E}|\beta_{ij}|^2 = \frac{(\gamma - 1)(p_i + p_j)}{p_i - p_j}, \quad \mathbb{E}\beta_{ij}^2 = 0. \quad (150)$$

Here the diagonal Gaussian has covariance $(\gamma - 1)(I - \sqrt{p}\sqrt{p}^\top)$. Multiplying its real parts by $2\sqrt{p_i}$ gives the first identity on the trace-zero hyperplane. Each β_{ij} is circular complex Gaussian. Averaging its displacement multiplies the Weyl characteristic function by $\exp(-\mathbb{E}|\beta_{ij}|^2|\xi|^2)$, leaving a thermal state whose mean photon number increases by $\mathbb{E}|\beta_{ij}|^2$:

$$\bar{n}_{ij}^{\text{PCT}} = \frac{p_j + (\gamma - 1)(p_i + p_j)}{p_i - p_j} = \frac{(\gamma - 1)p_i + \gamma p_j}{p_i - p_j}, \quad (151)$$

whose geometric parameter is (144). Consequently,

$$\Theta_p^{\text{PCT}} := \int_{\psi_p^\perp} \Theta(Z) d\mu_{\gamma-1}(Z) = \mathbf{N}_{0,(2\gamma-1)\Sigma_p} \otimes \bigotimes_{i < j} \tau_{q_{ij}^{\text{PCT}}}. \quad (152)$$

Integrating both LAN approximations by dominated convergence and using (143) gives

$$\left\| T_{m_n}^p(\mathcal{C}_{n,m_n}^{\text{PCT},d}(\rho_p^{\otimes n})) - \Theta_p^{\text{PCT}} \right\|_1 \longrightarrow 0, \quad (153)$$

$$\left\| \mathcal{C}_{n,m_n}^{\text{PCT},d}(\rho_p^{\otimes n}) - S_{m_n}^p(\Theta_p^{\text{PCT}}) \right\|_1 \longrightarrow 0. \quad (154)$$

The target has the same two-sided approximations with

$$\Theta_p = \Theta(0) = \mathbf{N}_{0,\Sigma_p} \otimes \bigotimes_{i < j} \tau_{q_{ij}}. \quad (155)$$

Step 2: Fidelity and strict comparison. Monotonicity under $T_{m_n}^p$ and $S_{m_n}^p$, together with trace-norm continuity, now gives the two-sided bound

$$F(\Theta_p^{\text{PCT}}, \Theta_p) - o(1) \leq F(\mathcal{C}_{n,m_n}^{\text{PCT},d}(\rho_p^{\otimes n}), \rho_p^{\otimes m_n}) \leq F(\Theta_p^{\text{PCT}}, \Theta_p) + o(1).$$

The limiting fidelity factorizes, with

$$F(\mathbf{N}_{0,(2\gamma-1)\Sigma_p}, \mathbf{N}_{0,\Sigma_p}) = \left(\frac{\sqrt{2\gamma-1}}{\gamma} \right)^{(d-1)/2} \quad (156)$$

and $F(\tau_q, \tau_x) = \sqrt{(1-q)(1-x)}/(1-\sqrt{qx})$. This proves the exact value (145).

A thermal state τ_x has mean photon number $x/(1-x)$. For one mode with $q \in (0,1)$, the optimal amplifier has mean $\bar{n}^{\text{opt}} = \gamma q/(1-q) + \gamma - 1$, hence

$$\bar{n}^{\text{opt}} > \frac{q}{1-q}, \quad \bar{n}^{\text{PCT}} - \bar{n}^{\text{opt}} = \frac{(\gamma-1)q}{1-q} > 0.$$

Thus $q < q^{\text{opt}} < q^{\text{PCT}}$, where $q^{\text{opt}} = (\gamma-1+q)/\gamma$ and $f_\gamma(q) = F(\tau_q, \tau_{q^{\text{opt}}})$. Since

$$\frac{d}{dx} \log F(\tau_q, \tau_x) = \frac{\sqrt{q} - \sqrt{x}}{2\sqrt{x}(1-x)(1-\sqrt{qx})} < 0, \quad x > q,$$

every PCT quantum factor is strictly smaller. PCT's classical factor is also smaller: $c \mapsto 2\sqrt{c}/(1+c)$ decreases for $c > 1$, and $2\gamma-1 > \gamma > 1$. Multiplying the factors proves $F_{\text{PCT}}(\gamma, p) < F_{\text{univ}}(\gamma, p)$. $\square$

D.3 Projector states

For projector states, we upper-bound PCT's actual fidelity to compare it with the optimum in Theorem 1.3. Its published data-processing lower bound does not suffice for this comparison.

Fix $1 \leq r < d$ and $P \in \text{Gr}(r, d)$, choose an isometry $V_P : \mathbb{C}^r \rightarrow \mathbb{C}^d$ with $V_P V_P^* = P$, and let

$$|\psi_P\rangle = \frac{1}{\sqrt{r}} \sum_{i=1}^r V_P |i\rangle \otimes |i\rangle, \quad |\psi_r\rangle = \frac{1}{\sqrt{r}} \sum_{i=1}^r |ii\rangle.$$

Proposition D.2 (Finite-sample PCT factorization for projector states). *Define*

$$s_{n,m}^{(r,d)} = \frac{\mathbf{d}_{rd}(n)/\mathbf{d}_{rd}(m)}{\mathbf{d}_{r^2}(n)/\mathbf{d}_{r^2}(m)} = \prod_{j=r^2}^{rd-1} \frac{n+j}{m+j}. \quad (157)$$

There is an exact finite-sample factorization

$$F\left(\mathcal{C}_{n,m}^{\text{PCT},r}((P/r)^{\otimes n}), (P/r)^{\otimes m}\right) = \sqrt{s_{n,m}^{(r,d)}} F\left(\zeta_{n,m}^{(r)}, (I_r/r)^{\otimes m}\right), \quad (158)$$

where

$$\zeta_{n,m}^{(r)} = \text{Tr}_{E^m} \mathcal{W}_{n,m}^{(r^2)}(|\psi_r\rangle\langle\psi_r|^{\otimes n}). \quad (159)$$

Proof. By (137), use the fixed purification ψ_P . Werner's channel is the one-row Cartan channel, so (57) applies with $\mu = (n)$, $\nu = (m)$, and $V_P \otimes I_r : \mathbb{C}^{r^2} \rightarrow \mathbb{C}^{rd}$. On the symmetric output, its compression projection is $(P \otimes I_r)^{\otimes m}$ and its dimension factor is $s_{n,m}^{(r,d)}$ by (157). Tracing out E^m gives

$$P^{\otimes m} \mathcal{C}_{n,m}^{\text{PCT},r}((P/r)^{\otimes n}) P^{\otimes m} = s_{n,m}^{(r,d)} V_P^{\otimes m} \zeta_{n,m}^{(r)} (V_P^*)^{\otimes m}. \quad (160)$$

Compressing to the target support $P^{\otimes m}$ leaves fidelity unchanged. Homogeneity then proves (158). $\square$

Proof of Theorem 1.4(b). By Proposition D.2, it remains to bound the internal fidelity. Assume $r > 1$, let

$$\mathbf{H}_r = \left\{ h \in \mathbb{R}^r : \sum_{i=1}^r h_i = 0 \right\}, \quad \Sigma_r = \left(\frac{1}{r} I_r - \frac{1}{r^2} \mathbf{1}_r \mathbf{1}_r^\top \right) \Big|_{\mathbf{H}_r},$$

and measure every system output in the computational basis. If $C_m = (C_{m,1}, \dots, C_{m,r})$ is the resulting count vector, its centered and rescaled version belongs to $\mathbf{H}_r$. Under the target $(I_r/r)^{\otimes m}$, the multinomial central limit theorem gives

$$\frac{C_m - (m/r) \mathbf{1}_r}{\sqrt{m}} \longrightarrow \mathbf{N}_{0, \Sigma_r} \quad \text{in distribution.} \quad (161)$$

Under $\zeta_{n,m_n}^{(r)}$, we will show

$$\frac{C_{m_n} - (m_n/r) \mathbf{1}_r}{\sqrt{m_n}} \longrightarrow \mathbf{N}_{0, (2\gamma-1)\Sigma_r} \quad \text{in distribution.} \quad (162)$$

Apply (143) to the internal state, with condensate ψ_r , dimension r^2 , and $Z \sim \mu_{\gamma-1}$ on $\psi_r^\perp$. Identify Z with a traceless complex $r \times r$ matrix. Conditional on Z , measuring the system part of the product of the normalized vector $\psi_r + m^{-1/2} Z$ gives multinomial probabilities

$$\begin{aligned} p_i^{(m)}(Z) &= \frac{r^{-1} + 2\Re Z_{ii}/\sqrt{rm} + m^{-1} \sum_j |Z_{ij}|^2}{1 + m^{-1} \|Z\|^2} \\ &= \frac{1}{r} + \frac{h_i(Z)}{\sqrt{m}} + O_Z(m^{-1}), \quad h_i(Z) = \frac{2\Re Z_{ii}}{\sqrt{r}}. \end{aligned} \quad (163)$$

The multinomial central limit theorem therefore gives the conditional limit $\mathbf{N}_{h(Z), \Sigma_r}$. Since $\text{Tr } Z = 0$, the mean $h(Z)$ belongs to $\mathbf{H}_r$. Under $\mu_{\gamma-1}$ it is centered Gaussian, with

$$\text{Cov}(h(Z))_{ij} = \frac{2(\gamma-1)}{r} \left(\delta_{ij} - \frac{1}{r} \right) = 2(\gamma-1)(\Sigma_r)_{ij}.$$

Dominated convergence of the conditional characteristic functions gives the Gaussian limit with covariance $\Sigma_r + 2(\gamma - 1)\Sigma_r = (2\gamma - 1)\Sigma_r$. The trace-norm approximation (143) and contractivity of measurement transfer this limit to $\zeta_{n,m_n}^{(r)}$, proving (162).

Let L_n, T_n be the laws of the centered, rescaled count vectors, converging weakly to their Gaussian limits L, T , respectively. For any finite partition $\mathcal{P}$ whose boundaries have zero Gaussian measure, coarse-graining and weak convergence give

$$\limsup_n F(L_n, T_n) \leq \sum_{A \in \mathcal{P}} \sqrt{L(A)T(A)}.$$

Along a refining sequence generating the Borel sets, the right-hand side decreases to $F(L, T)$. This upper semicontinuity, fidelity monotonicity under the count measurement, and (161)–(162) give

$$\limsup_{n \rightarrow \infty} F(\zeta_{n,m_n}^{(r)}, (I_r/r)^{\otimes m_n}) \leq \left(\frac{\sqrt{2\gamma - 1}}{\gamma} \right)^{(r-1)/2}. \quad (164)$$

Finally, (157) gives

$$s_{n,m_n}^{(r,d)} \longrightarrow \gamma^{-r(d-r)}.$$

Combining this with (158) and (164) proves (12). Its right-hand side is strictly smaller than $\gamma^{-r(d-r)/2}$ when $1 < r < d$, because $2\gamma - 1 < \gamma^2$. $\square$

Proposition D.3 (PCT's leading projector infidelity coefficient). *Fix $1 < r < d$ and $P \in \text{Gr}(r, d)$, let $a = r(d - r)$, and, for each fixed $0 < \delta < 1/(2r)$, take $m_n/n \rightarrow 1 + \delta$. Define*

$$\mathcal{E}_n(\delta) = 1 - F(\mathcal{C}_{n,m_n}^{\text{PCT},r}((P/r)^{\otimes n}), (P/r)^{\otimes m_n})^2, \quad C_r(\delta) = (1 - 4\delta^2)^{-(r^2-1)/2} - 1.$$

Then

$$1 - (1 + \delta)^{-a} \leq \liminf_n \mathcal{E}_n(\delta) \leq \limsup_n \mathcal{E}_n(\delta) \leq 1 - (1 + \delta)^{-a} + (1 + \delta)^{-a} C_r(\delta). \quad (165)$$

The restriction $\delta < 1/(2r)$ is used only for domination in the proof. Thus both envelopes equal $a\delta + O_{r,d}(\delta^2)$ as $\delta \downarrow 0$. In this iterated limit, PCT has the same leading coefficient of $1 - F^2$ as the optimal projector cloner.

Proof. It suffices to bound the internal squared infidelity in (158). Specialize the mixture (143) to the condensate ψ_r , and write

$$\tilde{\zeta}_L = \int_{\psi_r^\perp} \rho_{Z,L}^{\otimes L} d\mu_\delta(Z), \quad \rho_{Z,L} = \frac{I_r}{r} + \frac{AZ}{\sqrt{L}} + O_Z(L^{-1}), \quad A_Z = \frac{Z + Z^*}{\sqrt{r}}.$$

Here Z is a traceless complex $r \times r$ matrix, and $\|\zeta_{n,m_n}^{(r)} - \tilde{\zeta}_{m_n}\|_1 \rightarrow 0$. For independent $Z, W \sim \mu_\delta$, the scaled purity satisfies

$$r^L \text{Tr} \tilde{\zeta}_L^2 = \mathbb{E} \left[(r \text{Tr}(\rho_{Z,L} \rho_{W,L}))^L \right], \\ (r \text{Tr}(\rho_{Z,L} \rho_{W,L}))^L \longrightarrow e^{r \text{Tr}(A_Z A_W)}.$$

To justify exchanging limit and expectation, contraction under partial trace and the pure-state trace-distance formula give

$$\left\| \rho_{Z,L} - \frac{I_r}{r} \right\|_2 \leq \| |\psi_{Z,L}\rangle\langle\psi_{Z,L}| - |\psi_r\rangle\langle\psi_r| \|_1 = 2\sqrt{\frac{\|Z\|^2}{L + \|Z\|^2}}.$$

Since both perturbations have trace zero, Cauchy–Schwarz and $1 + x \leq e^x$ yield

$$0 \leq (r \operatorname{Tr}(\rho_{Z,L} \rho_{W,L}))^L \leq e^{4r\|Z\|\|W\|} \leq e^{2r(\|Z\|^2 + \|W\|^2)}.$$

The last expression is integrable when $\delta < 1/(2r)$. In a Hilbert–Schmidt orthonormal basis of traceless Hermitian matrices, the real parts x_j, y_j of the coordinates of Z, W are independent $\mathcal{N}_{0,\delta/2}$ variables and $r \operatorname{Tr}(A_Z A_W) = 4 \sum_{j=1}^{r^2-1} x_j y_j$. Dominated convergence and the Gaussian cross-moment therefore give

$$\lim_{L \rightarrow \infty} r^L \operatorname{Tr} \tilde{\zeta}_L^2 = \prod_{j=1}^{r^2-1} \mathbb{E} e^{4x_j y_j} = (1 - 4\delta^2)^{-(r^2-1)/2} = 1 + C_r(\delta). \quad (166)$$

For any state σ on $(\mathbb{C}^r)^{\otimes L}$,

$$1 - F\left(\sigma, (I_r/r)^{\otimes L}\right)^2 \leq r^L \operatorname{Tr} \sigma^2 - 1.$$

Indeed, apply $(\sqrt{x} - 1)^2 \leq (x - 1)^2$ to the relative eigenvalues $x = r^L \lambda_j(\sigma)$, average with weights r^{-L} , and use $1 - F^2 \leq 2(1 - F)$. Apply this to $\tilde{\zeta}_L$ and use fidelity continuity to transfer the bound to the PCT output:

$$\limsup_n \left[1 - F\left(\zeta_{n,m_n}^{(r)}, (I_r/r)^{\otimes m_n}\right)^2 \right] \leq C_r(\delta). \quad (167)$$

Finally, the exact support factorization and $s_{n,m_n}^{(r,d)} \rightarrow (1 + \delta)^{-a}$ imply (165). Since $C_r(\delta) = 2(r^2 - 1)\delta^2 + O_r(\delta^4)$, the asserted first-order coefficient follows. For any fixed $0 < \eta < 1$ and sufficiently small $\epsilon > 0$, both envelopes lie below ϵ at $\delta = (1 - \eta)\epsilon/a$ and above it at $\delta = (1 + \eta)\epsilon/a$. Letting $\eta \downarrow 0$ gives the leading sample coefficient a in Section 6.3, without assuming convergence of the fidelity at fixed δ . $\square$

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